

Diagnosing Mitigation Need: Benchmark Results for Controlled Shortages and Signal-Matched Interventions in Histopathology Patch Classification

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Abstract

Status. This is a legacy update-matched execution; its recovery and cost conclusions are descriptive pending the exposure-matched rerun.

A class can be underserved for several distinct reasons, and the count that marks it as rare does not say which. This report presents the outcomes of the controlled patch-classification benchmark specified in the companion protocol (Queisler, 2026a), which manipulates only the allocation of a fixed training budget among classes and measures what that manipulation damages. CAMELYON16, TCGA-UT, BRACS, and PANDA are evaluated with frozen Virchow2 features under size-matched balanced, moderate, and severe allocations, against natural-prevalence validation and test cohorts held constant across conditions. Three questions are answered in order: which shortage the controlled allocation realizes and what it damages, whether a mitigation method that reads the shortage matching that profile recovers more of the damage than one reading a different signal, and which realized shortages explain variation in damage magnitude and recovery headroom. Mitigation recovery is interpreted only for conditions whose allocation-induced deficit passes the prespecified gate, and the confirmatory family holds the intervention mechanism fixed so that the signal a method reads is the only quantity contrasted. Controlled allocation opened the discrimination gate in 12 of 18 comparison units and the probability-quality gate in 6, all of them on BRACS, with balanced-accuracy deficits spanning 0.000,7 to 0.134; damage grew with achieved severity and with the number of classes among which the fixed budget was divided. Recovery separated sharply but not along the axis the design was built to test: the prevalence- and nominal-support-weighted members of the confirmatory family returned most of the deficit wherever a gate opened, while the independent-support, difficulty, and diversity members returned essentially none of it and degraded tail probability quality further. The closing discussion converts these outcomes into design requirements for a mitigation method that reads shortage rather than prevalence alone, which is the purpose the benchmark serves.

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	These legacy differences do not support a roster-wide cost conclusion; that comparison is deferred to the exposure-matched rerun.	

1 Introduction

Status. These are legacy update-matched results. Because methods consumed unequal examples per update, recovery and cost conclusions are descriptive only until the exposure-matched rerun.

This report presents the results of the controlled patch-classification benchmark whose design, endpoints, gating rule, and inference procedure are fixed in the companion protocol (Queisler, 2026a). Nothing in the analysis reported here was chosen after seeing an outcome: the estimands, the deficit thresholds, the confirmatory method family, the multiplicity correction, and the exploratory status of the association models were all locked before the first definitive model fit. The reporting order below follows that locked plan rather than the order in which results proved interesting, and any departure from the plan is recorded in Appendix A.

The benchmark exists to inform the design of a new mitigation method, not to rank the existing ones. Its premise is that an underserved class may lack nominal patch support, independent evidence from patients or slides, learnable signal, or representational diversity, and that these shortages are not interchangeable. Existing methods are therefore treated as instruments: each reads one signal, and the contrast among them measures whether reading the right signal matters. The report is consequently read forward into Section 5, where the outcomes are converted into requirements a method would have to satisfy. Dataset provenance is documented separately (Queisler, 2026b), as is the signal taxonomy that locates each method within it (Queisler, 2026c).

The protocol poses three questions, restated here so that the results can be read without the companion document at hand:

RQ1. Which shortage does controlled allocation create, and what does it damage?

Under otherwise fixed conditions, how do controlled changes in training allocation affect tail-class discrimination and calibration, and which signal profile does each condition realize?

RQ2. Does signal-matched mitigation recover more of the damage? Which methods recover performance lost specifically to training imbalance, and does recovery improve when a method reads the shortage identified in RQ1?

RQ3. Which realized signal shortages explain variation in damage and mitigation headroom? Across controlled conditions, are allocation severity, independent-evidence shortage, support–difficulty alignment, and diversity shortage associated with damage magnitude and subsequent recovery?

The three sections that follow answer them in that order. Section 2 establishes whether a recoverable deficit exists and which signals characterize the conditions that produced it; the deficit gate defined there determines which conditions Section 3 is permitted to interpret. Section 4 then examines why deficit size and recovery headroom vary, as an exploratory association analysis rather than a second method ranking.

Two conventions govern how the evidence is labelled throughout. A result is *confirmatory* only if it belongs to the five-member method family, arises from a condition that passed the relevant gate, and survives the Holm adjustment applied jointly within its dataset. Every other quantity is *exploratory* or *descriptive*, and is marked as such in the text and in the table caption rather than only in the surrounding prose.

2 Allocation-Induced Damage

The controlled conditions must first be shown to have executed as specified, because the inference machinery is conditional on that. Three properties matter before any endpoint is read:

whether constrained allocation reached its target severities within the prespecified tolerance bands, whether the label-only bootstrap preflight left any dataset designated descriptive-only, and whether every confirmation seed block and every expected comparison is present across all three partitions. A descriptive-only designation forces both gates closed for that dataset, which removes it from the confirmatory results entirely; an incomplete comparison halts aggregation rather than being averaged over whatever is available. Appendix A carries the full record together with the tuning selections actually used and any deviation from the locked protocol.

All three properties hold. Every moderate condition realized ρ_{achieved} within $[9, 11]$ and every severe condition within $[90, 110]$, the tightest case being the difficulty-aligned severe condition on TCGA-UT at $\rho_{\text{achieved}} = 90.11$ to 91.02 ; no condition collapsed to the balanced allocation, so none is degenerate. No class on any dataset failed the preflight concentration checks, so no dataset is descriptive-only and both gates remain available everywhere. All four datasets carry three completed partitions, five confirmation seeds in every method block, the full expected set of comparisons on each partition, and an achieved-severity spread across partitions of at most 1.0% of the smallest realized value. The controlled comparison therefore rests on 18 comparison units: six on BRACS and six on TCGA-UT, which retain three tail assignments each, four on CAMELYON16, and two on PANDA, whose binary target admits only the native assignment once duplicated mappings are omitted.

2.1 Realized signal profiles

Allocation severity is the manipulated variable, but it is not the explanatory one. Each controlled condition is characterized instead by the signal profile it realizes: achieved imbalance ratio ρ and normalized-entropy deficit $1 - H_{\text{norm}}$, nominal patch-support shortage, independent patient or slide support shortage, support–difficulty alignment, and representational-diversity shortage. These quantities are measured from the frozen construction rather than inferred from the nominal target, so a condition built to a severe target may still present a mild independent-support shortage, or the reverse.

Table 1 reports the standardized shortage scores per comparison unit together with the dominant shortage each unit presents. The dominant shortage is what makes the matched and unmatched arms of Section 3 interpretable, so units whose two largest scores fall within 0.25 standard deviations are designated ambiguous and are reported descriptively only, never as evidence for or against signal matching. Six of the 18 units carry that designation: BRACS under the native moderate and difficulty-aligned severe conditions, CAMELYON16 under difficulty-reversed moderate, PANDA under native moderate, and TCGA-UT under native moderate and difficulty-aligned severe.

One column of that table requires comment before any of it is interpreted, because it constrains what the rest of the report can conclude. The standardized independent-support shortage is exactly zero in every one of the 18 units. This is a consequence of the design rather than a property of the datasets: the protocol fixes one per-class pool of partitioning units and slides and makes the same pool available to the balanced condition and to every imbalanced condition, so reallocating patches among classes changes how many patches a class draws but not the independent evidence it may draw from. Three consequences follow and are carried through the remainder of the report. Independent-support weighted cross-entropy can never be matched to a shortage that the design actually created. A unit whose four shortage scores are all negative will be labelled with independent support as its dominant shortage purely because zero is then the largest of them, which is what happens in the difficulty-reversed moderate conditions on BRACS and TCGA-UT. And the independent-support coefficient of the RQ3 models in Section 4 is fixed at zero by a predictor with no variance, not estimated from data. The benchmark manipulates allocation within a fixed pool of patients and slides by construction, and the price of that control is that one of the four taxonomy axes is held constant rather than varied.

Dataset	Assignment	Severity	ρ	$1 - H_{\text{norm}}$	S_{nom}	S_{ind}	S_{div}	A	Dominant
BRACS	Native	Moderate	10.00	0.128	-0.86	0.00	-0.86	0.36	Ambiguous
BRACS	Aligned	Moderate	10.00	0.128	-0.86	0.00	-0.38	-0.92	Difficulty
BRACS	Reversed	Moderate	10.00	0.128	-0.86	0.00	-0.72	0.92	Independent support
BRACS	Native	Severe	100.38	0.353	-0.01	0.00	0.89	0.36	Diversity
BRACS	Aligned	Severe	100.38	0.353	-0.01	0.00	1.42	-0.92	Ambiguous
BRACS	Reversed	Severe	100.38	0.353	-0.01	0.00	0.88	0.92	Diversity
CAMELYON16	Native	Moderate	10.00	0.560	-0.05	0.00	-0.78	-1.00	Difficulty
CAMELYON16	Reversed	Moderate	10.00	0.560	-0.05	0.00	-1.38	1.00	Ambiguous
CAMELYON16	Native	Severe	99.98	0.920	2.07	0.00	1.16	-1.00	Nominal support
CAMELYON16	Reversed	Severe	99.98	0.920	2.07	0.00	0.14	1.00	Nominal support
PANDA	Native	Moderate	10.00	0.561	-0.05	0.00	-1.42	1.00	Ambiguous
PANDA	Native	Severe	100.00	0.920	2.07	0.00	-1.35	1.00	Nominal support
TCGA-UT	Native	Moderate	10.00	0.061	-1.00	0.00	-0.50	0.09	Ambiguous
TCGA-UT	Aligned	Moderate	10.00	0.061	-1.00	0.00	-0.31	-0.99	Difficulty
TCGA-UT	Reversed	Moderate	10.00	0.061	-1.00	0.00	-0.40	0.99	Independent support
TCGA-UT	Native	Severe	100.17	0.179	-0.14	0.00	1.15	0.09	Diversity
TCGA-UT	Aligned	Severe	90.43	0.174	-0.19	0.00	1.31	-0.99	Ambiguous
TCGA-UT	Reversed	Severe	100.17	0.179	-0.14	0.00	1.33	0.99	Diversity

Table 1: Realized signal profiles per comparison unit: achieved ρ , normalized-entropy deficit, standardized nominal, independent-support and diversity shortages, and support–difficulty alignment, with the dominant shortage and the ambiguous designation applied where the two largest scores fall within 0.25 standard deviations.

2.2 Discrimination deficit and gate status

The discrimination deficit D_{BA} is the case-macro balanced accuracy of the size-matched balanced cross-entropy reference minus that of the imbalanced cross-entropy model in the same condition, estimated with the patient-clustered Bayesian bootstrap and averaged with equal weight across the three partitions. The discrimination gate opens for a comparison unit only when D_{BA} reaches at least 0.014,43 and its paired 95% interval excludes zero, a threshold fixed at twice the observed seed-level standard deviation so that a deficit indistinguishable from initialization noise cannot open it.

Table 2 gives the deficit, its interval, and the gate outcome for every combination of dataset, severity, and tail assignment. Deficits are reported for all units, including those that fail the threshold, the interval condition, or both, because a condition that produced no measurable damage is itself a finding about when mitigation is warranted.

Twelve of the 18 units opened the gate. Every one of the 18 deficits is positive, and only two intervals cover zero, on the CAMELYON16 native moderate and TCGA-UT difficulty-aligned moderate conditions, so redistributing a fixed budget toward head classes never improved case-macro balanced accuracy. Three regularities organize the table. Deficits grow with achieved severity in every dataset and tail assignment: BRACS under the difficulty-reversed assignment moves from 0.081 at $\rho = 10$ to 0.134 at $\rho = 100$, and PANDA from 0.026 to 0.146. Deficits grow with the difficulty-reversed assignment, which gives the least support to the classes that are easiest to separate and therefore concentrates scarcity where it cannot be compensated: on BRACS the reversed assignment damages roughly three times as much as the difficulty-aligned one at both severities. Finally, the two binary datasets and the two multiclass datasets do not behave alike, but not in the direction class count alone would suggest: PANDA, with two classes, records the largest severe deficit of all (0.146), while TCGA-UT, with 30, records deficits of at most 0.016. What separates them is how much of the fixed budget the tail retains. A binary severe allocation places 99% of the budget in one class, whereas a 30-class allocation on a logarithmic support profile leaves the smallest class with a share that is small relative to the head but still absolutely large, which is also visible in the normalized-entropy deficit: 0.920 for PANDA severe against 0.179 for TCGA-UT severe at the same nominal $\rho = 100$.

Dataset	Assignment	Severity	D_{BA}	95% CI	Gate
BRACS	Native	Moderate	0.0325	[0.0152, 0.0529]	Open
BRACS	Aligned	Moderate	0.0264	[0.0083, 0.0461]	Open
BRACS	Reversed	Moderate	0.0810	[0.0592, 0.1054]	Open
BRACS	Native	Severe	0.0604	[0.0383, 0.0834]	Open
BRACS	Aligned	Severe	0.0445	[0.0244, 0.0650]	Open
BRACS	Reversed	Severe	0.1337	[0.1073, 0.1610]	Open
CAMELYON16	Native	Moderate	0.0022	[-0.0008, 0.0071]	Below threshold
CAMELYON16	Reversed	Moderate	0.0033	[0.0015, 0.0051]	Below threshold
CAMELYON16	Native	Severe	0.0170	[0.0086, 0.0278]	Open
CAMELYON16	Reversed	Severe	0.0182	[0.0137, 0.0232]	Open
PANDA	Native	Moderate	0.0260	[0.0233, 0.0287]	Open
PANDA	Native	Severe	0.1458	[0.1396, 0.1510]	Open
TCGA-UT	Native	Moderate	0.0043	[0.0015, 0.0072]	Below threshold
TCGA-UT	Aligned	Moderate	0.0007	[-0.0015, 0.0030]	Below threshold
TCGA-UT	Reversed	Moderate	0.0049	[0.0021, 0.0077]	Below threshold
TCGA-UT	Native	Severe	0.0161	[0.0115, 0.0208]	Open
TCGA-UT	Aligned	Severe	0.0110	[0.0067, 0.0150]	Below threshold
TCGA-UT	Reversed	Severe	0.0146	[0.0093, 0.0198]	Open

Table 2: Discrimination deficit D_{BA} with paired 95% bootstrap intervals and discrimination-gate outcome, by dataset, severity, and tail assignment. Units failing the 0.014,43 threshold or the interval condition are shown and marked as not opening the gate.

2.3 Probability-quality deficit

Discrimination and probability quality can be damaged independently, so the calibration axis carries its own deficit and its own gate. The co-primary endpoint is tail-group macro negative log likelihood in nats, computed over the classes that the condition places in the tail tier; its gate opens at a deficit of at least 0.182,37 nats with a paired interval excluding zero. Expected calibration error, multiclass Brier score, and the temperature-scaled variant of each are secondary and never gate.

Table 3 reports the tail-group deficit and gate outcome per unit alongside crossed patient-bootstrap intervals on expected calibration error. Temperature scaling is fitted on natural-prevalence validation data and reported only as a diagnostic of how much of the observed miscalibration a single global scalar removes; Appendix C carries the raw and scaled comparison in full.

The calibration axis behaves very differently from the discrimination axis, and the difference is the sharpest single contrast in the report. All six BRACS units opened the calibration gate, with deficits from 0.281 to 2.047 nats, and no unit on any other dataset came close: the largest deficit outside BRACS is 0.144 nats, below the 0.182,37 threshold. More striking, seven units record a *negative* tail-group deficit, meaning the imbalanced model assigns better-calibrated probabilities to its own tail classes than the size-matched balanced reference does. Five of those seven are on TCGA-UT, where the effect reaches -0.201 nats; the other two are CAMELYON16 and PANDA under their moderate conditions. The secondary expected calibration error moves in the same direction as the primary endpoint wherever the gate opened, rising from 0.159 to 0.317 across the BRACS units as severity and assignment adversity increase, and remains at or below 0.026 on CAMELYON16 and PANDA throughout.

The reading these numbers support is that tail-group probability quality is damaged only where the tail classes are genuinely hard under the frozen representation. BRACS is the one dataset whose balanced-condition balanced accuracy sits near 0.49 on seven classes, so its tail classes are poorly separated even with even support, and removing support from them collapses the probability mass they receive. On CAMELYON16 and PANDA the tail remains highly separable, and on TCGA-UT the balanced reference already reaches 0.755 across 30 classes.

Dataset	Assignment	Severity	D_{cal} (nats)	95% CI	Gate	ECE	ECE 95% CI
BRACS	Native	Moderate	0.281	[0.123, 0.465]	Open	0.159	[0.117, 0.214]
BRACS	Aligned	Moderate	0.699	[0.146, 1.466]	Open	0.169	[0.114, 0.242]
BRACS	Reversed	Moderate	0.739	[0.361, 1.229]	Open	0.231	[0.180, 0.297]
BRACS	Native	Severe	0.777	[0.284, 1.405]	Open	0.240	[0.184, 0.308]
BRACS	Aligned	Severe	1.531	[1.035, 2.021]	Open	0.252	[0.208, 0.310]
BRACS	Reversed	Severe	2.047	[1.517, 2.590]	Open	0.317	[0.267, 0.380]
CAMELYON16	Native	Moderate	-0.035	[-0.098, 0.032]	Below threshold	0.004	[0.002, 0.006]
CAMELYON16	Reversed	Moderate	0.051	[0.020, 0.105]	Below threshold	0.005	[0.003, 0.008]
CAMELYON16	Native	Severe	0.144	[-0.009, 0.463]	Below threshold	0.003	[0.002, 0.006]
CAMELYON16	Reversed	Severe	0.087	[0.034, 0.169]	Below threshold	0.008	[0.005, 0.013]
PANDA	Native	Moderate	-0.160	[-0.185, -0.134]	Below threshold	0.015	[0.014, 0.017]
PANDA	Native	Severe	0.087	[0.053, 0.131]	Below threshold	0.026	[0.023, 0.030]
TCGA-UT	Native	Moderate	-0.201	[-0.232, -0.166]	Below threshold	0.087	[0.074, 0.100]
TCGA-UT	Aligned	Moderate	-0.062	[-0.082, -0.043]	Below threshold	0.049	[0.040, 0.059]
TCGA-UT	Reversed	Moderate	-0.067	[-0.084, -0.053]	Below threshold	0.073	[0.063, 0.083]
TCGA-UT	Native	Severe	-0.185	[-0.245, -0.118]	Below threshold	0.142	[0.131, 0.153]
TCGA-UT	Aligned	Severe	0.088	[0.051, 0.126]	Below threshold	0.095	[0.084, 0.107]
TCGA-UT	Reversed	Severe	-0.025	[-0.059, 0.015]	Below threshold	0.144	[0.132, 0.156]

Table 3: Probability-quality deficit on tail-group macro negative log likelihood in nats, with paired 95% intervals and calibration-gate outcome. Expected calibration error and its crossed patient-bootstrap interval are secondary descriptors and do not gate.

Where a class is easy, concentrating the budget elsewhere costs accuracy on it without making the model confidently wrong about it, and the tail-group likelihood can even improve because a smaller tail allocation shrinks the model’s tendency to over-assign probability to the tail on head-class examples.

2.4 Where the damage falls

Aggregate deficits do not say which classes absorbed the loss, and a mitigation method has to be told that. Recall, F_1 , negative log likelihood, and Brier score are therefore also read per class and per support tier, with the head tier the first $\lceil K/3 \rceil$ classes by allocated support and the tail tier the last $\lceil K/3 \rceil$. Table 5 gives the tier-level endpoints under each condition; the per-class breakdown appears in Appendix B. The balanced reference and the natural anchor induce no support ordering and therefore carry no tier assignment, so they do not appear there.

Dataset	Assignment	Condition	Tier	Recall	NLL	Brier
BRACS	Aligned	Moderate	body	0.408	2.841	0.990
BRACS	Aligned	Moderate	head	0.730	1.386	0.555
BRACS	Aligned	Moderate	tail	0.205	2.283	0.920
BRACS	Aligned	Severe	body	0.393	3.702	1.080
BRACS	Aligned	Severe	head	0.753	1.724	0.565
BRACS	Aligned	Severe	tail	0.120	3.169	1.063
BRACS	Reversed	Moderate	body	0.195	4.141	1.047
BRACS	Reversed	Moderate	head	0.455	2.761	1.001
BRACS	Reversed	Moderate	tail	0.456	1.652	0.597
BRACS	Reversed	Severe	body	0.130	6.088	1.266
BRACS	Reversed	Severe	head	0.498	3.388	1.021
BRACS	Reversed	Severe	tail	0.343	2.738	0.759
BRACS	Native	Moderate	body	0.380	3.135	1.083
BRACS	Native	Moderate	head	0.492	1.672	0.765
BRACS	Native	Moderate	tail	0.465	1.576	0.657
BRACS	Native	Severe	body	0.407	3.362	1.098
BRACS	Native	Severe	head	0.484	2.326	0.858
BRACS	Native	Severe	tail	0.407	2.049	0.740
CAMELYON16	Reversed	Moderate	head	0.988	0.241	0.065

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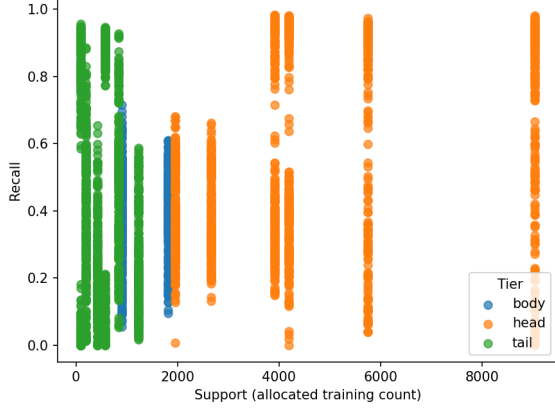
Dataset	Assignment	Condition	Tier	Recall	NLL	Brier
CAMELYON16	Reversed	Moderate	tail	0.981	0.054	0.010
CAMELYON16	Reversed	Severe	head	0.991	0.224	0.059
CAMELYON16	Reversed	Severe	tail	0.946	0.084	0.018
CAMELYON16	Native	Moderate	head	0.993	0.035	0.008
CAMELYON16	Native	Moderate	tail	0.973	0.335	0.067
CAMELYON16	Native	Severe	head	0.998	0.014	0.005
CAMELYON16	Native	Severe	tail	0.932	0.526	0.104
PANDA	Native	Moderate	head	0.976	0.133	0.068
PANDA	Native	Moderate	tail	0.721	0.580	0.348
PANDA	Native	Severe	head	0.994	0.115	0.059
PANDA	Native	Severe	tail	0.474	0.865	0.456
TCGA-UT	Aligned	Moderate	body	0.795	0.879	0.336
TCGA-UT	Aligned	Moderate	head	0.908	0.571	0.200
TCGA-UT	Aligned	Moderate	tail	0.558	1.320	0.584
TCGA-UT	Aligned	Severe	body	0.795	1.016	0.356
TCGA-UT	Aligned	Severe	head	0.912	0.694	0.204
TCGA-UT	Aligned	Severe	tail	0.522	1.472	0.619
TCGA-UT	Reversed	Moderate	body	0.777	0.810	0.315
TCGA-UT	Reversed	Moderate	head	0.590	1.633	0.637
TCGA-UT	Reversed	Moderate	tail	0.883	0.431	0.174
TCGA-UT	Reversed	Severe	body	0.771	0.988	0.332
TCGA-UT	Reversed	Severe	head	0.579	2.469	0.726
TCGA-UT	Reversed	Severe	tail	0.871	0.476	0.170
TCGA-UT	Native	Moderate	body	0.712	1.016	0.386
TCGA-UT	Native	Moderate	head	0.785	0.813	0.312
TCGA-UT	Native	Moderate	tail	0.753	0.939	0.393
TCGA-UT	Native	Severe	body	0.703	1.363	0.445
TCGA-UT	Native	Severe	head	0.789	1.195	0.365
TCGA-UT	Native	Severe	tail	0.722	0.947	0.363

Table 5: Head, body, and tail recall, negative log likelihood, and Brier score for the cross-entropy reference under each controlled condition, averaged with equal weight across the three partitions. Tier membership follows the allocation of the condition and is therefore not constant across conditions.

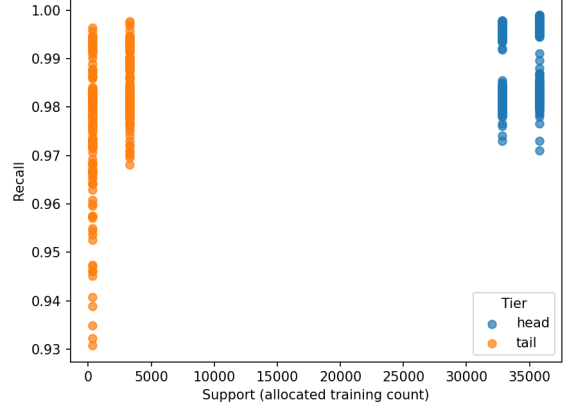
The tier view contradicts the simplest expectation, which is that damage lands on the tail. It does so under the difficulty-aligned assignment, where BRACS tail recall falls to 0.205 at moderate and 0.120 at severe severity against head recall above 0.73. Under the difficulty-reversed assignment the pattern inverts: BRACS severe tail recall is 0.343 while the *body* tier reaches only 0.130, below both the head and the tail. The reversed assignment gives the hardest classes the most support and the easiest classes the least, so the tail keeps a recall floor supplied by separability rather than by support, and the loss concentrates in the middle ranks, which have neither enough support nor enough intrinsic separability. A mitigation method keyed to support rank alone would target the wrong tier in exactly this case.

Figure 1 relates per-class recall to allocated training support directly. Its purpose is to separate two readings of the same aggregate deficit: a loss that varies smoothly with support implies that support is the binding constraint, whereas classes falling well below the support trend point to a shortage that allocation alone does not describe.

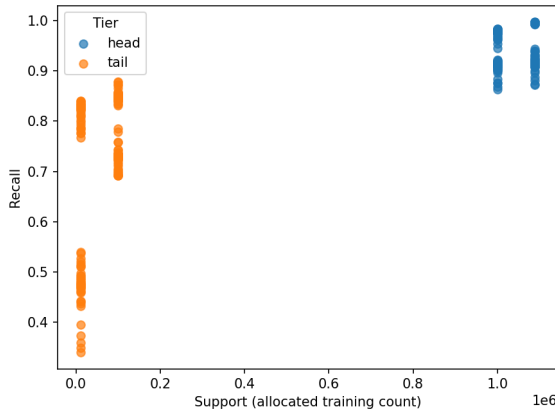
The scatter shows wide vertical spread at fixed allocated support on the two multiclass datasets, which is the visual form of the same point the tier table makes: at equal support, classes differ by more in recall than the support trend explains, so allocated count is not a sufficient statistic for how well a class is served.



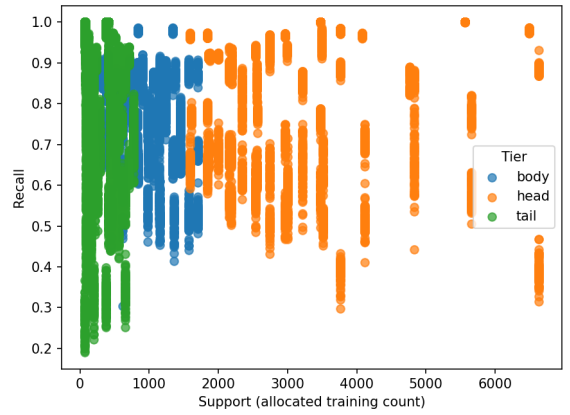
(a) BRACS



(b) CAMELYON16



(c) PANDA



(d) TCGA-UT

Figure 1: Per-class recall against allocated training patches, coloured by support tier, under the controlled conditions of the first locked partition. Classes falling below the support trend indicate a shortage that nominal allocation does not capture.

2.5 Gate routing and the natural anchor

The two gates partition the comparison units into four states: discrimination only, probability quality only, both, and neither. This partition is the single result that fixes the scope of Section 3, because recovery is expressed as a fraction of a deficit and is undefined where no deficit was established. Table 6 states the routing compactly for all units and is the table to consult before reading any recovery estimate.

Three of the four states are occupied. Six units, all on BRACS, opened both gates. Six further units opened the discrimination gate alone: both PANDA units, the two CAMELYON16 severe units, and the TCGA-UT severe units under the native and difficulty-reversed assignments. Six units opened neither gate, and every one of them fails on the threshold rather than on the interval, meaning the deficits there are estimated precisely enough to be distinguished from zero but are too small to be material. No unit opened the probability-quality gate alone, and no unit was closed by a descriptive-only designation.

The natural condition sits outside this partition. It trains on all eligible training patches and is therefore neither size-matched nor part of any estimand, but it is reported because it locates the controlled conditions against the imbalance the datasets present without intervention, which is the regime a deployed method would meet. Table 7 places its endpoints beside the size-matched conditions; differences involving the natural anchor confound allocation with training-

Dataset	Assignment	Severity	Discrimination	Probability quality	Routed to
BRACS	Native	Moderate	Open	Open	Discrimination, calibration
BRACS	Aligned	Moderate	Open	Open	Discrimination, calibration
BRACS	Reversed	Moderate	Open	Open	Discrimination, calibration
BRACS	Native	Severe	Open	Open	Discrimination, calibration
BRACS	Aligned	Severe	Open	Open	Discrimination, calibration
BRACS	Reversed	Severe	Open	Open	Discrimination, calibration
CAMELYON16	Native	Moderate	Below threshold	Below threshold	Neither
CAMELYON16	Reversed	Moderate	Below threshold	Below threshold	Neither
CAMELYON16	Native	Severe	Open	Below threshold	Discrimination
CAMELYON16	Reversed	Severe	Open	Below threshold	Discrimination
PANDA	Native	Moderate	Open	Below threshold	Discrimination
PANDA	Native	Severe	Open	Below threshold	Discrimination
TCGA-UT	Native	Moderate	Below threshold	Below threshold	Neither
TCGA-UT	Aligned	Moderate	Below threshold	Below threshold	Neither
TCGA-UT	Reversed	Moderate	Below threshold	Below threshold	Neither
TCGA-UT	Native	Severe	Open	Below threshold	Discrimination
TCGA-UT	Aligned	Severe	Below threshold	Below threshold	Neither
TCGA-UT	Reversed	Severe	Open	Below threshold	Discrimination

Table 6: Gate routing for every comparison unit: which of the discrimination and probability-quality gates opened, and the reason a closed gate did not open, distinguishing a deficit below threshold, an interval covering zero, and a descriptive-only designation.

set size and support no causal reading.

The anchor is worth one observation. Despite training on between three and 62 times as many patches at the dataset’s own prevalence, the natural condition does not separate itself from the size-matched balanced reference: it gains 0.006 balanced accuracy on BRACS and 0.001 on CAMELYON16, and loses 0.014 on PANDA, where the natural prevalence is mildly imbalanced at $\rho_{\text{achieved}} = 4.7$. Only TCGA-UT gains appreciably, from 0.755 to 0.792. Under frozen Virchow2 features the controlled budget is therefore already close to the point where additional patches stop buying discrimination, which is what the support pilot was designed to establish and which makes the controlled contrast a statement about allocation rather than about scale.

3 Mitigation Recovery

Everything in this section is restricted to the comparison units whose gates opened in Table 6. Recovery is defined as $R_M = [M(\text{method}) - M(\text{imbalanced CE})]/D_M$, the fraction of the established deficit that a method returns, so a unit with no established deficit has no defined recovery and is not reported here at any strength of evidence. Units designated descriptive-only by the bootstrap preflight are excluded on the same grounds; no unit was excluded on that ground.

3.1 The confirmatory five-family

The confirmatory family contains one method per signal, and all five apply a mean-one class weight to unmodified cross-entropy: prevalence-weighted cross-entropy, class-balanced cross-entropy, independent-support weighted cross-entropy, pilot-difficulty weighted cross-entropy, and semantic-scale weighted cross-entropy. Holding the intervention mechanism fixed in this way is what makes the comparison a contrast between signals rather than between loss surgeries of differing aggressiveness. Each method is tested against imbalanced cross-entropy by a paired partitioning-unit-block permutation test, and the Holm adjustment is applied jointly within a dataset over the five method tests plus the matched contrast for each gate-passing unit.

Table 9 reports recovery with its interval, the unadjusted and adjusted p -values, and the resulting status for each unit and method.

Dataset	Assignment	Condition	BA	Macro F_1	NLL	Macro NLL	ECE
BRACS	Aligned	Moderate	0.459	0.386	1.397	1.978	0.169
BRACS	Aligned	Severe	0.430	0.339	1.903	2.626	0.252
BRACS	Reversed	Moderate	0.418	0.379	1.843	2.483	0.231
BRACS	Reversed	Severe	0.379	0.340	2.642	3.495	0.317
BRACS	Native	Moderate	0.464	0.416	1.311	1.840	0.159
BRACS	Native	Severe	0.440	0.393	1.865	2.355	0.240
BRACS	—	Balanced	0.492	0.439	0.983	1.491	0.053
BRACS	—	Natural	0.498	0.485	0.949	1.451	0.080
CAMELYON16	Reversed	Moderate	0.984	0.838	0.058	0.148	0.005
CAMELYON16	Reversed	Severe	0.968	0.737	0.086	0.154	0.008
CAMELYON16	Native	Moderate	0.983	0.919	0.045	0.185	0.004
CAMELYON16	Native	Severe	0.965	0.936	0.035	0.270	0.003
CAMELYON16	—	Balanced	0.986	0.893	0.018	0.129	0.002
CAMELYON16	—	Natural	0.987	0.930	0.024	0.271	0.004
PANDA	Native	Moderate	0.848	0.874	0.216	0.356	0.015
PANDA	Native	Severe	0.734	0.787	0.254	0.490	0.026
PANDA	—	Balanced	0.882	0.854	0.185	0.355	0.007
PANDA	—	Natural	0.868	0.885	0.194	0.339	0.011
TCGA-UT	Aligned	Moderate	0.754	0.698	0.797	0.923	0.049
TCGA-UT	Aligned	Severe	0.743	0.682	0.960	1.061	0.095
TCGA-UT	Reversed	Moderate	0.750	0.706	0.839	0.958	0.073
TCGA-UT	Reversed	Severe	0.740	0.699	1.214	1.311	0.144
TCGA-UT	Native	Moderate	0.750	0.717	0.893	0.923	0.087
TCGA-UT	Native	Severe	0.738	0.714	1.244	1.169	0.142
TCGA-UT	—	Balanced	0.755	0.704	0.763	0.899	0.035
TCGA-UT	—	Natural	0.792	0.793	0.603	0.763	0.058

Table 7: Descriptive comparison of the natural anchor against the size-matched balanced, moderate, and severe conditions, for the cross-entropy reference. The natural condition is not size-matched, so these differences confound allocation with training-set size and enter no estimand.

Dataset	Assignment	Severity	Signal	R_M	95% CI	p	Holm p	Status
BRACS	Native	Moderate	Prevalence	0.937	[0.612, 1.244]	0.0042	0.1232	Tested
BRACS	Native	Moderate	Nominal support	0.920	[0.623, 1.191]	0.0034	0.1026	Tested
BRACS	Native	Moderate	Independent support	0.139	[-0.612, 0.573]	0.5853	1.0000	Tested
BRACS	Native	Moderate	Difficulty	-0.083	[-0.818, 0.281]	0.5242	1.0000	Tested
BRACS	Native	Moderate	Diversity	0.093	[-0.539, 0.485]	0.6671	1.0000	Tested
BRACS	Aligned	Moderate	Prevalence	0.921	[0.461, 1.453]	0.0063	0.1698	Tested
BRACS	Aligned	Moderate	Nominal support	0.997	[0.627, 1.659]	0.0019	0.0583	Tested
BRACS	Aligned	Moderate	Independent support	-0.029	[-0.973, 0.463]	0.9217	1.0000	Tested
BRACS	Aligned	Moderate	Difficulty	0.080	[-0.505, 0.483]	0.5825	1.0000	Tested
BRACS	Aligned	Moderate	Diversity	0.370	[-0.501, 1.038]	0.1540	1.0000	Tested
BRACS	Reversed	Moderate	Prevalence	0.878	[0.750, 0.998]	0.0001	0.0050	Tested
BRACS	Reversed	Moderate	Nominal support	0.842	[0.726, 0.953]	<0.0001	0.0033	Tested
BRACS	Reversed	Moderate	Independent support	-0.030	[-0.237, 0.161]	0.7092	1.0000	Tested
BRACS	Reversed	Moderate	Difficulty	-0.099	[-0.334, 0.084]	0.1235	1.0000	Tested
BRACS	Reversed	Moderate	Diversity	0.052	[-0.148, 0.243]	0.5209	1.0000	Tested
BRACS	Native	Severe	Prevalence	0.819	[0.542, 1.051]	0.0004	0.0148	Tested
BRACS	Native	Severe	Nominal support	0.859	[0.622, 1.081]	<0.0001	0.0031	Tested
BRACS	Native	Severe	Independent support	0.147	[-0.105, 0.385]	0.0516	1.0000	Tested
BRACS	Native	Severe	Difficulty	0.006	[-0.199, 0.201]	0.9617	1.0000	Tested
BRACS	Native	Severe	Diversity	0.148	[-0.044, 0.332]	0.0399	0.9578	Tested
BRACS	Aligned	Severe	Prevalence	0.768	[0.505, 1.153]	0.0003	0.0092	Tested
BRACS	Aligned	Severe	Nominal support	0.881	[0.605, 1.378]	<0.0001	0.0008	Tested
BRACS	Aligned	Severe	Independent support	0.006	[-0.239, 0.214]	0.9444	1.0000	Tested
BRACS	Aligned	Severe	Difficulty	-0.065	[-0.262, 0.075]	0.2933	1.0000	Tested
BRACS	Aligned	Severe	Diversity	0.090	[-0.158, 0.385]	0.3458	1.0000	Tested
BRACS	Reversed	Severe	Prevalence	0.785	[0.686, 0.882]	<0.0001	0.0007	Tested
BRACS	Reversed	Severe	Nominal support	0.743	[0.634, 0.850]	<0.0001	0.0007	Tested
BRACS	Reversed	Severe	Independent support	0.084	[-0.001, 0.168]	0.0894	1.0000	Tested
BRACS	Reversed	Severe	Difficulty	0.012	[-0.095, 0.112]	0.6817	1.0000	Tested
BRACS	Reversed	Severe	Diversity	-0.022	[-0.100, 0.048]	0.3862	1.0000	Tested
CAMELYON16	Native	Severe	Prevalence	0.893	[0.544, 1.093]	0.0003	0.0020	Tested
CAMELYON16	Native	Severe	Nominal support	0.838	[0.444, 1.037]	0.0050	0.0200	Tested
CAMELYON16	Native	Severe	Independent support	-0.157	[-1.030, 0.352]	0.1560	0.3120	Tested
CAMELYON16	Native	Severe	Difficulty	0.185	[-0.326, 0.515]	0.0252	0.0757	Tested
CAMELYON16	Native	Severe	Diversity	0.655	[0.222, 0.935]	0.0001	0.0008	Tested
CAMELYON16	Reversed	Severe	Prevalence	0.924	[0.819, 1.036]	<0.0001	0.0001	Tested
CAMELYON16	Reversed	Severe	Nominal support	0.822	[0.538, 0.964]	<0.0001	0.0001	Tested

Continued on next page

Dataset	Assignment	Severity	Signal	R_M	95% CI	p	Holm p	Status
CAMELYON16	Reversed	Severe	Independent support	0.480	[0.385, 0.581]	<0.0001	0.0001	Tested
CAMELYON16	Reversed	Severe	Difficulty	0.037	[-0.222, 0.232]	0.4238	0.4238	Tested
CAMELYON16	Reversed	Severe	Diversity	0.509	[0.375, 0.618]	<0.0001	0.0001	Tested
PANDA	Native	Moderate	Prevalence	1.162	[1.089, 1.253]	<0.0001	0.0001	Tested
PANDA	Native	Moderate	Nominal support	0.014	[-0.057, 0.077]	0.2574	0.3515	Tested
PANDA	Native	Moderate	Independent support	0.025	[-0.061, 0.093]	0.0325	0.0975	Tested
PANDA	Native	Moderate	Difficulty	-0.143	[-0.230, -0.068]	<0.0001	0.0001	Tested
PANDA	Native	Moderate	Diversity	0.758	[0.692, 0.823]	<0.0001	0.0001	Tested
PANDA	Native	Severe	Prevalence	0.968	[0.953, 0.984]	<0.0001	0.0001	Tested
PANDA	Native	Severe	Nominal support	0.118	[0.081, 0.151]	<0.0001	0.0001	Tested
PANDA	Native	Severe	Independent support	-0.018	[-0.050, 0.005]	<0.0001	0.0001	Tested
PANDA	Native	Severe	Difficulty	-0.069	[-0.129, -0.031]	<0.0001	0.0001	Tested
PANDA	Native	Severe	Diversity	-0.004	[-0.057, 0.028]	0.1757	0.3515	Tested
TCGA-UT	Native	Severe	Prevalence	0.812	[0.659, 0.968]	<0.0001	0.0001	Tested
TCGA-UT	Native	Severe	Nominal support	0.792	[0.639, 0.947]	<0.0001	0.0001	Tested
TCGA-UT	Native	Severe	Independent support	-0.003	[-0.063, 0.062]	0.9048	1.0000	Tested
TCGA-UT	Native	Severe	Difficulty	-0.047	[-0.130, 0.037]	0.1382	0.6911	Tested
TCGA-UT	Native	Severe	Diversity	-0.041	[-0.148, 0.058]	0.3448	1.0000	Tested
TCGA-UT	Reversed	Severe	Prevalence	0.828	[0.689, 0.939]	<0.0001	0.0001	Tested
TCGA-UT	Reversed	Severe	Nominal support	0.807	[0.653, 0.935]	<0.0001	0.0001	Tested
TCGA-UT	Reversed	Severe	Independent support	0.010	[-0.059, 0.070]	0.7143	1.0000	Tested
TCGA-UT	Reversed	Severe	Difficulty	-0.073	[-0.161, -0.005]	0.0076	0.0454	Tested
TCGA-UT	Reversed	Severe	Diversity	-0.025	[-0.162, 0.091]	0.4829	1.0000	Tested

Table 9: Confirmatory recovery R_M for the five-member method family on discrimination-gate-passing units, with paired 95% intervals, permutation p -values, and Holm-adjusted p -values within each dataset. Signal labels name the family member: prevalence is prevalence-weighted cross-entropy, nominal support is class-balanced cross-entropy, and the remaining three are the independent-support, pilot-difficulty, and semantic-scale weighted variants.

The family separates into two groups, and the separation is the same in all 12 gate-passing units. Prevalence-weighted and class-balanced cross-entropy recover most of the deficit nearly everywhere: on BRACS they return between 0.743 and 0.997 of it, on CAMELYON16 between 0.822 and 0.924, and on TCGA-UT between 0.792 and 0.828. The independent-support, difficulty, and diversity members recover essentially nothing on BRACS and TCGA-UT, with point estimates between -0.099 and 0.370 across those 24 cells and intervals covering zero in almost every one of them. The one dataset where the non-prevalence signals do move an endpoint is CAMELYON16, where semantic-scale weighting recovers 0.655 under the native severe condition and 0.509 under the reversed one, and independent-support weighting recovers 0.480 under the reversed condition; on a two-class target, however, the diversity and independent-support weights are two-point orderings and cannot be distinguished from a coarse prevalence correction on the mechanism they apply.

PANDA is the informative exception within the working group. There prevalence weighting recovers 1.162 at moderate and 0.968 at severe severity, while class-balanced cross-entropy, which reads nominal support, recovers only 0.014 and 0.118. The two methods differ solely in whether the weight is proportional to inverse frequency or to the effective-number reweighting of nominal counts, and at $\rho = 100$ on a two-class problem the effective-number transform saturates and returns nearly uniform weights. The two members of the family that share the prevalence and nominal-support corner of the taxonomy therefore do not behave interchangeably, which matters directly for the matched contrast below.

Holm adjustment within a dataset changes the picture only at its margin, and it bites hardest where the family is largest. BRACS opened both gates in all six units, so its family holds 68 tests: five methods in each of the six units on each of the two axes, plus the matched contrast wherever the unit is not ambiguous. Under that correction four of the six prevalence-weighted discrimination tests survive and two do not, despite recovery point estimates above 0.92, because the two moderate conditions that fail have wide recovery intervals. On CAMELYON16, PANDA, and TCGA-UT, whose families hold 12, 11, and 12 tests because only the discrimination gate opened, every prevalence-weighted discrimination test survives at adjusted $p \leq 0.02$.

3.2 Matched against unmatched signals

The claim RQ2 actually tests is not that some method recovers damage but that the method reading the shortage a condition presents recovers more of it than a method reading a different shortage. Each gate-passing unit is therefore split into a matched arm, containing the family member whose signal corresponds to the dominant shortage from Table 1, and an unmatched arm containing the remainder. Ambiguous units, where the two largest shortage scores fall within 0.25 standard deviations, contribute descriptively and carry no confirmatory weight.

Table 11 gives the direct matched-versus-unmatched contrast per unit with its interval and adjusted p -value, and records which family member in fact recovered most. Two further quantities qualify how that contrast should be read. The first is the matched- β diagnostic arm, in which independent-support weighted cross-entropy is refitted at the effective weight strength selected for class-balanced cross-entropy; it separates an advantage attributable to the signal from one attributable to a more aggressive tuned intervention, and appears in Table 12. The second is agreement between the dominant shortage and the method that in fact recovered most, which is reported per unit because a systematic mismatch would indicate that the shortage taxonomy, not the methods, is what fails to describe these conditions.

Dataset	Assignment	Severity	Gate	Dominant	Contrast	95% CI	Holm p	Best recoverer	Agrees
BRACS	Native	Moderate	Discrimination	Ambiguous	—	—	—	Prevalence	—
BRACS	Aligned	Moderate	Discrimination	Difficulty	-0.0128	[-0.0251, -0.0024]	0.2597	Nominal support	No
BRACS	Reversed	Moderate	Discrimination	Independent support	-0.0363	[-0.0532, -0.0221]	0.0007	Prevalence	No
BRACS	Native	Severe	Discrimination	Diversity	-0.0187	[-0.0315, -0.0073]	0.0342	Nominal support	No
BRACS	Aligned	Severe	Discrimination	Ambiguous	—	—	—	Nominal support	—
BRACS	Reversed	Severe	Discrimination	Diversity	-0.0572	[-0.0711, -0.0436]	0.0007	Prevalence	No
CAMELYON16	Native	Severe	Discrimination	Nominal support	0.0108	[0.0043, 0.0180]	0.0130	Prevalence	Yes
CAMELYON16	Reversed	Severe	Discrimination	Nominal support	0.0097	[0.0067, 0.0124]	0.0001	Prevalence	Yes
PANDA	Native	Moderate	Discrimination	Ambiguous	—	—	—	Prevalence	—
PANDA	Native	Severe	Discrimination	Nominal support	0.0835	[0.0809, 0.0862]	0.0001	Prevalence	Yes
TCGA-UT	Native	Severe	Discrimination	Diversity	-0.0069	[-0.0093, -0.0046]	0.0001	Prevalence	No
TCGA-UT	Reversed	Severe	Discrimination	Diversity	-0.0061	[-0.0087, -0.0034]	0.0001	Prevalence	No
BRACS	Native	Moderate	Calibration	Ambiguous	—	—	—	Prevalence	—
BRACS	Aligned	Moderate	Calibration	Difficulty	-0.4299	[-0.5656, -0.3362]	0.0007	Prevalence	No
BRACS	Reversed	Moderate	Calibration	Independent support	-0.8625	[-1.5058, -0.2598]	0.0007	Prevalence	No
BRACS	Native	Severe	Calibration	Diversity	-0.7399	[-1.1265, -0.4189]	0.0007	Prevalence	No
BRACS	Aligned	Severe	Calibration	Ambiguous	—	—	—	Prevalence	—
BRACS	Reversed	Severe	Calibration	Diversity	-2.6952	[-3.2275, -2.1190]	0.0007	Prevalence	No

Table 11: Direct contrast between the signal-matched arm and the unmatched arm on gate-passing units, with paired 95% intervals and Holm-adjusted p -values. Ambiguous units carry no matched arm and are descriptive only. “Best recoverer” names the family member with the largest recovery in that unit, and “Agrees” records whether it belongs to the matched arm.

Dataset	Assignment	Severity	CE	Independent	Independent at matched β	Nominal
BRACS	Native	Moderate	0.464	0.467	0.470	0.495
BRACS	Aligned	Moderate	0.459	0.451	0.450	0.486
BRACS	Reversed	Moderate	0.418	0.418	0.416	0.489
BRACS	Native	Severe	0.440	0.442	0.447	0.496
BRACS	Aligned	Severe	0.430	0.428	0.429	0.474
BRACS	Reversed	Severe	0.379	0.387	0.388	0.486
CAMELYON16	Native	Moderate	0.983	0.981	0.979	0.985
CAMELYON16	Reversed	Moderate	0.984	0.983	0.985	0.986
CAMELYON16	Native	Severe	0.965	0.962	0.962	0.983
CAMELYON16	Reversed	Severe	0.968	0.977	0.979	0.985
PANDA	Native	Moderate	0.848	0.848	0.838	0.848
PANDA	Native	Severe	0.734	0.733	0.715	0.751
TCGA-UT	Native	Moderate	0.750	0.749	0.749	0.755
TCGA-UT	Aligned	Moderate	0.754	0.753	0.753	0.754
TCGA-UT	Reversed	Moderate	0.750	0.751	0.750	0.754
TCGA-UT	Native	Severe	0.738	0.738	0.738	0.751
TCGA-UT	Aligned	Severe	0.743	0.742	0.742	0.747
TCGA-UT	Reversed	Severe	0.740	0.740	0.741	0.752

Table 12: Matched- β diagnostic: case-macro balanced accuracy of independent-support weighted cross-entropy refitted at the effective weight strength selected for class-balanced cross-entropy, beside the imbalanced cross-entropy reference and the two anchors it separates. Descriptive, and not part of the Holm family.

The contrast is signed in opposite directions on different datasets, and the sign is fully predicted by which shortage the unit’s profile makes dominant. Where the dominant shortage is nominal support, the matched arm contains prevalence-weighted and class-balanced cross-entropy, and the contrast is positive: 0.011 and 0.010 on the two CAMELYON16 severe units and 0.084 on PANDA severe, all surviving Holm adjustment. Where the dominant shortage is difficulty, diversity, or independent support, the matched arm contains exactly one method from the group that recovers nothing while the unmatched arm retains both methods that work, and the contrast is negative in every such unit: from -0.006 to -0.057 on discrimination, across four BRACS and two TCGA-UT units, and from -0.430 to -2.695 nats on probability quality.

The agreement column makes the same point without the arithmetic of the contrast. In all 18 gate-passing rows the family member that recovered most is either prevalence-weighted or class-balanced cross-entropy. The dominant shortage and the best recoverer therefore agree exactly when, and only when, the dominant shortage is nominal support. This is not evidence that signal matching works; it is evidence that on these datasets only one corner of the taxonomy is served by a method that recovers anything, so the matched contrast measures whether the unit happened to land in that corner.

The matched- β diagnostic rules out the most obvious alternative explanation. If independent-support weighting failed because its tuned weight strength was too timid rather than because its signal is uninformative, refitting it at the strength selected for class-balanced cross-entropy should close part of the gap. It does not. Across all 18 units the refit moves balanced accuracy by at most 0.018 from the independent-support arm, and the two largest movements, both on PANDA, are downward; on 13 of the 18 units the movement is at most 0.002. The gap to class-balanced cross-entropy meanwhile remains as large as 0.098, on BRACS under the difficulty-reversed severe condition. The failure of the non-prevalence signals is a property of what those weights encode, not of how hard they were applied.

3.3 Recovery of probability quality

A method may return discrimination while leaving probabilities no better calibrated, or return calibration without moving balanced accuracy, and a mitigation method that must serve both would need to be designed for that trade-off explicitly. Recovery on the tail-group negative log likelihood axis is therefore reported separately in Table 14, over the units whose calibration

gate opened, which need not be the units whose discrimination gate opened. Where both axes were damaged in the same unit, the two recovery fractions are read together, and any method recovering on one axis while degrading the other is identified explicitly.

Dataset	Assignment	Severity	Signal	R_M	95% CI	p	Holm p	Status	Discrimination R_M
BRACS	Native	Moderate	Prevalence	1.106	[0.922, 1.438]	<0.0001	0.0031	Tested	0.937
BRACS	Native	Moderate	Nominal support	1.004	[0.789, 1.311]	0.0002	0.0063	Tested	0.920
BRACS	Native	Moderate	Independent support	-1.142	[-4.267, -0.128]	0.0052	0.1467	Tested	0.139
BRACS	Native	Moderate	Difficulty	-1.155	[-3.672, -0.218]	0.0103	0.2597	Tested	-0.083
BRACS	Native	Moderate	Diversity	-0.647	[-2.870, 0.088]	0.0871	1.0000	Tested	0.093
BRACS	Aligned	Moderate	Prevalence	1.261	[1.113, 2.237]	<0.0001	0.0007	Tested	0.921
BRACS	Aligned	Moderate	Nominal support	1.211	[1.079, 2.065]	<0.0001	0.0007	Tested	0.997
BRACS	Aligned	Moderate	Independent support	-0.225	[-3.762, 0.382]	0.2157	1.0000	Tested	-0.029
BRACS	Aligned	Moderate	Difficulty	0.012	[-3.074, 0.485]	0.9388	1.0000	Tested	0.080
BRACS	Aligned	Moderate	Diversity	0.260	[-2.464, 0.646]	0.2420	1.0000	Tested	0.370
BRACS	Reversed	Moderate	Prevalence	0.928	[0.771, 1.019]	<0.0001	0.0007	Tested	0.878
BRACS	Reversed	Moderate	Nominal support	0.887	[0.701, 0.977]	<0.0001	0.0007	Tested	0.842
BRACS	Reversed	Moderate	Independent support	-1.086	[-3.560, 0.074]	<0.0001	0.0007	Tested	-0.030
BRACS	Reversed	Moderate	Difficulty	-1.544	[-4.045, -0.656]	<0.0001	0.0007	Tested	-0.099
BRACS	Reversed	Moderate	Diversity	0.053	[-1.084, 0.491]	0.7199	1.0000	Tested	0.052
BRACS	Native	Severe	Prevalence	0.913	[0.708, 1.003]	<0.0001	0.0007	Tested	0.819
BRACS	Native	Severe	Nominal support	0.881	[0.602, 0.977]	<0.0001	0.0007	Tested	0.859
BRACS	Native	Severe	Independent support	-1.043	[-4.107, -0.322]	<0.0001	0.0007	Tested	0.147
BRACS	Native	Severe	Difficulty	-1.487	[-4.283, -0.613]	<0.0001	0.0007	Tested	0.006
BRACS	Native	Severe	Diversity	-1.136	[-4.909, -0.256]	<0.0001	0.0007	Tested	0.148
BRACS	Aligned	Severe	Prevalence	1.044	[0.995, 1.103]	<0.0001	0.0007	Tested	0.768
BRACS	Aligned	Severe	Nominal support	1.007	[0.958, 1.057]	<0.0001	0.0007	Tested	0.881
BRACS	Aligned	Severe	Independent support	-0.650	[-1.671, 0.015]	<0.0001	0.0007	Tested	0.006
BRACS	Aligned	Severe	Difficulty	-0.912	[-1.651, -0.541]	<0.0001	0.0007	Tested	-0.065
BRACS	Aligned	Severe	Diversity	-0.454	[-1.120, -0.015]	<0.0001	0.0007	Tested	0.090
BRACS	Reversed	Severe	Prevalence	0.947	[0.898, 0.981]	<0.0001	0.0007	Tested	0.785
BRACS	Reversed	Severe	Nominal support	0.926	[0.888, 0.959]	<0.0001	0.0007	Tested	0.743
BRACS	Reversed	Severe	Independent support	-0.158	[-0.692, 0.203]	0.1837	1.0000	Tested	0.084
BRACS	Reversed	Severe	Difficulty	-1.025	[-1.921, -0.499]	<0.0001	0.0007	Tested	0.012
BRACS	Reversed	Severe	Diversity	-1.144	[-1.913, -0.693]	<0.0001	0.0007	Tested	-0.022

Table 14: Recovery of tail-group macro negative log likelihood on the six calibration-gate-passing units, all on BRACS, with paired 95% intervals and Holm-adjusted p -values, reported alongside the discrimination recovery of the same method and unit, since both gates opened in every one of these units.

All six calibration-gate units are on BRACS, and all six also opened the discrimination gate, so the two axes can be read against each other directly. Prevalence-weighted and class-balanced cross-entropy recover between 0.881 and 1.261 of the tail-group likelihood deficit, all at adjusted $p \leq 0.007$, and recover discrimination in the same cells at between 0.743 and 0.997. On these two methods the axes do not trade off: whatever repairs the tail’s discrimination also repairs its probability mass.

The other three members do trade off, and in the damaging direction. Independent-support, difficulty, and diversity weighting return recovery fractions between -1.544 and 0.260 on the calibration axis, with 15 of the 18 corresponding estimates negative and 10 of those surviving Holm adjustment at $p \leq 0.007$. A negative recovery fraction here means the method left tail-group probabilities materially worse than imbalanced cross-entropy had left them, while moving discrimination by an amount indistinguishable from zero. Difficulty weighting under the BRACS difficulty-reversed moderate condition is the clearest instance: tail-group recovery -1.544 against discrimination recovery -0.099 . A weight that reads a signal unrelated to the damage does not merely fail to help; it redistributes probability mass across classes and degrades the endpoint that the gate identified as damaged.

3.4 The wider method roster

The remaining roster methods change the training objective, the sampling distribution, the decision threshold, or the classifier head, and so cannot be compared to the five-family at a fixed intervention mechanism. They are reported with intervals but without confirmatory status: balanced sampling, train-time and post-hoc logit adjustment, classifier retraining, focal loss, the prototype-affinity head, the soft- F_1 and soft-MCC hybrids, and set learning. Their role in this report is to bound what is achievable on these conditions by any tested means, which is what Section 5 needs in order to distinguish a signal that no method exploits from a deficit that no mechanism repairs.

Dataset	Assignment	Severity	Gate	Method	R_M	95% CI
BRACS	Native	Moderate	Discrimination	Balanced sampling	0.872	[0.448, 1.135]
BRACS	Native	Moderate	Discrimination	CE + soft F_1	0.782	[0.386, 1.067]
BRACS	Native	Moderate	Discrimination	CE + soft MCC	0.775	[0.366, 1.057]
BRACS	Native	Moderate	Discrimination	CFAL	0.268	[-0.544, 0.712]
BRACS	Native	Moderate	Discrimination	cRT	0.125	[-0.431, 0.398]
BRACS	Native	Moderate	Discrimination	Focal loss	-0.037	[-0.708, 0.333]
BRACS	Native	Moderate	Discrimination	Logit adjustment (train)	0.895	[0.558, 1.214]
BRACS	Native	Moderate	Discrimination	OKO	0.357	[-0.417, 0.816]
BRACS	Native	Moderate	Discrimination	Logit adjustment (post-hoc)	0.193	[0.062, 0.362]
BRACS	Aligned	Moderate	Discrimination	Balanced sampling	0.676	[-0.120, 1.171]
BRACS	Aligned	Moderate	Discrimination	CE + soft F_1	0.626	[-0.198, 1.117]
BRACS	Aligned	Moderate	Discrimination	CE + soft MCC	0.613	[-0.212, 1.097]
BRACS	Aligned	Moderate	Discrimination	CFAL	0.506	[-0.282, 1.275]
BRACS	Aligned	Moderate	Discrimination	cRT	-0.359	[-1.911, 0.100]
BRACS	Aligned	Moderate	Discrimination	Focal loss	0.091	[-0.595, 0.611]
BRACS	Aligned	Moderate	Discrimination	Logit adjustment (train)	0.552	[-0.515, 1.168]
BRACS	Aligned	Moderate	Discrimination	OKO	0.415	[-0.522, 1.115]
BRACS	Aligned	Moderate	Discrimination	Logit adjustment (post-hoc)	-0.149	[-0.885, 0.117]
BRACS	Reversed	Moderate	Discrimination	Balanced sampling	0.841	[0.714, 0.953]
BRACS	Reversed	Moderate	Discrimination	CE + soft F_1	0.844	[0.718, 0.955]
BRACS	Reversed	Moderate	Discrimination	CE + soft MCC	0.836	[0.706, 0.955]
BRACS	Reversed	Moderate	Discrimination	CFAL	0.319	[0.085, 0.529]
BRACS	Reversed	Moderate	Discrimination	cRT	0.215	[0.058, 0.355]
BRACS	Reversed	Moderate	Discrimination	Focal loss	-0.014	[-0.200, 0.157]
BRACS	Reversed	Moderate	Discrimination	Logit adjustment (train)	0.849	[0.722, 0.958]
BRACS	Reversed	Moderate	Discrimination	OKO	0.632	[0.439, 0.796]
BRACS	Reversed	Moderate	Discrimination	Logit adjustment (post-hoc)	0.215	[0.134, 0.317]
BRACS	Native	Severe	Discrimination	Balanced sampling	0.801	[0.565, 0.998]
BRACS	Native	Severe	Discrimination	CE + soft F_1	0.790	[0.530, 1.027]
BRACS	Native	Severe	Discrimination	CE + soft MCC	0.775	[0.481, 1.081]
BRACS	Native	Severe	Discrimination	CFAL	0.513	[0.277, 0.761]
BRACS	Native	Severe	Discrimination	cRT	0.402	[0.142, 0.595]
BRACS	Native	Severe	Discrimination	Focal loss	-0.031	[-0.259, 0.172]
BRACS	Native	Severe	Discrimination	Logit adjustment (train)	0.687	[0.425, 0.894]
BRACS	Native	Severe	Discrimination	OKO	0.587	[0.311, 0.810]
BRACS	Native	Severe	Discrimination	Logit adjustment (post-hoc)	0.365	[0.223, 0.546]
BRACS	Aligned	Severe	Discrimination	Balanced sampling	0.747	[0.447, 1.092]
BRACS	Aligned	Severe	Discrimination	CE + soft F_1	0.736	[0.414, 1.082]
BRACS	Aligned	Severe	Discrimination	CE + soft MCC	0.718	[0.440, 1.086]
BRACS	Aligned	Severe	Discrimination	CFAL	0.484	[0.106, 0.926]
BRACS	Aligned	Severe	Discrimination	cRT	0.091	[-0.311, 0.348]
BRACS	Aligned	Severe	Discrimination	Focal loss	0.044	[-0.230, 0.283]
BRACS	Aligned	Severe	Discrimination	Logit adjustment (train)	0.168	[-0.293, 0.485]

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Dataset	Assignment	Severity	Gate	Method	R_M	95% CI
BRACS	Aligned	Severe	Discrimination	OKO	0.560	[0.256, 0.902]
BRACS	Aligned	Severe	Discrimination	Logit adjustment (post-hoc)	0.058	[-0.182, 0.241]
BRACS	Reversed	Severe	Discrimination	Balanced sampling	0.787	[0.685, 0.878]
BRACS	Reversed	Severe	Discrimination	CE + soft F_1	0.787	[0.683, 0.879]
BRACS	Reversed	Severe	Discrimination	CE + soft MCC	0.871	[0.729, 1.021]
BRACS	Reversed	Severe	Discrimination	CFAL	0.436	[0.328, 0.546]
BRACS	Reversed	Severe	Discrimination	cRT	0.296	[0.219, 0.385]
BRACS	Reversed	Severe	Discrimination	Focal loss	0.044	[-0.056, 0.138]
BRACS	Reversed	Severe	Discrimination	Logit adjustment (train)	0.712	[0.613, 0.830]
BRACS	Reversed	Severe	Discrimination	OKO	0.811	[0.712, 0.907]
BRACS	Reversed	Severe	Discrimination	Logit adjustment (post-hoc)	0.193	[0.125, 0.282]
CAMELYON16	Native	Severe	Discrimination	Balanced sampling	0.865	[0.586, 1.108]
CAMELYON16	Native	Severe	Discrimination	CE + soft F_1	0.953	[0.699, 1.169]
CAMELYON16	Native	Severe	Discrimination	CE + soft MCC	0.897	[0.639, 1.058]
CAMELYON16	Native	Severe	Discrimination	CFAL	0.609	[0.172, 0.895]
CAMELYON16	Native	Severe	Discrimination	cRT	0.490	[0.125, 0.805]
CAMELYON16	Native	Severe	Discrimination	Focal loss	-0.161	[-1.330, 0.308]
CAMELYON16	Native	Severe	Discrimination	Logit adjustment (train)	0.918	[0.756, 1.043]
CAMELYON16	Native	Severe	Discrimination	OKO	1.003	[0.788, 1.216]
CAMELYON16	Native	Severe	Discrimination	Logit adjustment (post-hoc)	0.502	[0.148, 0.751]
CAMELYON16	Reversed	Severe	Discrimination	Balanced sampling	0.956	[0.862, 1.074]
CAMELYON16	Reversed	Severe	Discrimination	CE + soft F_1	0.997	[0.921, 1.111]
CAMELYON16	Reversed	Severe	Discrimination	CE + soft MCC	0.846	[0.724, 0.937]
CAMELYON16	Reversed	Severe	Discrimination	CFAL	0.615	[0.462, 0.727]
CAMELYON16	Reversed	Severe	Discrimination	cRT	0.422	[0.223, 0.619]
CAMELYON16	Reversed	Severe	Discrimination	Focal loss	0.111	[-0.067, 0.291]
CAMELYON16	Reversed	Severe	Discrimination	Logit adjustment (train)	0.932	[0.857, 1.042]
CAMELYON16	Reversed	Severe	Discrimination	OKO	0.946	[0.862, 1.063]
CAMELYON16	Reversed	Severe	Discrimination	Logit adjustment (post-hoc)	0.772	[0.653, 0.909]
PANDA	Native	Moderate	Discrimination	Balanced sampling	1.091	[1.031, 1.168]
PANDA	Native	Moderate	Discrimination	CE + soft F_1	1.211	[1.142, 1.297]
PANDA	Native	Moderate	Discrimination	CE + soft MCC	1.211	[1.142, 1.297]
PANDA	Native	Moderate	Discrimination	CFAL	-0.683	[-0.833, -0.562]
PANDA	Native	Moderate	Discrimination	cRT	1.118	[1.048, 1.205]
PANDA	Native	Moderate	Discrimination	Focal loss	-0.041	[-0.241, 0.133]
PANDA	Native	Moderate	Discrimination	Logit adjustment (train)	1.162	[1.093, 1.246]
PANDA	Native	Moderate	Discrimination	OKO	1.153	[1.078, 1.243]
PANDA	Native	Moderate	Discrimination	Logit adjustment (post-hoc)	0.925	[0.846, 1.017]
PANDA	Native	Severe	Discrimination	Balanced sampling	0.975	[0.963, 0.989]
PANDA	Native	Severe	Discrimination	CE + soft F_1	0.979	[0.967, 0.993]
PANDA	Native	Severe	Discrimination	CE + soft MCC	0.983	[0.970, 0.998]
PANDA	Native	Severe	Discrimination	CFAL	-0.463	[-0.541, -0.402]
PANDA	Native	Severe	Discrimination	cRT	0.925	[0.911, 0.940]

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Dataset	Assignment	Severity	Gate	Method	R_M	95% CI
PANDA	Native	Severe	Discrimination	Focal loss	-0.011	[-0.032, 0.007]
PANDA	Native	Severe	Discrimination	Logit adjustment (train)	0.972	[0.960, 0.986]
PANDA	Native	Severe	Discrimination	OKO	0.965	[0.952, 0.980]
PANDA	Native	Severe	Discrimination	Logit adjustment (post-hoc)	0.910	[0.893, 0.926]
TCGA-UT	Native	Severe	Discrimination	Balanced sampling	0.821	[0.687, 0.963]
TCGA-UT	Native	Severe	Discrimination	CE + soft F_1	0.847	[0.697, 1.012]
TCGA-UT	Native	Severe	Discrimination	CE + soft MCC	0.883	[0.730, 1.058]
TCGA-UT	Native	Severe	Discrimination	CFAL	0.973	[0.707, 1.301]
TCGA-UT	Native	Severe	Discrimination	cRT	0.270	[0.112, 0.442]
TCGA-UT	Native	Severe	Discrimination	Focal loss	-0.094	[-0.254, 0.050]
TCGA-UT	Native	Severe	Discrimination	Logit adjustment (train)	0.584	[0.420, 0.783]
TCGA-UT	Native	Severe	Discrimination	OKO	1.646	[1.326, 2.142]
TCGA-UT	Native	Severe	Discrimination	Logit adjustment (post-hoc)	0.407	[0.276, 0.583]
TCGA-UT	Reversed	Severe	Discrimination	Balanced sampling	0.846	[0.703, 0.962]
TCGA-UT	Reversed	Severe	Discrimination	CE + soft F_1	0.919	[0.776, 1.060]
TCGA-UT	Reversed	Severe	Discrimination	CE + soft MCC	0.958	[0.806, 1.120]
TCGA-UT	Reversed	Severe	Discrimination	CFAL	1.064	[0.785, 1.538]
TCGA-UT	Reversed	Severe	Discrimination	cRT	0.167	[0.031, 0.309]
TCGA-UT	Reversed	Severe	Discrimination	Focal loss	-0.125	[-0.287, -0.020]
TCGA-UT	Reversed	Severe	Discrimination	Logit adjustment (train)	0.500	[0.346, 0.670]
TCGA-UT	Reversed	Severe	Discrimination	OKO	1.784	[1.423, 2.447]
TCGA-UT	Reversed	Severe	Discrimination	Logit adjustment (post-hoc)	0.344	[0.227, 0.535]
BRACS	Native	Moderate	Calibration	Balanced sampling	1.022	[0.718, 1.408]
BRACS	Native	Moderate	Calibration	CE + soft F_1	0.939	[0.640, 1.333]
BRACS	Native	Moderate	Calibration	CE + soft MCC	0.896	[0.594, 1.248]
BRACS	Native	Moderate	Calibration	CFAL	-0.222	[-1.893, 0.376]
BRACS	Native	Moderate	Calibration	cRT	-0.712	[-1.880, -0.212]
BRACS	Native	Moderate	Calibration	Focal loss	-0.406	[-2.148, 0.162]
BRACS	Native	Moderate	Calibration	Logit adjustment (train)	1.296	[1.117, 1.756]
BRACS	Native	Moderate	Calibration	OKO	0.724	[0.097, 1.135]
BRACS	Native	Moderate	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
BRACS	Aligned	Moderate	Calibration	Balanced sampling	1.184	[1.046, 2.023]
BRACS	Aligned	Moderate	Calibration	CE + soft F_1	1.159	[1.018, 1.955]
BRACS	Aligned	Moderate	Calibration	CE + soft MCC	1.150	[1.012, 1.885]
BRACS	Aligned	Moderate	Calibration	CFAL	0.999	[0.774, 1.270]
BRACS	Aligned	Moderate	Calibration	cRT	-0.088	[-1.827, 0.194]
BRACS	Aligned	Moderate	Calibration	Focal loss	0.530	[-0.616, 0.768]
BRACS	Aligned	Moderate	Calibration	Logit adjustment (train)	0.934	[0.381, 1.026]
BRACS	Aligned	Moderate	Calibration	OKO	1.077	[0.950, 1.579]
BRACS	Aligned	Moderate	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
BRACS	Reversed	Moderate	Calibration	Balanced sampling	0.895	[0.711, 0.976]
BRACS	Reversed	Moderate	Calibration	CE + soft F_1	0.888	[0.700, 0.971]
BRACS	Reversed	Moderate	Calibration	CE + soft MCC	0.872	[0.649, 0.968]

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Dataset	Assignment	Severity	Gate	Method	R_M	95% CI
BRACS	Reversed	Moderate	Calibration	CFAL	0.251	[-0.594, 0.581]
BRACS	Reversed	Moderate	Calibration	cRT	0.007	[-0.464, 0.217]
BRACS	Reversed	Moderate	Calibration	Focal loss	-0.135	[-1.339, 0.315]
BRACS	Reversed	Moderate	Calibration	Logit adjustment (train)	1.060	[1.012, 1.151]
BRACS	Reversed	Moderate	Calibration	OKO	0.721	[0.415, 0.861]
BRACS	Reversed	Moderate	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
BRACS	Native	Severe	Calibration	Balanced sampling	0.902	[0.684, 0.992]
BRACS	Native	Severe	Calibration	CE + soft F_1	0.886	[0.715, 0.983]
BRACS	Native	Severe	Calibration	CE + soft MCC	0.026	[-1.822, 0.495]
BRACS	Native	Severe	Calibration	CFAL	0.529	[-0.375, 0.777]
BRACS	Native	Severe	Calibration	cRT	0.062	[-0.551, 0.358]
BRACS	Native	Severe	Calibration	Focal loss	-0.753	[-2.948, -0.204]
BRACS	Native	Severe	Calibration	Logit adjustment (train)	0.979	[0.782, 1.068]
BRACS	Native	Severe	Calibration	OKO	0.785	[0.332, 0.921]
BRACS	Native	Severe	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
BRACS	Aligned	Severe	Calibration	Balanced sampling	1.058	[0.989, 1.139]
BRACS	Aligned	Severe	Calibration	CE + soft F_1	1.054	[0.976, 1.139]
BRACS	Aligned	Severe	Calibration	CE + soft MCC	0.313	[0.055, 0.515]
BRACS	Aligned	Severe	Calibration	CFAL	0.970	[0.904, 1.034]
BRACS	Aligned	Severe	Calibration	cRT	0.206	[0.096, 0.295]
BRACS	Aligned	Severe	Calibration	Focal loss	0.427	[0.138, 0.607]
BRACS	Aligned	Severe	Calibration	Logit adjustment (train)	0.802	[0.723, 0.860]
BRACS	Aligned	Severe	Calibration	OKO	0.864	[0.709, 0.965]
BRACS	Aligned	Severe	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
BRACS	Reversed	Severe	Calibration	Balanced sampling	0.928	[0.883, 0.958]
BRACS	Reversed	Severe	Calibration	CE + soft F_1	0.926	[0.881, 0.956]
BRACS	Reversed	Severe	Calibration	CE + soft MCC	0.890	[0.746, 1.001]
BRACS	Reversed	Severe	Calibration	CFAL	0.711	[0.597, 0.790]
BRACS	Reversed	Severe	Calibration	cRT	0.488	[0.409, 0.566]
BRACS	Reversed	Severe	Calibration	Focal loss	-0.026	[-0.384, 0.194]
BRACS	Reversed	Severe	Calibration	Logit adjustment (train)	1.021	[0.989, 1.059]
BRACS	Reversed	Severe	Calibration	OKO	0.934	[0.898, 0.967]
BRACS	Reversed	Severe	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]

Table 16: Exploratory recovery estimates for the full method roster on gate-passing units, with paired 95% intervals. These methods vary the intervention mechanism as well as the signal, carry no confirmatory status, and are not part of the Holm family.

The roster establishes that no gate-passing unit is beyond repair. In every one of the 12 discrimination-gate units at least one roster method recovers more than 0.67 of the deficit, and in 10 of them at least one recovers more than 0.80. The best exploratory recoverers are balanced sampling, train-time logit adjustment, and the soft-MCC hybrid on BRACS; set learning and the soft- F_1 hybrid on CAMELYON16; the soft-MCC hybrid on PANDA; and set learning on TCGA-UT, where it reaches 1.646 and 1.784 and therefore overshoots the size-matched balanced reference. Three of these mechanisms resample or reweight at the level of the batch rather than the loss term, which is the family for which full recovery is mechanically attainable, so overshoot is not anomalous. The practical consequence for Section 5 is that the recovery ceiling on these conditions is set by neither the data nor the representation: where damage exists, some tested mechanism reaches most of it.

4 Explaining Heterogeneity

Damage magnitude and recovery headroom vary across conditions, and RQ3 asks which realized shortages track that variation. Two association models are fitted, one for signed balanced-accuracy damage over the cross-entropy cells and one for recovery over the gate-passing method cells. Both use the same linear predictor over standardized quantities,

$$\mu_j = \alpha + u_{g[j]} + \beta_\rho \log \rho_j + \beta_{\text{ind}} S_{\text{ind},j} + \beta_A A_j + \beta_{\text{div}} S_{\text{div},j}, \quad (1)$$

where ρ_j is achieved severity, $S_{\text{ind},j}$ the independent-support shortage, A_j the support-difficulty alignment, $S_{\text{div},j}$ the diversity shortage, and $u_{g[j]}$ a random intercept over dataset-target groups that absorbs level differences between datasets. Observation errors come from the bootstrap distributions of the corresponding endpoints.

The damage model is fitted over the 18 cross-entropy cells and the recovery model over the gate-passing method cells with a defined recovery fraction; the matched-versus-unmatched contrast rows have no deficit denominator and therefore no recovery, so they enter neither. These models are exploratory by construction and by fitting procedure. They are estimated at the posterior mode and return point coefficients with no standard errors and no posterior intervals, so a coefficient reported here indicates direction and relative magnitude but licenses no claim of statistical significance. The only accuracy evidence available for them is the leave-one-group-out check, which refits each model with one dataset-target group held out and reports the held-out root mean squared error; a group whose held-out error is much larger than the rest indicates that the association does not transfer across datasets. Coefficients and that diagnostic appear in Tables 17 and 18.

Model	α	$\beta_{\log \rho}$	β_{ind}	β_A	β_{div}	σ_u	σ
Damage	0.0308	0.0471	0.0000	-0.0008	-0.0424	0.0484	0.0184
Recovery	0.5596	-0.0115	0.0000	-0.0242	-0.0290	0.0404	0.4372

Table 17: Exploratory association models for signed balanced-accuracy damage and for gated recovery: standardized coefficients from Equation (1), together with the group and residual scale. Estimated at the posterior mode; no intervals are available.

Three readings follow from the damage model, each qualified by the absence of intervals. Achieved severity is the largest signal predictor, at $\beta_{\log \rho} = 0.047,1$ on standardized scale against a residual scale of $\sigma = 0.018,4$, which restates in model form the severity ordering already visible in Table 2. Diversity shortage enters with a comparable magnitude and the opposite sign, $\beta_{\text{div}} = -0.042,4$; because severe conditions have both higher ρ and higher measured diversity shortage, the two predictors are strongly collinear across only 18 units, and the negative coefficient is better read as the diversity term absorbing part of a severity effect the model has

Held-out group	Cells	Held-out RMSE
BRACS (roi subtype)	6	0.0649
CAMELYON16 (tumor presence)	4	0.0481
PANDA (cancer presence)	2	0.0861
TCGA-UT (cancer type)	6	0.0131

Table 18: Leave-one-group-out check on the damage model: each dataset–target group is held out in turn, the model refitted on the remainder, and the held-out root mean squared error reported alongside the number of cells in the held-out group. Exploratory.

already assigned to $\log \rho$ than as evidence that losing representational diversity reduces damage. Support–difficulty alignment contributes almost nothing, $\beta_A = -0.000,8$, which sits awkwardly beside the large assignment effects in Table 2; the alignment predictor is a single correlation summarizing an entire condition, and on the two binary datasets it can take only the values ± 1 , so it is a coarse encoding of the assignment contrast rather than a null result about it. The independent-support coefficient is exactly zero, for the reason established in Section 2.1: its predictor has no variance.

The variance components are as informative as the slopes. The dataset–target random-intercept scale, $\sigma_u = 0.048,4$, is more than two and a half times the residual scale, so more of the variation in damage lies between datasets than the four signal predictors explain within them. The leave-one-group-out errors are consistent with that: holding out TCGA-UT gives a held-out error of 0.013,1, comparable to the residual scale, whereas holding out PANDA gives 0.086,1, larger than any deficit that dataset records except its own severe condition. PANDA contributes only two cells and has the largest severe deficit in the benchmark, so the model cannot place it from the other three groups. The association therefore does not transfer across datasets, and no coefficient here should be carried into a method design without a within-dataset check.

The recovery model, fitted over the 252 gate-passing method cells that have a defined recovery fraction, answers RQ3’s second half in the negative. Its intercept of 0.559,6 places average recovery a little above half the established deficit, but none of the four realized shortages accounts for the spread around that average: the three estimable slopes are $\beta_{\log \rho} = -0.011,5$, $\beta_A = -0.024,2$, and $\beta_{\text{div}} = -0.029,0$, all an order of magnitude smaller than the residual scale $\sigma = 0.437,2$. The variance components make the reason plain. The residual scale exceeds the dataset–target intercept scale $\sigma_u = 0.040,4$ by a factor of eleven, exactly reversing the ordering in the damage model, so almost all variation in recovery lies within a dataset rather than between datasets. What varies within a dataset in this model is the method, which the model deliberately does not include as a predictor. Recovery headroom on these conditions is therefore a property of which method was applied, not of which shortage the condition realized, which is the same conclusion Section 3.1 reaches from the confirmatory family directly.

5 Discussion

The preceding sections establish what the controlled allocation damaged, how much of that damage the tested methods returned, and which realized shortages tracked the variation. This section reads those outcomes as design evidence. The question it answers is not which existing method won, but what a mitigation method would have to read, and to do, in order to address the shortages these datasets actually present.

5.1 What the evidence supports

The findings are summarized here by research question rather than by dataset, since the design intends datasets to be replicates of a mechanism rather than objects of interest in themselves. Three statements structure the summary, and every statement is bound to the gate status of the units it rests on, so that a claim about recovery is never carried by a unit in which no damage was established.

On RQ1, controlled allocation produced a material discrimination deficit in 12 of 18 comparison units and a material probability-quality deficit in six, all six of the latter on BRACS. Damage grows with achieved severity and with adversity of the tail assignment, and it is largest where the fixed budget is split between few classes, so that the tail retains almost none of it. The six units that opened neither gate all fail on magnitude rather than on precision: at $\rho = 10$ on CAMELYON16 and TCGA-UT, and at $\rho = 100$ on the difficulty-aligned TCGA-UT condition, redistributing the budget measurably but immaterially changes case-macro balanced accuracy. Damage did not fall where support rank alone predicts: under the difficulty-reversed assignment the body tier absorbs more loss than the tail.

On RQ2, recovery separated widely across the confirmatory family at a fixed intervention mechanism, but along a single split. Prevalence-weighted and class-balanced cross-entropy recover most of both the discrimination and the probability-quality deficit wherever a gate opened; independent-support, difficulty, and diversity weighting recover essentially none of the discrimination deficit and actively degrade tail probability quality in 15 of 18 calibration cells. The matched-versus-unmatched contrast is positive only in the three units whose dominant shortage is nominal support, and negative in every unit whose dominant shortage is one of the other three, because in those units the matched arm holds exactly the member that does not work. The matched- β diagnostic excludes tuned weight strength as the explanation.

On RQ3, achieved severity is the strongest signal-side predictor of damage magnitude, diversity shortage enters with a comparable magnitude but is collinear with severity across only 18 units, support–difficulty alignment contributes almost nothing under its single-correlation encoding, and independent-support shortage is fixed at zero by the design. Between-dataset variation exceeds within-dataset residual variation in the damage model, and the leave-one-group-out check shows the association does not transfer to PANDA. The recovery model reverses that ordering by a factor of eleven and gives all four shortages coefficients an order of magnitude below its residual scale, so no realized shortage explains recovery headroom; what varies is the method, which the model does not include. Both models carry point estimates only.

5.2 Which shortage a mitigation method must read

A method must estimate the shortage that binds, which makes the empirical question whether the dominant shortage was identifiable and whether it mattered. On these datasets the answer is that the dominant shortage was identifiable in 12 of 18 units and did not carry the recovery. Neither independent-support nor diversity shortage tracked damage that prevalence failed to track, and the two methods that read prevalence and nominal support recovered most of every deficit that was established. Under the outcome the protocol anticipated for this case, the simpler weighting is the correct default and the added shortage-estimation machinery buys nothing.

That conclusion has to be stated with the design boundary attached, because the boundary is what produced part of it. Independent-support shortage was not merely uninformative here; it was never created. The shared per-class evidence pool that makes the allocation contrast clean also holds independent support fixed across conditions, so the benchmark tests independent-support weighting on conditions in which the quantity it reads does not vary. The correct inference is that a prevalence-only method is sufficient for allocation-induced shortage under a

fixed evidence pool, not that independent support is an unproductive signal in general. Testing that would require cohorts that differ in how many patients back a class, which this design deliberately excludes.

The frequency of the ambiguous designation qualifies the reading further. Six of 18 units, four of them at moderate severity, have their two largest shortage scores within a quarter of a standard deviation. A taxonomy that cannot separate the top two axes in a third of its conditions does not partition sharply enough to route a method, and in that case the design implication is a method that combines signals rather than one that selects among them.

5.3 Signal against mechanism

Because the five confirmatory methods share one intervention mechanism and differ only in the signal their weights read, the spread of recovery across them isolates how much leverage the choice of signal carries. That spread is wide, running from below zero to above 0.99 within single units, so signal choice does determine whether a weighted-loss method works at all. The spread is not, however, evidence that a better shortage estimate would help, because it is bimodal rather than graded: two members work almost everywhere and three work almost nowhere, with little in between. The productive direction that this pattern indicates is not a finer estimate of difficulty or diversity but an explanation of why weights derived from those signals fail to move the endpoint at all.

The comparison between prevalence-weighted and class-balanced cross-entropy on PANDA sharpens that point, since those two occupy adjacent corners of the taxonomy yet differ by a full deficit's worth of recovery, 0.968 against 0.118 at severe severity. What separates them is not the signal in the taxonomy's sense but the transform applied to it: the effective-number reweighting saturates at high imbalance on a binary target and returns nearly uniform weights. A signal taxonomy that treats these two as neighbours therefore under-describes the design space that actually determines recovery.

The matched- β diagnostic separates these readings from a third possibility. A method may appear to win on its signal while in fact winning on a more aggressive tuned weight strength; refitting independent-support weighting at the class-balanced weight strength removes that confound. Here the refit changes balanced accuracy by at most 0.018 in any unit, and its largest movement, on PANDA at severe severity, is downward; the gap to class-balanced cross-entropy remains as large as 0.098. The advantage that class-balanced weighting holds is therefore attributable to what its weights encode and not to how aggressively they are applied.

5.4 Where recovery ceilings

The units that define an unmet need would be those in which a gate opened and no method, confirmatory or exploratory, returned a substantial fraction of the deficit. There are none. In every gate-passing unit at least one tested method recovers more than 0.67 of the deficit, and the full roster reaches above 0.80 in nine of the 12 discrimination-gate units. The ceiling on these conditions therefore belongs to neither the data nor the frozen representation, and a new method cannot be motivated by damage that existing mechanisms leave unrepaired. It has to be motivated instead by what those mechanisms cost, by the endpoints they degrade while repairing another, or by the regimes in which they should not be applied.

The complementary case is equally informative for scoping. Six units opened no gate at all, and all six failed on magnitude rather than precision. A method that intervenes there spends capacity and risks degrading endpoints that were not damaged, and the calibration results in Section 3.3 show that this risk is real rather than hypothetical: in cells where the wrong signal was applied, tail-group probability quality fell well below the untreated imbalanced reference. Knowing when not to act is part of the specification, not an afterthought.

5.5 Design requirements and open questions

The preceding subsections resolve into a small set of requirements for a mitigation method, each traceable to a specific result and annotated with the strength of evidence behind it.

1. The method must read prevalence or nominal support, and must not rely on a difficulty or diversity estimate as its primary signal. *Confirmatory*, from the five-family separation in Table 9 on all 12 discrimination-gate units.
2. It must specify the transform applied to that signal, not only the signal itself, because the two prevalence-corner members differ by a full deficit’s worth of recovery on PANDA where the effective-number transform saturates. *Confirmatory*, from the same table.
3. The estimate must be formed per class and per condition rather than transferred across datasets, since the dataset–target intercept scale exceeds the residual scale and the leave-one-group-out error on PANDA is six times that on TCGA-UT. *Exploratory*, from Tables 17 and 18.
4. It must not degrade tail-group probability quality while repairing discrimination. *Confirmatory*, from the 15 negative calibration-recovery cells in Table 14, 10 of which survive Holm adjustment.
5. It must decline to intervene where allocation produced no material deficit, which on these datasets is every moderate condition on CAMELYON16 and TCGA-UT and the difficulty-aligned severe condition on TCGA-UT. *Confirmatory*, from the gate routing in Table 6.
6. It must not require access to test prevalence, labelled data beyond the training partition, or per-condition tuning that a deployment could not perform. *Design constraint*, carried from the protocol rather than established by a result here.

The exploratory models of Section 4 supply point estimates with no intervals and may therefore motivate a design direction but never justify one on their own.

Three questions remain outside this benchmark. It manipulates allocation within a fixed pool, so it says nothing about patient shortage as such. It also fixes Virchow2 features. Whether any gate-passing regime has unmet need is deferred to the exposure-matched rerun.

6 Conclusion and Scope

This report answers three questions in the order the protocol fixed: what controlled allocation damaged, how much of that damage signal-matched mitigation returned, and which realized shortages tracked the variation in both. Its contribution is the design evidence assembled in Section 5, which states what a mitigation method must read, at what granularity, under what constraints, and where it should decline to act.

The legacy run cannot settle which signal recovers damage or whether a gate-passing condition resists repair. Those conclusions await the exposure-matched rerun.

Four scope conditions bound every claim above. The three partitions are locked design repetitions of one dataset, not independent cohorts, so the intervals quantify variation across partitionings and initializations rather than across populations. The association models of Section 4 are exploratory and carry no causal interpretation, and their coefficients come without intervals. The benchmark covers patch classification only, where each prediction unit carries its own label; at whole-slide level, reallocating labelled slides changes nominal and independent support together, so the two signals could not be separated in the way this design requires. Temperature scaling is reported as a diagnostic throughout and never gates a decision.

A Realized Conditions and Analysis Validity

This appendix carries the execution record that Section 2 summarizes in prose. It documents, per dataset, what the frozen construction realized and whether the conditions required for the confirmatory inference to be valid were met.

A.1 Realized construction

Table 20 reports the achieved imbalance ratio and normalized-entropy deficit for each dataset, partition, and controlled condition, together with the realized partitioning-unit, slide, and patch counts. Conditions falling outside the tolerance bands, $\rho \in [9, 11]$ for moderate and $\rho \in [90, 110]$ for severe, are flagged as off-target, and allocations that collapsed to balanced are flagged as degenerate and excluded from all estimands. No condition carries either flag. Partitioning-unit and slide counts are unique within a realization: one unit may supply patches to more than one class, so per-class counts are not summed.

Two classes are absent from the TCGA-UT target set, for independent reasons recorded in the frozen provenance. Diffuse Large B-cell Lymphoma is excluded because no shared total is both tolerance-feasible and compatible with the per-unit and per-slide contribution caps for it. Cholangiocarcinoma is excluded because its held-out participant count cannot support the split-unit-clustered inference the confirmatory analysis requires. The retained target is therefore 30 cancer types, and every quantity reported for TCGA-UT in this report refers to that 30-class task.

Dataset	Split	Condition	Assignment	Units	Slides	Patches	ρ	$1 - H_{\text{norm}}$	Tolerance
BRACS	0	Natural	—	103	271	217399	5.901	0.1202	—
BRACS	0	Balanced	—	85	204	16796	1.000	0.0000	—
BRACS	0	Moderate	Aligned	85	204	16796	10.007	0.1280	On target
BRACS	0	Severe	Aligned	85	194	16796	100.467	0.3530	On target
BRACS	0	Moderate	Reversed	85	204	16796	10.007	0.1280	On target
BRACS	0	Severe	Reversed	85	203	16796	100.467	0.3530	On target
BRACS	0	Moderate	Native	85	204	16796	10.007	0.1280	On target
BRACS	0	Severe	Native	85	204	16796	100.467	0.3530	On target
BRACS	1	Natural	—	103	273	193010	5.722	0.1178	—
BRACS	1	Balanced	—	89	233	27202	1.000	0.0000	—
BRACS	1	Moderate	Aligned	89	233	27202	10.003	0.1280	On target
BRACS	1	Severe	Aligned	89	232	27202	100.301	0.3530	On target
BRACS	1	Moderate	Reversed	89	233	27202	10.003	0.1280	On target
BRACS	1	Severe	Reversed	89	233	27202	100.301	0.3530	On target
BRACS	1	Moderate	Native	89	233	27202	10.003	0.1280	On target
BRACS	1	Severe	Native	89	233	27202	100.301	0.3530	On target
BRACS	2	Natural	—	103	262	217704	8.297	0.1432	—
BRACS	2	Balanced	—	78	179	13982	1.001	0.0000	—
BRACS	2	Moderate	Aligned	78	179	13982	10.004	0.1280	On target
BRACS	2	Severe	Aligned	78	173	13982	100.360	0.3530	On target
BRACS	2	Moderate	Reversed	78	179	13982	10.004	0.1280	On target
BRACS	2	Severe	Reversed	78	179	13982	100.360	0.3530	On target
BRACS	2	Moderate	Native	78	179	13982	10.004	0.1280	On target
BRACS	2	Severe	Native	78	179	13982	100.360	0.3530	On target
CAMELYON16	0	Natural	—	264	264	2234906	22.453	0.7457	—
CAMELYON16	0	Balanced	—	109	109	36127	1.000	0.0000	—
CAMELYON16	0	Moderate	Reversed	109	109	36127	10.001	0.5605	On target
CAMELYON16	0	Severe	Reversed	109	109	36127	99.913	0.9198	On target
CAMELYON16	0	Moderate	Native	109	109	36127	10.001	0.5605	On target
CAMELYON16	0	Severe	Native	109	109	36127	99.913	0.9198	On target
CAMELYON16	1	Natural	—	264	264	2139913	33.608	0.8112	—
CAMELYON16	1	Balanced	—	118	118	28810	1.000	0.0000	—
CAMELYON16	1	Moderate	Reversed	118	118	28810	10.000	0.5605	On target
CAMELYON16	1	Severe	Reversed	118	118	28810	100.088	0.9199	On target
CAMELYON16	1	Moderate	Native	118	118	28810	10.000	0.5605	On target
CAMELYON16	1	Severe	Native	118	118	28810	100.088	0.9199	On target
CAMELYON16	2	Natural	—	264	264	2193242	28.943	0.7889	—
CAMELYON16	2	Balanced	—	103	103	30685	1.000	0.0000	—
CAMELYON16	2	Moderate	Reversed	103	103	30685	9.998	0.5605	On target
CAMELYON16	2	Severe	Reversed	103	103	30685	99.938	0.9198	On target
CAMELYON16	2	Moderate	Native	103	103	30685	9.998	0.5605	On target
CAMELYON16	2	Severe	Native	103	103	30685	99.938	0.9198	On target
PANDA	0	Natural	—	7358	7358	3153535	4.735	0.3324	—

Continued on next page

Dataset	Split	Condition	Assignment	Units	Slides	Patches	ρ	$1 - H_{\text{norm}}$	Tolerance
PANDA	0	Balanced	—	6025	6025	1099847	1.000	0.0000	—
PANDA	0	Moderate	Native	6025	6025	1099847	10.000	0.5605	On target
PANDA	0	Severe	Native	6025	6025	1099847	99.996	0.9199	On target
PANDA	1	Natural	—	7358	7358	3161828	4.645	0.3262	—
PANDA	1	Balanced	—	6079	6079	1120125	1.000	0.0000	—
PANDA	1	Moderate	Native	6079	6079	1120125	10.000	0.5605	On target
PANDA	1	Severe	Native	6079	6079	1120125	100.003	0.9199	On target
PANDA	2	Natural	—	7358	7358	3141372	4.645	0.3262	—
PANDA	2	Balanced	—	6080	6080	1112957	1.000	0.0000	—
PANDA	2	Moderate	Native	6080	6080	1112957	10.000	0.5605	On target
PANDA	2	Severe	Native	6080	6080	1112957	100.003	0.9199	On target
TCGA-UT	0	Natural	—	4983	6055	1116020	18.254	0.0765	—
TCGA-UT	0	Balanced	—	917	933	44792	1.001	0.0000	—
TCGA-UT	0	Moderate	Aligned	917	933	44792	9.992	0.0609	On target
TCGA-UT	0	Severe	Aligned	917	933	44792	90.181	0.1735	On target
TCGA-UT	0	Moderate	Reversed	917	933	44792	9.992	0.0609	On target
TCGA-UT	0	Severe	Reversed	917	930	44792	100.500	0.1792	On target
TCGA-UT	0	Moderate	Native	917	933	44792	9.992	0.0609	On target
TCGA-UT	0	Severe	Native	917	933	44792	100.500	0.1792	On target
TCGA-UT	1	Natural	—	4983	6079	1120590	17.850	0.0754	—
TCGA-UT	1	Balanced	—	752	821	52204	1.001	0.0000	—
TCGA-UT	1	Moderate	Aligned	752	821	52204	10.000	0.0609	On target
TCGA-UT	1	Severe	Aligned	752	821	52204	90.107	0.1736	On target
TCGA-UT	1	Moderate	Reversed	752	821	52204	10.000	0.0609	On target
TCGA-UT	1	Severe	Reversed	752	818	52204	100.403	0.1792	On target
TCGA-UT	1	Moderate	Native	752	821	52204	10.000	0.0609	On target
TCGA-UT	1	Severe	Native	752	821	52204	100.403	0.1792	On target
TCGA-UT	2	Natural	—	4983	6100	1121580	18.395	0.0764	—
TCGA-UT	2	Balanced	—	671	699	36988	1.001	0.0000	—
TCGA-UT	2	Moderate	Aligned	671	699	36988	10.003	0.0609	On target
TCGA-UT	2	Severe	Aligned	671	699	36988	91.017	0.1740	On target
TCGA-UT	2	Moderate	Reversed	671	699	36988	10.003	0.0609	On target
TCGA-UT	2	Severe	Reversed	671	699	36988	99.600	0.1792	On target
TCGA-UT	2	Moderate	Native	671	699	36988	10.003	0.0609	On target
TCGA-UT	2	Severe	Native	671	699	36988	99.600	0.1792	On target

Table 20: Realized training support per dataset, partition, and controlled condition: unique partitioning units, slides, patches, achieved ρ , and normalized-entropy deficit, with tolerance status. The natural anchor and the balanced reference carry no severity tolerance.

A.2 Resampling validity and completeness

The label-only preflight resamples partitioning units before any model is fitted and checks whether class-level endpoints can be estimated stably at all. Three concentration checks apply per class: whether the 2.5th percentile of the Kish effective count falls below five, whether a single unit carries more than half the weight in more than 5% of replicates, and whether a pooled cell draws on fewer than five contributing units. A class failing any check designates its dataset descriptive-only, which forces both gates closed. Table 22 reports these quantities and the resulting designation, taking the most adverse reading across the three partitions for each class.

Dataset	Class	Kish n_{eff}	2.5th pct.	Max unit weight	Dominant-unit share	Descriptive-only
BRACS	ADH	10.8	7.2	0.180	0.000	No
BRACS	DCIS	8.8	5.5	0.211	0.000	No
BRACS	FEA	9.3	6.0	0.201	0.000	No
BRACS	IC	14.3	9.7	0.145	0.000	No
BRACS	N	14.4	9.9	0.144	0.000	No
BRACS	PB	18.9	13.5	0.116	0.000	No
BRACS	UDH	17.9	12.6	0.121	0.000	No
CAMELYON16	normal	71.5	60.0	0.039	0.000	No
CAMELYON16	tumor	23.9	17.7	0.096	0.000	No
PANDA	benign	2019.6	1956.0	0.002	0.000	No
PANDA	cancer	1416.7	1363.5	0.003	0.000	No
TCGA-UT	Adrenocortical carcinoma	10.8	7.1	0.180	0.000	No
TCGA-UT	Bladder Urothelial Carcinoma	56.9	46.4	0.047	0.000	No
TCGA-UT	Brain Lower Grade Glioma	91.0	77.4	0.032	0.000	No
TCGA-UT	Breast invasive carcinoma	145.0	128.3	0.022	0.000	No
TCGA-UT	Cervical squamous cell carcinoma and endocervical adenocarcinoma	38.0	29.6	0.066	0.000	No
TCGA-UT	Colon adenocarcinoma	42.4	33.5	0.060	0.000	No
TCGA-UT	Esophageal carcinoma	21.9	16.0	0.103	0.000	No
TCGA-UT	Glioblastoma multiforme	70.4	58.7	0.040	0.000	No
TCGA-UT	Head and Neck squamous cell carcinoma	66.5	55.1	0.042	0.000	No
TCGA-UT	Kidney Chromophobe	16.8	11.8	0.127	0.000	No
TCGA-UT	Kidney renal clear cell carcinoma	76.0	64.3	0.037	0.000	No
TCGA-UT	Kidney renal papillary cell carcinoma	42.9	34.1	0.060	0.000	No
TCGA-UT	Liver hepatocellular carcinoma	54.5	44.3	0.049	0.000	No
TCGA-UT	Lung adenocarcinoma	85.9	72.6	0.034	0.000	No
TCGA-UT	Lung squamous cell carcinoma	88.0	75.1	0.033	0.000	No
TCGA-UT	Mesothelioma	12.8	8.5	0.157	0.000	No
TCGA-UT	Ovarian serous cystadenocarcinoma	17.4	12.2	0.124	0.000	No
TCGA-UT	Pancreatic adenocarcinoma	24.4	18.1	0.094	0.000	No
TCGA-UT	Pheochromocytoma and Paraganglioma	10.9	7.1	0.180	0.000	No
TCGA-UT	Prostate adenocarcinoma	60.3	49.8	0.045	0.000	No
TCGA-UT	Rectum adenocarcinoma	13.9	9.4	0.148	0.000	No
TCGA-UT	Sarcoma	40.9	32.4	0.062	0.000	No
TCGA-UT	Skin Cutaneous Melanoma	64.5	53.5	0.043	0.000	No
TCGA-UT	Stomach adenocarcinoma	51.5	41.7	0.051	0.000	No
TCGA-UT	Testicular Germ Cell Tumors	26.4	19.7	0.089	0.000	No
TCGA-UT	Thymoma	16.3	11.4	0.130	0.000	No
TCGA-UT	Thyroid carcinoma	70.5	59.1	0.040	0.000	No
TCGA-UT	Uterine Carcinosarcoma	10.3	6.6	0.188	0.000	No
TCGA-UT	Uterine Corpus Endometrial Carcinoma	76.4	64.5	0.037	0.000	No
TCGA-UT	Uveal Melanoma	11.3	7.4	0.174	0.000	No

Table 22: Label-only bootstrap preflight per dataset and class: Kish effective count and its 2.5th percentile, maximum single-unit weight fraction, share of replicates with a dominant unit, and the resulting descriptive-only designation. Values are the most adverse across the three partitions.

No class on any dataset is designated descriptive-only. The tightest margins are on BRACS, whose ductal carcinoma in situ class reaches a 2.5th-percentile Kish effective count of 5.5 against a floor of five, and no class on any dataset records a dominant unit in any replicate.

Completeness is enforced rather than reported after the fact. Every confirmation block must contain all five initialization seeds, every expected combination of tail assignment, severity, method, and gate must appear in all three partitions, and the achieved severity of a condition must not vary by more than half across partitions. Table 23 records the outcome of each guard per dataset, and Section A.3 states what was done where a guard did not pass.

Dataset	Splits	Seeds per block	Comparisons	Max severity spread	Descriptive-only	Guards
BRACS	3	5	188	0.002	No	All passed
CAMELYON16	3	5	126	0.002	No	All passed
PANDA	3	5	62	0.000	No	All passed
TCGA-UT	3	5	188	0.010	No	All passed

Table 23: Completeness and consistency guards per dataset: confirmation-seed blocks, expected comparison coverage across the three partitions, and achieved-severity spread. These guards halt aggregation rather than degrade it, so a written result implies every guard passed.

A.3 Tuning selections and deviations

The legacy selections optimize balanced accuracy, pooled across tail assignments, with macro F1 and NLL tie-breakers. The rerun reports frozen exposure budgets, derived updates, and the expanded learning-rate envelope.

Table 25 records the legacy selected controls, update budgets, and boundary hits.

Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Legacy updates
BRACS	Balanced	—	Balanced sampling	1e-05	0.2500	Yes	3960
BRACS	Balanced	—	CE	3e-05	—	No	3960
BRACS	Balanced	—	CE + soft F_1	3e-05	0.0000	No	3960
BRACS	Balanced	—	CE + soft MCC	3e-05	64.0000	No	3960
BRACS	Balanced	—	CFAL	0.001	0.2500	No	3960
BRACS	Balanced	—	Nominal support	3e-05	0.9999	No	3960
BRACS	Balanced	—	cRT	0.001	—	No	3960
BRACS	Balanced	—	Focal loss	3e-05	0.0000	No	3960
BRACS	Balanced	—	Independent support	3e-05	0.8000	No	3960
BRACS	Balanced	—	Logit adjustment (train)	3e-05	2.0000	No	3960
BRACS	Balanced	—	OKO	0.003	2.0000	No	3960
BRACS	Balanced	—	Difficulty	3e-05	0.2500	No	3960
BRACS	Balanced	—	Logit adjustment (post-hoc)	—	1.0000	No	3960
BRACS	Balanced	—	Diversity	3e-05	0.2500	No	3960
BRACS	Balanced	—	Prevalence	3e-05	0.2500	No	3960
BRACS	Moderate	Aligned	Balanced sampling	3e-05	1.0000	No	3960
BRACS	Moderate	Aligned	CE	0.003	—	No	3960
BRACS	Moderate	Aligned	CE + soft F_1	3e-05	0.2500	No	3960
BRACS	Moderate	Aligned	CE + soft MCC	3e-05	0.2500	No	3960
BRACS	Moderate	Aligned	CFAL	0.01	1.0000	Yes	3960
BRACS	Moderate	Aligned	Nominal support	3e-05	0.9999	No	3960
BRACS	Moderate	Aligned	cRT	0.003	—	No	3960
BRACS	Moderate	Aligned	Focal loss	0.001	1.5000	No	3960
BRACS	Moderate	Aligned	Independent support	0.001	0.9900	No	3960
BRACS	Moderate	Aligned	Logit adjustment (train)	3e-05	1.0000	No	3960
BRACS	Moderate	Aligned	OKO	0.001	4.0000	No	3960
BRACS	Moderate	Aligned	Difficulty	0.001	0.2500	No	3960
BRACS	Moderate	Aligned	Logit adjustment (post-hoc)	—	0.5000	No	3960
BRACS	Moderate	Aligned	Diversity	0.0003	0.7500	No	3960
BRACS	Moderate	Aligned	Prevalence	3e-05	1.0000	No	3960
BRACS	Moderate	Reversed	Balanced sampling	3e-05	1.0000	No	3960
BRACS	Moderate	Reversed	CE	0.003	—	No	3960
BRACS	Moderate	Reversed	CE + soft F_1	3e-05	0.2500	No	3960
BRACS	Moderate	Reversed	CE + soft MCC	3e-05	0.2500	No	3960
BRACS	Moderate	Reversed	CFAL	0.01	1.0000	Yes	3960
BRACS	Moderate	Reversed	Nominal support	3e-05	0.9999	No	3960
BRACS	Moderate	Reversed	cRT	0.003	—	No	3960
BRACS	Moderate	Reversed	Focal loss	0.001	1.5000	No	3960
BRACS	Moderate	Reversed	Independent support	0.001	0.9900	No	3960
BRACS	Moderate	Reversed	Logit adjustment (train)	3e-05	1.0000	No	3960
BRACS	Moderate	Reversed	OKO	0.001	4.0000	No	3960
BRACS	Moderate	Reversed	Difficulty	0.001	0.2500	No	3960
BRACS	Moderate	Reversed	Logit adjustment (post-hoc)	—	0.5000	No	3960

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Legacy updates
BRACS	Moderate	Reversed	Diversity	0.0003	0.7500	No	3960
BRACS	Moderate	Reversed	Prevalence	3e-05	1.0000	No	3960
BRACS	Moderate	Native	Balanced sampling	3e-05	1.0000	No	3960
BRACS	Moderate	Native	CE	0.003	—	No	3960
BRACS	Moderate	Native	CE + soft F_1	3e-05	0.2500	No	3960
BRACS	Moderate	Native	CE + soft MCC	3e-05	0.2500	No	3960
BRACS	Moderate	Native	CFAL	0.01	1.0000	Yes	3960
BRACS	Moderate	Native	Nominal support	3e-05	0.9999	No	3960
BRACS	Moderate	Native	cRT	0.003	—	No	3960
BRACS	Moderate	Native	Focal loss	0.001	1.5000	No	3960
BRACS	Moderate	Native	Independent support	0.001	0.9900	No	3960
BRACS	Moderate	Native	Logit adjustment (train)	3e-05	1.0000	No	3960
BRACS	Moderate	Native	OKO	0.001	4.0000	No	3960
BRACS	Moderate	Native	Difficulty	0.001	0.2500	No	3960
BRACS	Moderate	Native	Logit adjustment (post-hoc)	—	0.5000	No	3960
BRACS	Moderate	Native	Diversity	0.0003	0.7500	No	3960
BRACS	Moderate	Native	Prevalence	3e-05	1.0000	No	3960
BRACS	Natural	—	Balanced sampling	0.0001	0.5000	No	50970
BRACS	Natural	—	CE	0.0001	—	No	50970
BRACS	Natural	—	CE + soft F_1	0.0003	0.2500	No	50970
BRACS	Natural	—	CE + soft MCC	0.0003	0.2500	No	50970
BRACS	Natural	—	CFAL	0.01	2.0000	Yes	50970
BRACS	Natural	—	cRT	0.003	—	No	50970
BRACS	Natural	—	Focal loss	0.0001	4.0000	No	50970
BRACS	Natural	—	Logit adjustment (train)	0.0001	0.5000	No	50970
BRACS	Natural	—	OKO	0.0003	1.0000	No	50970
BRACS	Natural	—	Logit adjustment (post-hoc)	—	0.5000	No	50970
BRACS	Natural	—	Prevalence	0.0001	1.0000	No	50970
BRACS	Severe	Aligned	Balanced sampling	1e-05	1.0000	Yes	3960
BRACS	Severe	Aligned	CE	0.001	—	No	3960
BRACS	Severe	Aligned	CE + soft F_1	1e-05	0.2500	Yes	3960
BRACS	Severe	Aligned	CE + soft MCC	3e-05	64.0000	No	3960
BRACS	Severe	Aligned	CFAL	0.001	1.0000	No	3960
BRACS	Severe	Aligned	Nominal support	3e-05	0.9999	No	3960
BRACS	Severe	Aligned	cRT	0.001	—	No	3960
BRACS	Severe	Aligned	Focal loss	0.001	2.0000	No	3960
BRACS	Severe	Aligned	Independent support	0.001	0.9990	No	3960
BRACS	Severe	Aligned	Logit adjustment (train)	0.0003	1.0000	No	3960
BRACS	Severe	Aligned	OKO	0.0001	1.0000	No	3960
BRACS	Severe	Aligned	Difficulty	0.001	0.2500	No	3960
BRACS	Severe	Aligned	Logit adjustment (post-hoc)	—	0.5000	No	3960
BRACS	Severe	Aligned	Diversity	0.001	0.2500	No	3960
BRACS	Severe	Aligned	Prevalence	3e-05	1.0000	No	3960

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Legacy updates
BRACS	Severe	Reversed	Balanced sampling	1e-05	1.0000	Yes	3960
BRACS	Severe	Reversed	CE	0.001	—	No	3960
BRACS	Severe	Reversed	CE + soft F_1	1e-05	0.2500	Yes	3960
BRACS	Severe	Reversed	CE + soft MCC	3e-05	64.0000	No	3960
BRACS	Severe	Reversed	CFAL	0.001	1.0000	No	3960
BRACS	Severe	Reversed	Nominal support	3e-05	0.9999	No	3960
BRACS	Severe	Reversed	cRT	0.001	—	No	3960
BRACS	Severe	Reversed	Focal loss	0.001	2.0000	No	3960
BRACS	Severe	Reversed	Independent support	0.001	0.9990	No	3960
BRACS	Severe	Reversed	Logit adjustment (train)	0.0003	1.0000	No	3960
BRACS	Severe	Reversed	OKO	0.0001	1.0000	No	3960
BRACS	Severe	Reversed	Difficulty	0.001	0.2500	No	3960
BRACS	Severe	Reversed	Logit adjustment (post-hoc)	—	0.5000	No	3960
BRACS	Severe	Reversed	Diversity	0.001	0.2500	No	3960
BRACS	Severe	Reversed	Prevalence	3e-05	1.0000	No	3960
BRACS	Severe	Native	Balanced sampling	1e-05	1.0000	Yes	3960
BRACS	Severe	Native	CE	0.001	—	No	3960
BRACS	Severe	Native	CE + soft F_1	1e-05	0.2500	Yes	3960
BRACS	Severe	Native	CE + soft MCC	3e-05	64.0000	No	3960
BRACS	Severe	Native	CFAL	0.001	1.0000	No	3960
BRACS	Severe	Native	Nominal support	3e-05	0.9999	No	3960
BRACS	Severe	Native	cRT	0.001	—	No	3960
BRACS	Severe	Native	Focal loss	0.001	2.0000	No	3960
BRACS	Severe	Native	Independent support	0.001	0.9990	No	3960
BRACS	Severe	Native	Logit adjustment (train)	0.0003	1.0000	No	3960
BRACS	Severe	Native	OKO	0.0001	1.0000	No	3960
BRACS	Severe	Native	Difficulty	0.001	0.2500	No	3960
BRACS	Severe	Native	Logit adjustment (post-hoc)	—	0.5000	No	3960
BRACS	Severe	Native	Diversity	0.001	0.2500	No	3960
BRACS	Severe	Native	Prevalence	3e-05	1.0000	No	3960
CAMELYON16	Balanced	—	Balanced sampling	0.001	0.5000	No	8490
CAMELYON16	Balanced	—	CE	0.001	—	No	8490
CAMELYON16	Balanced	—	CE + soft F_1	0.01	64.0000	Yes	8490
CAMELYON16	Balanced	—	CE + soft MCC	0.003	64.0000	No	8490
CAMELYON16	Balanced	—	CFAL	0.001	0.2500	No	8490
CAMELYON16	Balanced	—	Nominal support	0.001	0.9000	No	8490
CAMELYON16	Balanced	—	cRT	0.01	—	Yes	8490
CAMELYON16	Balanced	—	Focal loss	0.001	0.5000	No	8490
CAMELYON16	Balanced	—	Independent support	0.01	0.9900	Yes	8490
CAMELYON16	Balanced	—	Logit adjustment (train)	0.001	0.2500	No	8490
CAMELYON16	Balanced	—	OKO	0.003	1.0000	No	8490
CAMELYON16	Balanced	—	Difficulty	0.01	0.7500	Yes	8490
CAMELYON16	Balanced	—	Logit adjustment (post-hoc)	—	0.2500	No	8490

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Legacy updates
CAMELYON16	Balanced	—	Diversity	0.001	1.0000	No	8490
CAMELYON16	Balanced	—	Prevalence	0.001	0.5000	No	8490
CAMELYON16	Moderate	Reversed	Balanced sampling	0.003	0.7500	No	8490
CAMELYON16	Moderate	Reversed	CE	0.01	—	Yes	8490
CAMELYON16	Moderate	Reversed	CE + soft F_1	0.001	64.0000	No	8490
CAMELYON16	Moderate	Reversed	CE + soft MCC	0.003	64.0000	No	8490
CAMELYON16	Moderate	Reversed	CFAL	0.001	0.2500	No	8490
CAMELYON16	Moderate	Reversed	Nominal support	0.001	0.9999	No	8490
CAMELYON16	Moderate	Reversed	cRT	0.01	—	Yes	8490
CAMELYON16	Moderate	Reversed	Focal loss	0.01	0.0000	Yes	8490
CAMELYON16	Moderate	Reversed	Independent support	0.01	0.8000	Yes	8490
CAMELYON16	Moderate	Reversed	Logit adjustment (train)	0.001	1.0000	No	8490
CAMELYON16	Moderate	Reversed	OKO	0.003	1.0000	No	8490
CAMELYON16	Moderate	Reversed	Difficulty	0.003	0.5000	No	8490
CAMELYON16	Moderate	Reversed	Logit adjustment (post-hoc)	—	0.5000	No	8490
CAMELYON16	Moderate	Reversed	Diversity	0.01	0.5000	Yes	8490
CAMELYON16	Moderate	Reversed	Prevalence	0.003	1.0000	No	8490
CAMELYON16	Moderate	Native	Balanced sampling	0.003	0.7500	No	8490
CAMELYON16	Moderate	Native	CE	0.01	—	Yes	8490
CAMELYON16	Moderate	Native	CE + soft F_1	0.001	64.0000	No	8490
CAMELYON16	Moderate	Native	CE + soft MCC	0.003	64.0000	No	8490
CAMELYON16	Moderate	Native	CFAL	0.001	0.2500	No	8490
CAMELYON16	Moderate	Native	Nominal support	0.001	0.9999	No	8490
CAMELYON16	Moderate	Native	cRT	0.01	—	Yes	8490
CAMELYON16	Moderate	Native	Focal loss	0.01	0.0000	Yes	8490
CAMELYON16	Moderate	Native	Independent support	0.01	0.8000	Yes	8490
CAMELYON16	Moderate	Native	Logit adjustment (train)	0.001	1.0000	No	8490
CAMELYON16	Moderate	Native	OKO	0.003	1.0000	No	8490
CAMELYON16	Moderate	Native	Difficulty	0.003	0.5000	No	8490
CAMELYON16	Moderate	Native	Logit adjustment (post-hoc)	—	0.5000	No	8490
CAMELYON16	Moderate	Native	Diversity	0.01	0.5000	Yes	8490
CAMELYON16	Moderate	Native	Prevalence	0.003	1.0000	No	8490
CAMELYON16	Natural	—	CE	0.01	—	Yes	523830
CAMELYON16	Severe	Reversed	Balanced sampling	1e-05	1.0000	Yes	8490
CAMELYON16	Severe	Reversed	CE	0.001	—	No	8490
CAMELYON16	Severe	Reversed	CE + soft F_1	3e-05	64.0000	No	8490
CAMELYON16	Severe	Reversed	CE + soft MCC	0.01	64.0000	Yes	8490
CAMELYON16	Severe	Reversed	CFAL	0.01	0.2500	Yes	8490
CAMELYON16	Severe	Reversed	Nominal support	0.003	0.9999	No	8490
CAMELYON16	Severe	Reversed	cRT	0.001	—	No	8490
CAMELYON16	Severe	Reversed	Focal loss	0.01	1.5000	Yes	8490
CAMELYON16	Severe	Reversed	Independent support	0.001	0.9990	No	8490
CAMELYON16	Severe	Reversed	Logit adjustment (train)	0.0001	1.0000	No	8490

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Legacy updates
CAMELYON16	Severe	Reversed	OKO	0.001	1.0000	No	8490
CAMELYON16	Severe	Reversed	Difficulty	0.003	0.2500	No	8490
CAMELYON16	Severe	Reversed	Logit adjustment (post-hoc)	—	1.0000	No	8490
CAMELYON16	Severe	Reversed	Diversity	0.01	1.0000	Yes	8490
CAMELYON16	Severe	Reversed	Prevalence	0.0003	1.0000	No	8490
CAMELYON16	Severe	Native	Balanced sampling	1e-05	1.0000	Yes	8490
CAMELYON16	Severe	Native	CE	0.001	—	No	8490
CAMELYON16	Severe	Native	CE + soft F_1	3e-05	64.0000	No	8490
CAMELYON16	Severe	Native	CE + soft MCC	0.01	64.0000	Yes	8490
CAMELYON16	Severe	Native	CFAL	0.01	0.2500	Yes	8490
CAMELYON16	Severe	Native	Nominal support	0.003	0.9999	No	8490
CAMELYON16	Severe	Native	cRT	0.001	—	No	8490
CAMELYON16	Severe	Native	Focal loss	0.01	1.5000	Yes	8490
CAMELYON16	Severe	Native	Independent support	0.001	0.9990	No	8490
CAMELYON16	Severe	Native	Logit adjustment (train)	0.0001	1.0000	No	8490
CAMELYON16	Severe	Native	OKO	0.001	1.0000	No	8490
CAMELYON16	Severe	Native	Difficulty	0.003	0.2500	No	8490
CAMELYON16	Severe	Native	Logit adjustment (post-hoc)	—	1.0000	No	8490
CAMELYON16	Severe	Native	Diversity	0.01	1.0000	Yes	8490
CAMELYON16	Severe	Native	Prevalence	0.0003	1.0000	No	8490
PANDA	Balanced	—	Balanced sampling	1e-05	0.0000	Yes	257790
PANDA	Balanced	—	CE	1e-05	—	Yes	257790
PANDA	Balanced	—	CE + soft F_1	1e-05	16.0000	Yes	257790
PANDA	Balanced	—	CE + soft MCC	0.003	4.0000	No	257790
PANDA	Balanced	—	CFAL	0.0001	0.2500	No	257790
PANDA	Balanced	—	Nominal support	1e-05	0.9000	Yes	257790
PANDA	Balanced	—	cRT	1e-05	—	Yes	257790
PANDA	Balanced	—	Focal loss	1e-05	0.0000	Yes	257790
PANDA	Balanced	—	Independent support	1e-05	0.9990	Yes	257790
PANDA	Balanced	—	Logit adjustment (train)	1e-05	1.0000	Yes	257790
PANDA	Balanced	—	OKO	0.001	1.0000	No	257790
PANDA	Balanced	—	Difficulty	1e-05	0.2500	Yes	257790
PANDA	Balanced	—	Logit adjustment (post-hoc)	—	0.2500	No	257790
PANDA	Balanced	—	Diversity	1e-05	0.7500	Yes	257790
PANDA	Balanced	—	Prevalence	1e-05	0.7500	Yes	257790
PANDA	Moderate	Native	Balanced sampling	0.003	1.0000	No	257790
PANDA	Moderate	Native	CE	3e-05	—	No	257790
PANDA	Moderate	Native	CE + soft F_1	1e-05	16.0000	Yes	257790
PANDA	Moderate	Native	CE + soft MCC	1e-05	16.0000	Yes	257790
PANDA	Moderate	Native	CFAL	0.001	2.0000	No	257790
PANDA	Moderate	Native	Nominal support	3e-05	0.9999	No	257790
PANDA	Moderate	Native	cRT	1e-05	—	Yes	257790
PANDA	Moderate	Native	Focal loss	0.01	1.5000	Yes	257790

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Legacy updates
PANDA	Moderate	Native	Independent support	3e-05	0.8000	No	257790
PANDA	Moderate	Native	Logit adjustment (train)	1e-05	1.0000	Yes	257790
PANDA	Moderate	Native	OKO	0.001	1.0000	No	257790
PANDA	Moderate	Native	Difficulty	3e-05	0.2500	No	257790
PANDA	Moderate	Native	Logit adjustment (post-hoc)	—	1.0000	No	257790
PANDA	Moderate	Native	Diversity	0.01	1.0000	Yes	257790
PANDA	Moderate	Native	Prevalence	1e-05	1.0000	Yes	257790
PANDA	Natural	—	CE	0.001	—	No	739110
PANDA	Severe	Native	Balanced sampling	1e-05	1.0000	Yes	257790
PANDA	Severe	Native	CE	3e-05	—	No	257790
PANDA	Severe	Native	CE + soft F_1	1e-05	4.0000	Yes	257790
PANDA	Severe	Native	CE + soft MCC	1e-05	4.0000	Yes	257790
PANDA	Severe	Native	CFAL	0.003	4.0000	No	257790
PANDA	Severe	Native	Nominal support	3e-05	0.9999	No	257790
PANDA	Severe	Native	cRT	1e-05	—	Yes	257790
PANDA	Severe	Native	Focal loss	3e-05	1.0000	No	257790
PANDA	Severe	Native	Independent support	3e-05	0.8000	No	257790
PANDA	Severe	Native	Logit adjustment (train)	3e-05	1.0000	No	257790
PANDA	Severe	Native	OKO	0.003	1.0000	No	257790
PANDA	Severe	Native	Difficulty	3e-05	0.2500	No	257790
PANDA	Severe	Native	Logit adjustment (post-hoc)	—	1.0000	No	257790
PANDA	Severe	Native	Diversity	3e-05	0.5000	No	257790
PANDA	Severe	Native	Prevalence	1e-05	1.0000	Yes	257790
TCGA-UT	Balanced	—	Balanced sampling	3e-05	1.0000	No	10500
TCGA-UT	Balanced	—	CE	1e-05	—	Yes	10500
TCGA-UT	Balanced	—	CE + soft F_1	3e-05	16.0000	No	10500
TCGA-UT	Balanced	—	CE + soft MCC	1e-05	16.0000	Yes	10500
TCGA-UT	Balanced	—	CFAL	0.0001	0.2500	No	10500
TCGA-UT	Balanced	—	Nominal support	1e-05	0.9990	Yes	10500
TCGA-UT	Balanced	—	cRT	0.0003	—	No	10500
TCGA-UT	Balanced	—	Focal loss	1e-05	0.0000	Yes	10500
TCGA-UT	Balanced	—	Independent support	1e-05	0.9500	Yes	10500
TCGA-UT	Balanced	—	Logit adjustment (train)	1e-05	0.2500	Yes	10500
TCGA-UT	Balanced	—	OKO	3e-05	4.0000	No	10500
TCGA-UT	Balanced	—	Difficulty	1e-05	0.2500	Yes	10500
TCGA-UT	Balanced	—	Logit adjustment (post-hoc)	—	2.0000	No	10500
TCGA-UT	Balanced	—	Diversity	1e-05	1.0000	Yes	10500
TCGA-UT	Balanced	—	Prevalence	1e-05	0.2500	Yes	10500
TCGA-UT	Moderate	Aligned	Balanced sampling	1e-05	1.0000	Yes	10500
TCGA-UT	Moderate	Aligned	CE	3e-05	—	No	10500
TCGA-UT	Moderate	Aligned	CE + soft F_1	1e-05	16.0000	Yes	10500
TCGA-UT	Moderate	Aligned	CE + soft MCC	1e-05	16.0000	Yes	10500
TCGA-UT	Moderate	Aligned	CFAL	3e-05	0.2500	No	10500

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Legacy updates
TCGA-UT	Moderate	Aligned	Nominal support	1e-05	0.9999	Yes	10500
TCGA-UT	Moderate	Aligned	cRT	0.0003	—	No	10500
TCGA-UT	Moderate	Aligned	Focal loss	3e-05	0.0000	No	10500
TCGA-UT	Moderate	Aligned	Independent support	3e-05	0.9900	No	10500
TCGA-UT	Moderate	Aligned	Logit adjustment (train)	3e-05	0.5000	No	10500
TCGA-UT	Moderate	Aligned	OKO	3e-05	4.0000	No	10500
TCGA-UT	Moderate	Aligned	Difficulty	3e-05	0.2500	No	10500
TCGA-UT	Moderate	Aligned	Logit adjustment (post-hoc)	—	0.5000	No	10500
TCGA-UT	Moderate	Aligned	Diversity	3e-05	1.0000	No	10500
TCGA-UT	Moderate	Aligned	Prevalence	1e-05	1.0000	Yes	10500
TCGA-UT	Moderate	Reversed	Balanced sampling	1e-05	1.0000	Yes	10500
TCGA-UT	Moderate	Reversed	CE	3e-05	—	No	10500
TCGA-UT	Moderate	Reversed	CE + soft F_1	1e-05	16.0000	Yes	10500
TCGA-UT	Moderate	Reversed	CE + soft MCC	1e-05	16.0000	Yes	10500
TCGA-UT	Moderate	Reversed	CFAL	3e-05	0.2500	No	10500
TCGA-UT	Moderate	Reversed	Nominal support	1e-05	0.9999	Yes	10500
TCGA-UT	Moderate	Reversed	cRT	0.0003	—	No	10500
TCGA-UT	Moderate	Reversed	Focal loss	3e-05	0.0000	No	10500
TCGA-UT	Moderate	Reversed	Independent support	3e-05	0.9900	No	10500
TCGA-UT	Moderate	Reversed	Logit adjustment (train)	3e-05	0.5000	No	10500
TCGA-UT	Moderate	Reversed	OKO	3e-05	4.0000	No	10500
TCGA-UT	Moderate	Reversed	Difficulty	3e-05	0.2500	No	10500
TCGA-UT	Moderate	Reversed	Logit adjustment (post-hoc)	—	0.5000	No	10500
TCGA-UT	Moderate	Reversed	Diversity	3e-05	1.0000	No	10500
TCGA-UT	Moderate	Reversed	Prevalence	1e-05	1.0000	Yes	10500
TCGA-UT	Moderate	Native	Balanced sampling	1e-05	1.0000	Yes	10500
TCGA-UT	Moderate	Native	CE	3e-05	—	No	10500
TCGA-UT	Moderate	Native	CE + soft F_1	1e-05	16.0000	Yes	10500
TCGA-UT	Moderate	Native	CE + soft MCC	1e-05	16.0000	Yes	10500
TCGA-UT	Moderate	Native	CFAL	3e-05	0.2500	No	10500
TCGA-UT	Moderate	Native	Nominal support	1e-05	0.9999	Yes	10500
TCGA-UT	Moderate	Native	cRT	0.0003	—	No	10500
TCGA-UT	Moderate	Native	Focal loss	3e-05	0.0000	No	10500
TCGA-UT	Moderate	Native	Independent support	3e-05	0.9900	No	10500
TCGA-UT	Moderate	Native	Logit adjustment (train)	3e-05	0.5000	No	10500
TCGA-UT	Moderate	Native	OKO	3e-05	4.0000	No	10500
TCGA-UT	Moderate	Native	Difficulty	3e-05	0.2500	No	10500
TCGA-UT	Moderate	Native	Logit adjustment (post-hoc)	—	0.5000	No	10500
TCGA-UT	Moderate	Native	Diversity	3e-05	1.0000	No	10500
TCGA-UT	Moderate	Native	Prevalence	1e-05	1.0000	Yes	10500
TCGA-UT	Natural	—	CE	3e-05	—	No	261570
TCGA-UT	Severe	Aligned	Balanced sampling	1e-05	1.0000	Yes	10500
TCGA-UT	Severe	Aligned	CE	0.0001	—	No	10500

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Legacy updates
TCGA-UT	Severe	Aligned	CE + soft F_1	1e-05	4.0000	Yes	10500
TCGA-UT	Severe	Aligned	CE + soft MCC	1e-05	4.0000	Yes	10500
TCGA-UT	Severe	Aligned	CFAL	0.0003	4.0000	No	10500
TCGA-UT	Severe	Aligned	Nominal support	1e-05	0.9999	Yes	10500
TCGA-UT	Severe	Aligned	cRT	0.001	—	No	10500
TCGA-UT	Severe	Aligned	Focal loss	0.0001	0.0000	No	10500
TCGA-UT	Severe	Aligned	Independent support	0.0001	0.9500	No	10500
TCGA-UT	Severe	Aligned	Logit adjustment (train)	3e-05	0.5000	No	10500
TCGA-UT	Severe	Aligned	OKO	3e-05	4.0000	No	10500
TCGA-UT	Severe	Aligned	Difficulty	0.0001	0.2500	No	10500
TCGA-UT	Severe	Aligned	Logit adjustment (post-hoc)	—	0.5000	No	10500
TCGA-UT	Severe	Aligned	Diversity	0.0001	0.5000	No	10500
TCGA-UT	Severe	Aligned	Prevalence	1e-05	1.0000	Yes	10500
TCGA-UT	Severe	Reversed	Balanced sampling	1e-05	1.0000	Yes	10500
TCGA-UT	Severe	Reversed	CE	0.0001	—	No	10500
TCGA-UT	Severe	Reversed	CE + soft F_1	1e-05	4.0000	Yes	10500
TCGA-UT	Severe	Reversed	CE + soft MCC	1e-05	4.0000	Yes	10500
TCGA-UT	Severe	Reversed	CFAL	0.0003	4.0000	No	10500
TCGA-UT	Severe	Reversed	Nominal support	1e-05	0.9999	Yes	10500
TCGA-UT	Severe	Reversed	cRT	0.001	—	No	10500
TCGA-UT	Severe	Reversed	Focal loss	0.0001	0.0000	No	10500
TCGA-UT	Severe	Reversed	Independent support	0.0001	0.9500	No	10500
TCGA-UT	Severe	Reversed	Logit adjustment (train)	3e-05	0.5000	No	10500
TCGA-UT	Severe	Reversed	OKO	3e-05	4.0000	No	10500
TCGA-UT	Severe	Reversed	Difficulty	0.0001	0.2500	No	10500
TCGA-UT	Severe	Reversed	Logit adjustment (post-hoc)	—	0.5000	No	10500
TCGA-UT	Severe	Reversed	Diversity	0.0001	0.5000	No	10500
TCGA-UT	Severe	Reversed	Prevalence	1e-05	1.0000	Yes	10500
TCGA-UT	Severe	Native	Balanced sampling	1e-05	1.0000	Yes	10500
TCGA-UT	Severe	Native	CE	0.0001	—	No	10500
TCGA-UT	Severe	Native	CE + soft F_1	1e-05	4.0000	Yes	10500
TCGA-UT	Severe	Native	CE + soft MCC	1e-05	4.0000	Yes	10500
TCGA-UT	Severe	Native	CFAL	0.0003	4.0000	No	10500
TCGA-UT	Severe	Native	Nominal support	1e-05	0.9999	Yes	10500
TCGA-UT	Severe	Native	cRT	0.001	—	No	10500
TCGA-UT	Severe	Native	Focal loss	0.0001	0.0000	No	10500
TCGA-UT	Severe	Native	Independent support	0.0001	0.9500	No	10500
TCGA-UT	Severe	Native	Logit adjustment (train)	3e-05	0.5000	No	10500
TCGA-UT	Severe	Native	OKO	3e-05	4.0000	No	10500
TCGA-UT	Severe	Native	Difficulty	0.0001	0.2500	No	10500
TCGA-UT	Severe	Native	Logit adjustment (post-hoc)	—	0.5000	No	10500
TCGA-UT	Severe	Native	Diversity	0.0001	0.5000	No	10500
TCGA-UT	Severe	Native	Prevalence	1e-05	1.0000	Yes	10500

Table 25: Legacy selected controls, update budgets, and learning-rate boundary hits. The rerun uses the expanded envelope.

Four deviations from the locked protocol were made, each before any definitive gate row was read, and each is recorded here with its reason. Every selection in Table 25 that terminated at the 1×10^{-5} or 1×10^{-2} envelope boundary is flagged there, since a boundary selection means the envelope rather than the validation data determined the choice.

The two deficit-gate thresholds were derived rather than assumed. The protocol had fixed $D_{\text{BA}} \geq 0.02$ and $D_{\text{cal}} \geq 0.05$ nats without a stated derivation, and the discrimination value conflicted with the support pilot’s own materiality constant of 0.01 for the same endpoint under the same kind of manipulation. Both were replaced by quantities the protocol already fixes: the discrimination threshold became $\max(0.01, 2\sigma_{\text{seed}}) = 0.014,43$ and the calibration threshold $2\sigma_{\text{seed}} = 0.182,37$ nats, where σ_{seed} is the largest across datasets and partitions of the confirmation-seed standard deviation of the corresponding balanced-condition endpoint. The change was applied identically to all four datasets and frozen before any gate row was read.

The gate-defining estimator was changed from patch-micro to case-macro, and the evidence cell from a per-partition cell to a cell pooled across a case’s partition appearances. Both are corrections of a mismatch between a diagnostic and the estimator it certifies: the preflight weighted a case by its patch count, so one large annotated region or one macro-metastasis slide could carry more than half of a class’s weight, and the per-partition cell certified an estimand that no reported quantity uses, since the bootstrap draws one shared weight per case over the union of the three test frames.

Cholangiocarcinoma was excluded from the TCGA-UT target set, reducing it from 31 eligible classes to 30, because its held-out participant count could not support split-unit-clustered inference. The exclusion uses the existing construction-time mechanism and removes the class from training as well as evaluation, which is stronger than the evidence problem requires and was accepted in exchange for a uniform 30-type task definition.

The legacy validation cadence was tied to update count. The rerun uses about 170 checkpoints per method-specific derived update count.

Three execution defects are recorded here rather than as protocol changes, because none of them altered an estimand, a threshold, or a selection rule.

The cross-dataset matching record, which standardizes the four shortage scores across all 18 units and assigns each its dominant shortage, is written per dataset root. It was produced for BRACS, CAMELYON16, and PANDA before their confirmatory aggregation ran, but for TCGA-UT it was produced afterwards, so the first aggregation of that dataset emitted no matched-versus-unmatched rows. The record was regenerated for TCGA-UT from the same four roots and the aggregation repeated; the dominant-shortage and ambiguity labels it assigns are identical to those the other three datasets used, and their standardized scores agree to within 0.005. The other three datasets retain their original records, since re-deriving them would change no label. The repeated aggregation reproduced every TCGA-UT cross-entropy deficit exactly and added the eight matched-contrast rows the frozen matching record implies.

The recovery association model of Section 4 initially returned no estimate. The matched-versus-unmatched contrast rows are contrasts between two arms rather than comparisons against the reference, so they carry no deficit denominator and no recovery fraction; including them among the model’s observations propagated an undefined value through the whole fit. They are now excluded from the recovery cells, which is what the estimand in Equation (1) always implied. Thirteen of the 265 gate-passing method cells are excluded on this ground, and the damage model is unaffected.

The final aggregation pass for TCGA-UT exhausted its wall-clock allocation while writing the secondary interval tables, after the crossed comparison file on which every gate and recovery estimate in this report depends had already been written. Those interval tables are computed from the fitted runs and the partition databases alone, neither of which the repeated aggregation changed, so the tables carried over from the preceding complete pass are the same tables the interrupted pass would have written. Nothing reported here is drawn from a partially written

artifact.

B Per-Class and Tier Endpoints

Section 2.4 reports damage at tier level; this appendix gives the per-class detail behind it. For each dataset, recall, F_1 , negative log likelihood, and Brier score are listed per class and condition for the cross-entropy reference, averaged within a partition and then with equal weight across the three partitions. The tier assignment of each class is given alongside, since tier membership follows the allocation of the condition and is therefore not constant across conditions. The table is restricted to cross-entropy because that is the reference against which every deficit in Section 2 is defined; per-class endpoints for the remaining methods are available in the analysis outputs but would multiply the table by the size of the roster without informing a claim made here.

Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
BRACS	Aligned	Moderate	ADH	body	0.206	0.109	3.125	1.179
BRACS	Aligned	Moderate	ADH	tail	0.075	0.093	3.459	1.290
BRACS	Aligned	Moderate	DCIS	body	0.549	0.507	1.087	0.457
BRACS	Aligned	Moderate	DCIS	head	0.533	0.381	1.704	0.614
BRACS	Aligned	Moderate	DCIS	tail	0.350	0.451	1.085	0.521
BRACS	Aligned	Moderate	FEA	body	0.468	0.239	4.311	1.335
BRACS	Aligned	Moderate	FEA	head	0.516	0.272	2.456	0.910
BRACS	Aligned	Moderate	IC	head	0.877	0.888	0.728	0.242
BRACS	Aligned	Moderate	N	head	0.793	0.428	1.226	0.613
BRACS	Aligned	Moderate	PB	tail	0.207	0.318	1.887	0.786
BRACS	Aligned	Moderate	UDH	tail	0.242	0.263	2.293	0.939
BRACS	Aligned	Severe	ADH	body	0.242	0.130	4.371	1.281
BRACS	Aligned	Severe	ADH	tail	0.018	0.031	5.300	1.476
BRACS	Aligned	Severe	DCIS	body	0.519	0.474	1.827	0.600
BRACS	Aligned	Severe	DCIS	head	0.555	0.389	1.622	0.585
BRACS	Aligned	Severe	DCIS	tail	0.256	0.384	1.365	0.591
BRACS	Aligned	Severe	FEA	body	0.418	0.177	4.908	1.359
BRACS	Aligned	Severe	FEA	head	0.523	0.252	3.350	0.980
BRACS	Aligned	Severe	IC	head	0.880	0.882	1.099	0.280
BRACS	Aligned	Severe	N	head	0.846	0.405	1.299	0.567
BRACS	Aligned	Severe	PB	tail	0.127	0.212	2.540	0.954
BRACS	Aligned	Severe	UDH	tail	0.135	0.168	2.978	1.055
BRACS	Reversed	Moderate	ADH	body	0.058	0.063	2.861	1.098
BRACS	Reversed	Moderate	ADH	head	0.440	0.181	3.838	1.300
BRACS	Reversed	Moderate	DCIS	body	0.360	0.415	1.534	0.532
BRACS	Reversed	Moderate	DCIS	head	0.821	0.438	1.161	0.553
BRACS	Reversed	Moderate	DCIS	tail	0.402	0.365	2.237	0.705
BRACS	Reversed	Moderate	FEA	body	0.167	0.141	8.027	1.510
BRACS	Reversed	Moderate	FEA	tail	0.253	0.251	2.753	0.948
BRACS	Reversed	Moderate	IC	tail	0.712	0.804	0.872	0.266
BRACS	Reversed	Moderate	N	tail	0.352	0.353	1.505	0.657
BRACS	Reversed	Moderate	PB	head	0.349	0.452	2.607	0.940
BRACS	Reversed	Moderate	UDH	head	0.449	0.284	2.730	1.013
BRACS	Reversed	Severe	ADH	body	0.097	0.102	7.323	1.352
BRACS	Reversed	Severe	ADH	head	0.589	0.172	3.720	1.233
BRACS	Reversed	Severe	DCIS	body	0.183	0.251	2.986	0.847
BRACS	Reversed	Severe	DCIS	head	0.829	0.319	1.634	0.490
BRACS	Reversed	Severe	DCIS	tail	0.280	0.305	3.116	0.891
BRACS	Reversed	Severe	FEA	body	0.110	0.102	7.957	1.600
BRACS	Reversed	Severe	FEA	tail	0.177	0.204	4.829	1.138
BRACS	Reversed	Severe	IC	tail	0.590	0.727	1.240	0.355
BRACS	Reversed	Severe	N	tail	0.228	0.285	2.715	0.867
BRACS	Reversed	Severe	PB	head	0.375	0.477	3.609	1.012

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Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
BRACS	Reversed	Severe	UDH	head	0.448	0.280	3.531	1.066
BRACS	Native	Moderate	ADH	tail	0.126	0.119	3.087	1.241
BRACS	Native	Moderate	DCIS	tail	0.432	0.463	1.151	0.550
BRACS	Native	Moderate	FEA	body	0.380	0.309	3.135	1.083
BRACS	Native	Moderate	IC	tail	0.837	0.878	0.490	0.179
BRACS	Native	Moderate	N	head	0.738	0.419	1.121	0.576
BRACS	Native	Moderate	PB	head	0.289	0.404	1.891	0.838
BRACS	Native	Moderate	UDH	head	0.448	0.320	2.004	0.881
BRACS	Native	Severe	ADH	tail	0.104	0.111	3.762	1.294
BRACS	Native	Severe	DCIS	tail	0.353	0.411	1.735	0.704
BRACS	Native	Severe	FEA	body	0.407	0.297	3.362	1.098
BRACS	Native	Severe	IC	tail	0.762	0.833	0.649	0.221
BRACS	Native	Severe	N	head	0.747	0.408	1.421	0.616
BRACS	Native	Severe	PB	head	0.292	0.391	2.948	1.002
BRACS	Native	Severe	UDH	head	0.413	0.296	2.607	0.955
CAMELYON16	Reversed	Moderate	normal	tail	0.981	0.990	0.054	0.010
CAMELYON16	Reversed	Moderate	tumor	head	0.988	0.686	0.241	0.065
CAMELYON16	Reversed	Severe	normal	tail	0.946	0.972	0.084	0.018
CAMELYON16	Reversed	Severe	tumor	head	0.991	0.501	0.224	0.059
CAMELYON16	Native	Moderate	normal	head	0.993	0.996	0.035	0.008
CAMELYON16	Native	Moderate	tumor	tail	0.973	0.841	0.335	0.067
CAMELYON16	Native	Severe	normal	head	0.998	0.997	0.014	0.005
CAMELYON16	Native	Severe	tumor	tail	0.932	0.875	0.526	0.104
PANDA	Native	Moderate	benign	head	0.976	0.957	0.133	0.068
PANDA	Native	Moderate	cancer	tail	0.721	0.790	0.580	0.348
PANDA	Native	Severe	benign	head	0.994	0.941	0.115	0.059
PANDA	Native	Severe	cancer	tail	0.474	0.633	0.865	0.456
TCGA-UT	Aligned	Moderate	Adrenocortical carcinoma	head	0.763	0.768	1.711	0.445
TCGA-UT	Aligned	Moderate	Bladder Urothelial Carcinoma	tail	0.452	0.502	1.711	0.696
TCGA-UT	Aligned	Moderate	Brain Lower Grade Glioma	body	0.887	0.894	0.439	0.195
TCGA-UT	Aligned	Moderate	Brain Lower Grade Glioma	head	0.835	0.831	0.578	0.292
TCGA-UT	Aligned	Moderate	Breast invasive carcinoma	body	0.759	0.820	0.662	0.260
TCGA-UT	Aligned	Moderate	Cervical squamous cell carcinoma and endocervical adenocarcinoma	tail	0.500	0.490	1.465	0.676
TCGA-UT	Aligned	Moderate	Colon adenocarcinoma	tail	0.527	0.630	0.766	0.380
TCGA-UT	Aligned	Moderate	Esophageal carcinoma	tail	0.309	0.267	2.372	1.042
TCGA-UT	Aligned	Moderate	Glioblastoma multiforme	body	0.819	0.829	0.409	0.219
TCGA-UT	Aligned	Moderate	Head and Neck squamous cell carcinoma	body	0.736	0.668	0.919	0.380
TCGA-UT	Aligned	Moderate	Head and Neck squamous cell carcinoma	tail	0.745	0.703	0.884	0.387
TCGA-UT	Aligned	Moderate	Kidney Chromophobe	head	0.929	0.825	0.391	0.192
TCGA-UT	Aligned	Moderate	Kidney renal clear cell carcinoma	body	0.783	0.858	1.164	0.331
TCGA-UT	Aligned	Moderate	Kidney renal clear cell carcinoma	head	0.867	0.906	0.733	0.211
TCGA-UT	Aligned	Moderate	Kidney renal papillary cell carcinoma	body	0.898	0.833	0.385	0.160
TCGA-UT	Aligned	Moderate	Kidney renal papillary cell carcinoma	head	0.818	0.796	0.781	0.291

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Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
TCGA-UT	Aligned	Moderate	Liver hepatocellular carcinoma	body	0.864	0.861	0.552	0.189
TCGA-UT	Aligned	Moderate	Liver hepatocellular carcinoma	head	0.921	0.806	0.354	0.137
TCGA-UT	Aligned	Moderate	Lung adenocarcinoma	body	0.729	0.771	0.806	0.346
TCGA-UT	Aligned	Moderate	Lung squamous cell carcinoma	tail	0.543	0.596	1.008	0.483
TCGA-UT	Aligned	Moderate	Mesothelioma	body	0.864	0.508	0.864	0.363
TCGA-UT	Aligned	Moderate	Mesothelioma	tail	0.645	0.568	1.870	0.618
TCGA-UT	Aligned	Moderate	Ovarian serous cystadenocarcinoma	body	0.842	0.527	1.184	0.474
TCGA-UT	Aligned	Moderate	Pancreatic adenocarcinoma	body	0.752	0.618	1.603	0.525
TCGA-UT	Aligned	Moderate	Pheochromocytoma and Paraganglioma	head	0.958	0.834	0.389	0.126
TCGA-UT	Aligned	Moderate	Prostate adenocarcinoma	head	0.977	0.969	0.171	0.054
TCGA-UT	Aligned	Moderate	Rectum adenocarcinoma	tail	0.751	0.368	1.338	0.722
TCGA-UT	Aligned	Moderate	Sarcoma	body	0.850	0.860	0.390	0.153
TCGA-UT	Aligned	Moderate	Skin Cutaneous Melanoma	body	0.626	0.680	0.939	0.398
TCGA-UT	Aligned	Moderate	Skin Cutaneous Melanoma	tail	0.693	0.740	0.661	0.302
TCGA-UT	Aligned	Moderate	Stomach adenocarcinoma	tail	0.478	0.536	1.356	0.586
TCGA-UT	Aligned	Moderate	Testicular Germ Cell Tumors	head	0.917	0.897	0.454	0.148
TCGA-UT	Aligned	Moderate	Thymoma	body	0.977	0.864	0.151	0.070
TCGA-UT	Aligned	Moderate	Thymoma	head	0.940	0.837	0.422	0.158
TCGA-UT	Aligned	Moderate	Thyroid carcinoma	head	0.960	0.968	0.227	0.067
TCGA-UT	Aligned	Moderate	Uterine Carcinosarcoma	body	0.623	0.280	2.192	0.757
TCGA-UT	Aligned	Moderate	Uterine Carcinosarcoma	tail	0.711	0.443	1.603	0.604
TCGA-UT	Aligned	Moderate	Uterine Corpus Endometrial Carcinoma	tail	0.611	0.661	0.996	0.387
TCGA-UT	Aligned	Moderate	Uveal Melanoma	head	0.959	0.678	0.572	0.286
TCGA-UT	Aligned	Severe	Adrenocortical carcinoma	head	0.774	0.769	2.200	0.460
TCGA-UT	Aligned	Severe	Bladder Urothelial Carcinoma	tail	0.368	0.446	1.981	0.765
TCGA-UT	Aligned	Severe	Brain Lower Grade Glioma	body	0.873	0.885	0.681	0.230
TCGA-UT	Aligned	Severe	Brain Lower Grade Glioma	head	0.842	0.832	0.737	0.312
TCGA-UT	Aligned	Severe	Breast invasive carcinoma	body	0.749	0.810	0.828	0.302
TCGA-UT	Aligned	Severe	Cervical squamous cell carcinoma and endocervical adenocarcinoma	tail	0.447	0.449	1.519	0.689
TCGA-UT	Aligned	Severe	Colon adenocarcinoma	tail	0.522	0.622	0.985	0.489
TCGA-UT	Aligned	Severe	Esophageal carcinoma	tail	0.248	0.225	2.604	1.066
TCGA-UT	Aligned	Severe	Glioblastoma multiforme	body	0.816	0.825	0.520	0.239
TCGA-UT	Aligned	Severe	Head and Neck squamous cell carcinoma	body	0.757	0.662	1.115	0.431
TCGA-UT	Aligned	Severe	Head and Neck squamous cell carcinoma	tail	0.739	0.687	1.099	0.451
TCGA-UT	Aligned	Severe	Kidney Chromophobe	head	0.933	0.818	0.426	0.185
TCGA-UT	Aligned	Severe	Kidney renal clear cell carcinoma	body	0.781	0.855	1.459	0.351
TCGA-UT	Aligned	Severe	Kidney renal clear cell carcinoma	head	0.868	0.905	0.965	0.228
TCGA-UT	Aligned	Severe	Kidney renal papillary cell carcinoma	body	0.885	0.807	0.437	0.174
TCGA-UT	Aligned	Severe	Kidney renal papillary cell carcinoma	head	0.806	0.785	1.059	0.331
TCGA-UT	Aligned	Severe	Liver hepatocellular carcinoma	body	0.864	0.847	0.721	0.215
TCGA-UT	Aligned	Severe	Liver hepatocellular carcinoma	head	0.917	0.793	0.462	0.159
TCGA-UT	Aligned	Severe	Lung adenocarcinoma	body	0.709	0.757	1.058	0.409
TCGA-UT	Aligned	Severe	Lung squamous cell carcinoma	tail	0.536	0.582	1.195	0.522

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Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
TCGA-UT	Aligned	Severe	Mesothelioma	body	0.862	0.490	0.799	0.338
TCGA-UT	Aligned	Severe	Mesothelioma	tail	0.649	0.540	1.826	0.598
TCGA-UT	Aligned	Severe	Ovarian serous cystadenocarcinoma	body	0.860	0.507	1.206	0.460
TCGA-UT	Aligned	Severe	Pancreatic adenocarcinoma	body	0.769	0.596	1.809	0.527
TCGA-UT	Aligned	Severe	Pheochromocytoma and Paraganglioma	head	0.955	0.813	0.472	0.133
TCGA-UT	Aligned	Severe	Prostate adenocarcinoma	head	0.981	0.963	0.197	0.056
TCGA-UT	Aligned	Severe	Rectum adenocarcinoma	tail	0.682	0.354	1.334	0.680
TCGA-UT	Aligned	Severe	Sarcoma	body	0.836	0.851	0.475	0.186
TCGA-UT	Aligned	Severe	Skin Cutaneous Melanoma	body	0.616	0.669	1.076	0.443
TCGA-UT	Aligned	Severe	Skin Cutaneous Melanoma	tail	0.644	0.710	0.823	0.359
TCGA-UT	Aligned	Severe	Stomach adenocarcinoma	tail	0.455	0.519	1.514	0.631
TCGA-UT	Aligned	Severe	Testicular Germ Cell Tumors	head	0.929	0.888	0.554	0.156
TCGA-UT	Aligned	Severe	Thymoma	body	0.981	0.844	0.182	0.079
TCGA-UT	Aligned	Severe	Thymoma	head	0.956	0.803	0.419	0.143
TCGA-UT	Aligned	Severe	Thyroid carcinoma	head	0.958	0.960	0.316	0.082
TCGA-UT	Aligned	Severe	Uterine Carcinosarcoma	body	0.641	0.277	2.396	0.748
TCGA-UT	Aligned	Severe	Uterine Carcinosarcoma	tail	0.711	0.483	1.662	0.565
TCGA-UT	Aligned	Severe	Uterine Corpus Endometrial Carcinoma	tail	0.585	0.649	1.141	0.427
TCGA-UT	Aligned	Severe	Uveal Melanoma	head	0.971	0.662	0.503	0.235
TCGA-UT	Reversed	Moderate	Adrenocortical carcinoma	tail	0.723	0.755	1.259	0.401
TCGA-UT	Reversed	Moderate	Bladder Urothelial Carcinoma	head	0.482	0.498	2.516	0.808
TCGA-UT	Reversed	Moderate	Brain Lower Grade Glioma	body	0.871	0.893	0.439	0.206
TCGA-UT	Reversed	Moderate	Brain Lower Grade Glioma	tail	0.814	0.824	0.442	0.242
TCGA-UT	Reversed	Moderate	Breast invasive carcinoma	body	0.743	0.816	0.664	0.258
TCGA-UT	Reversed	Moderate	Cervical squamous cell carcinoma and endocervical adenocarcinoma	head	0.573	0.498	1.859	0.748
TCGA-UT	Reversed	Moderate	Colon adenocarcinoma	head	0.536	0.605	0.837	0.396
TCGA-UT	Reversed	Moderate	Esophageal carcinoma	head	0.487	0.273	2.942	1.046
TCGA-UT	Reversed	Moderate	Glioblastoma multiforme	body	0.824	0.829	0.430	0.240
TCGA-UT	Reversed	Moderate	Head and Neck squamous cell carcinoma	body	0.628	0.652	0.832	0.343
TCGA-UT	Reversed	Moderate	Head and Neck squamous cell carcinoma	head	0.667	0.681	0.903	0.392
TCGA-UT	Reversed	Moderate	Kidney Chromophobe	tail	0.914	0.838	0.364	0.198
TCGA-UT	Reversed	Moderate	Kidney renal clear cell carcinoma	body	0.787	0.865	1.016	0.324
TCGA-UT	Reversed	Moderate	Kidney renal clear cell carcinoma	tail	0.866	0.906	0.539	0.186
TCGA-UT	Reversed	Moderate	Kidney renal papillary cell carcinoma	body	0.887	0.843	0.391	0.178
TCGA-UT	Reversed	Moderate	Kidney renal papillary cell carcinoma	tail	0.801	0.804	0.703	0.298
TCGA-UT	Reversed	Moderate	Liver hepatocellular carcinoma	body	0.854	0.879	0.484	0.188
TCGA-UT	Reversed	Moderate	Liver hepatocellular carcinoma	tail	0.906	0.829	0.254	0.113
TCGA-UT	Reversed	Moderate	Lung adenocarcinoma	body	0.719	0.769	0.778	0.332
TCGA-UT	Reversed	Moderate	Lung squamous cell carcinoma	head	0.580	0.604	1.137	0.513
TCGA-UT	Reversed	Moderate	Mesothelioma	body	0.849	0.514	0.925	0.391
TCGA-UT	Reversed	Moderate	Mesothelioma	head	0.653	0.535	2.177	0.623
TCGA-UT	Reversed	Moderate	Ovarian serous cystadenocarcinoma	body	0.796	0.575	0.902	0.367
TCGA-UT	Reversed	Moderate	Pancreatic adenocarcinoma	body	0.707	0.665	1.337	0.432

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Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
TCGA-UT	Reversed	Moderate	Pheochromocytoma and Paraganglioma	tail	0.942	0.885	0.345	0.129
TCGA-UT	Reversed	Moderate	Prostate adenocarcinoma	tail	0.963	0.973	0.099	0.036
TCGA-UT	Reversed	Moderate	Rectum adenocarcinoma	head	0.730	0.343	1.564	0.804
TCGA-UT	Reversed	Moderate	Sarcoma	body	0.845	0.861	0.372	0.150
TCGA-UT	Reversed	Moderate	Skin Cutaneous Melanoma	body	0.658	0.713	0.996	0.433
TCGA-UT	Reversed	Moderate	Skin Cutaneous Melanoma	head	0.745	0.766	0.860	0.372
TCGA-UT	Reversed	Moderate	Stomach adenocarcinoma	head	0.489	0.536	1.856	0.691
TCGA-UT	Reversed	Moderate	Testicular Germ Cell Tumors	tail	0.863	0.899	0.335	0.130
TCGA-UT	Reversed	Moderate	Thymoma	body	0.969	0.902	0.091	0.040
TCGA-UT	Reversed	Moderate	Thymoma	tail	0.915	0.881	0.263	0.115
TCGA-UT	Reversed	Moderate	Thyroid carcinoma	tail	0.952	0.968	0.146	0.050
TCGA-UT	Reversed	Moderate	Uterine Carcinosarcoma	body	0.599	0.302	2.131	0.723
TCGA-UT	Reversed	Moderate	Uterine Carcinosarcoma	head	0.719	0.432	1.581	0.605
TCGA-UT	Reversed	Moderate	Uterine Corpus Endometrial Carcinoma	head	0.621	0.667	1.189	0.445
TCGA-UT	Reversed	Moderate	Uveal Melanoma	tail	0.904	0.739	0.384	0.197
TCGA-UT	Reversed	Severe	Adrenocortical carcinoma	tail	0.703	0.744	1.394	0.410
TCGA-UT	Reversed	Severe	Bladder Urothelial Carcinoma	head	0.477	0.483	3.897	0.912
TCGA-UT	Reversed	Severe	Brain Lower Grade Glioma	body	0.860	0.877	0.635	0.224
TCGA-UT	Reversed	Severe	Brain Lower Grade Glioma	tail	0.821	0.824	0.516	0.238
TCGA-UT	Reversed	Severe	Breast invasive carcinoma	body	0.739	0.809	0.983	0.314
TCGA-UT	Reversed	Severe	Cervical squamous cell carcinoma and endocervical adenocarcinoma	head	0.560	0.485	2.806	0.845
TCGA-UT	Reversed	Severe	Colon adenocarcinoma	head	0.538	0.596	1.631	0.570
TCGA-UT	Reversed	Severe	Esophageal carcinoma	head	0.456	0.276	4.538	1.128
TCGA-UT	Reversed	Severe	Glioblastoma multiforme	body	0.804	0.817	0.680	0.308
TCGA-UT	Reversed	Severe	Head and Neck squamous cell carcinoma	body	0.659	0.658	1.001	0.345
TCGA-UT	Reversed	Severe	Head and Neck squamous cell carcinoma	head	0.662	0.674	1.318	0.445
TCGA-UT	Reversed	Severe	Kidney Chromophobe	tail	0.895	0.831	0.432	0.198
TCGA-UT	Reversed	Severe	Kidney renal clear cell carcinoma	body	0.787	0.864	1.249	0.337
TCGA-UT	Reversed	Severe	Kidney renal clear cell carcinoma	tail	0.856	0.900	0.728	0.209
TCGA-UT	Reversed	Severe	Kidney renal papillary cell carcinoma	body	0.857	0.821	0.532	0.227
TCGA-UT	Reversed	Severe	Kidney renal papillary cell carcinoma	tail	0.787	0.790	0.890	0.319
TCGA-UT	Reversed	Severe	Liver hepatocellular carcinoma	body	0.841	0.878	0.610	0.215
TCGA-UT	Reversed	Severe	Liver hepatocellular carcinoma	tail	0.894	0.830	0.310	0.125
TCGA-UT	Reversed	Severe	Lung adenocarcinoma	body	0.712	0.755	1.129	0.387
TCGA-UT	Reversed	Severe	Lung squamous cell carcinoma	head	0.587	0.601	1.728	0.609
TCGA-UT	Reversed	Severe	Mesothelioma	body	0.844	0.488	0.928	0.368
TCGA-UT	Reversed	Severe	Mesothelioma	head	0.655	0.515	2.727	0.648
TCGA-UT	Reversed	Severe	Ovarian serous cystadenocarcinoma	body	0.800	0.577	0.767	0.298
TCGA-UT	Reversed	Severe	Pancreatic adenocarcinoma	body	0.691	0.651	1.598	0.431
TCGA-UT	Reversed	Severe	Pheochromocytoma and Paraganglioma	tail	0.928	0.883	0.307	0.118
TCGA-UT	Reversed	Severe	Prostate adenocarcinoma	tail	0.943	0.965	0.105	0.040
TCGA-UT	Reversed	Severe	Rectum adenocarcinoma	head	0.699	0.316	1.932	0.775
TCGA-UT	Reversed	Severe	Sarcoma	body	0.834	0.852	0.454	0.172

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Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
TCGA-UT	Reversed	Severe	Skin Cutaneous Melanoma	body	0.657	0.712	1.256	0.507
TCGA-UT	Reversed	Severe	Skin Cutaneous Melanoma	head	0.736	0.754	1.324	0.466
TCGA-UT	Reversed	Severe	Stomach adenocarcinoma	head	0.481	0.536	2.985	0.825
TCGA-UT	Reversed	Severe	Testicular Germ Cell Tumors	tail	0.843	0.889	0.416	0.138
TCGA-UT	Reversed	Severe	Thymoma	body	0.944	0.911	0.104	0.051
TCGA-UT	Reversed	Severe	Thymoma	tail	0.909	0.868	0.280	0.109
TCGA-UT	Reversed	Severe	Thyroid carcinoma	tail	0.938	0.962	0.175	0.055
TCGA-UT	Reversed	Severe	Uterine Carcinosarcoma	body	0.624	0.299	2.472	0.688
TCGA-UT	Reversed	Severe	Uterine Carcinosarcoma	head	0.707	0.434	1.743	0.600
TCGA-UT	Reversed	Severe	Uterine Corpus Endometrial Carcinoma	head	0.606	0.656	1.920	0.569
TCGA-UT	Reversed	Severe	Uveal Melanoma	tail	0.907	0.760	0.214	0.114
TCGA-UT	Native	Moderate	Adrenocortical carcinoma	body	0.721	0.760	1.438	0.411
TCGA-UT	Native	Moderate	Bladder Urothelial Carcinoma	body	0.476	0.502	2.270	0.746
TCGA-UT	Native	Moderate	Brain Lower Grade Glioma	head	0.843	0.851	0.645	0.268
TCGA-UT	Native	Moderate	Breast invasive carcinoma	head	0.781	0.826	0.861	0.301
TCGA-UT	Native	Moderate	Cervical squamous cell carcinoma and endocervical adenocarcinoma	body	0.505	0.499	1.493	0.661
TCGA-UT	Native	Moderate	Colon adenocarcinoma	body	0.621	0.677	0.812	0.416
TCGA-UT	Native	Moderate	Esophageal carcinoma	tail	0.368	0.285	2.279	0.939
TCGA-UT	Native	Moderate	Glioblastoma multiforme	head	0.834	0.831	0.486	0.234
TCGA-UT	Native	Moderate	Head and Neck squamous cell carcinoma	head	0.715	0.681	1.008	0.414
TCGA-UT	Native	Moderate	Kidney Chromophobe	tail	0.910	0.835	0.288	0.158
TCGA-UT	Native	Moderate	Kidney renal clear cell carcinoma	head	0.843	0.891	0.904	0.256
TCGA-UT	Native	Moderate	Kidney renal papillary cell carcinoma	body	0.833	0.816	0.597	0.237
TCGA-UT	Native	Moderate	Liver hepatocellular carcinoma	body	0.881	0.857	0.434	0.163
TCGA-UT	Native	Moderate	Lung adenocarcinoma	head	0.741	0.772	0.961	0.373
TCGA-UT	Native	Moderate	Lung squamous cell carcinoma	head	0.610	0.615	1.293	0.551
TCGA-UT	Native	Moderate	Mesothelioma	tail	0.743	0.587	1.043	0.410
TCGA-UT	Native	Moderate	Ovarian serous cystadenocarcinoma	tail	0.797	0.593	0.772	0.318
TCGA-UT	Native	Moderate	Pancreatic adenocarcinoma	tail	0.711	0.663	1.319	0.431
TCGA-UT	Native	Moderate	Pheochromocytoma and Paraganglioma	tail	0.933	0.876	0.332	0.121
TCGA-UT	Native	Moderate	Prostate adenocarcinoma	body	0.972	0.973	0.119	0.039
TCGA-UT	Native	Moderate	Rectum adenocarcinoma	tail	0.629	0.370	1.163	0.657
TCGA-UT	Native	Moderate	Sarcoma	head	0.869	0.870	0.463	0.178
TCGA-UT	Native	Moderate	Skin Cutaneous Melanoma	body	0.720	0.751	0.876	0.379
TCGA-UT	Native	Moderate	Stomach adenocarcinoma	body	0.494	0.548	1.768	0.681
TCGA-UT	Native	Moderate	Testicular Germ Cell Tumors	body	0.899	0.909	0.351	0.131
TCGA-UT	Native	Moderate	Thymoma	tail	0.936	0.883	0.235	0.102
TCGA-UT	Native	Moderate	Thyroid carcinoma	head	0.958	0.969	0.197	0.062
TCGA-UT	Native	Moderate	Uterine Carcinosarcoma	tail	0.594	0.393	1.632	0.617
TCGA-UT	Native	Moderate	Uterine Corpus Endometrial Carcinoma	head	0.657	0.681	1.317	0.482
TCGA-UT	Native	Moderate	Uveal Melanoma	tail	0.904	0.750	0.326	0.177
TCGA-UT	Native	Severe	Adrenocortical carcinoma	body	0.719	0.759	1.569	0.403
TCGA-UT	Native	Severe	Bladder Urothelial Carcinoma	body	0.466	0.489	3.151	0.818

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Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
TCGA-UT	Native	Severe	Brain Lower Grade Glioma	head	0.855	0.849	0.890	0.266
TCGA-UT	Native	Severe	Breast invasive carcinoma	head	0.791	0.821	1.319	0.362
TCGA-UT	Native	Severe	Cervical squamous cell carcinoma and endocervical adenocarcinoma	body	0.458	0.470	1.967	0.728
TCGA-UT	Native	Severe	Colon adenocarcinoma	body	0.619	0.680	1.343	0.609
TCGA-UT	Native	Severe	Esophageal carcinoma	tail	0.340	0.268	2.484	0.921
TCGA-UT	Native	Severe	Glioblastoma multiforme	head	0.819	0.824	0.816	0.291
TCGA-UT	Native	Severe	Head and Neck squamous cell carcinoma	head	0.722	0.675	1.423	0.479
TCGA-UT	Native	Severe	Kidney Chromophobe	tail	0.882	0.820	0.293	0.150
TCGA-UT	Native	Severe	Kidney renal clear cell carcinoma	head	0.847	0.887	1.211	0.276
TCGA-UT	Native	Severe	Kidney renal papillary cell carcinoma	body	0.820	0.811	0.720	0.259
TCGA-UT	Native	Severe	Liver hepatocellular carcinoma	body	0.872	0.857	0.567	0.189
TCGA-UT	Native	Severe	Lung adenocarcinoma	head	0.739	0.761	1.478	0.445
TCGA-UT	Native	Severe	Lung squamous cell carcinoma	head	0.625	0.615	1.888	0.637
TCGA-UT	Native	Severe	Mesothelioma	tail	0.691	0.599	1.068	0.407
TCGA-UT	Native	Severe	Ovarian serous cystadenocarcinoma	tail	0.764	0.594	0.714	0.272
TCGA-UT	Native	Severe	Pancreatic adenocarcinoma	tail	0.697	0.667	1.444	0.423
TCGA-UT	Native	Severe	Pheochromocytoma and Paraganglioma	tail	0.914	0.884	0.371	0.121
TCGA-UT	Native	Severe	Prostate adenocarcinoma	body	0.971	0.972	0.132	0.041
TCGA-UT	Native	Severe	Rectum adenocarcinoma	tail	0.580	0.353	0.967	0.511
TCGA-UT	Native	Severe	Sarcoma	head	0.875	0.870	0.672	0.230
TCGA-UT	Native	Severe	Skin Cutaneous Melanoma	body	0.716	0.747	1.208	0.460
TCGA-UT	Native	Severe	Stomach adenocarcinoma	body	0.488	0.543	2.556	0.798
TCGA-UT	Native	Severe	Testicular Germ Cell Tumors	body	0.895	0.907	0.421	0.146
TCGA-UT	Native	Severe	Thymoma	tail	0.921	0.877	0.281	0.109
TCGA-UT	Native	Severe	Thyroid carcinoma	head	0.956	0.965	0.253	0.072
TCGA-UT	Native	Severe	Uterine Carcinosarcoma	tail	0.556	0.431	1.637	0.601
TCGA-UT	Native	Severe	Uterine Corpus Endometrial Carcinoma	head	0.665	0.686	2.003	0.594
TCGA-UT	Native	Severe	Uveal Melanoma	tail	0.878	0.751	0.216	0.118

Table 27: Equal-weight three-split per-class test endpoints for the cross-entropy reference by condition, with the tier assignment each condition induces.

C Calibration Detail

The calibration gate reads tail-group macro negative log likelihood, but the shape of the miscalibration matters for method design and is not visible in a scalar. This appendix reports the raw and temperature-scaled expected calibration error, negative log likelihood, and Brier score per condition and method for the cross-entropy reference and the confirmatory family, and shows the reliability curves from which they are computed.

Dataset	Assignment	Condition	Method	NLL	Brier	ECE		Temp. NLL	Temp. Brier	Temp. ECE
BRACS	Aligned	Moderate	CE	1.397	0.572	0.169	[0.114, 0.242]	1.066	0.514	0.048
BRACS	Aligned	Moderate	Nominal support	1.037	0.514	0.055	[0.045, 0.088]	1.033	0.513	0.056
BRACS	Aligned	Moderate	Independent support	1.418	0.616	0.209	[0.157, 0.279]	1.115	0.537	0.072
BRACS	Aligned	Moderate	Difficulty	1.367	0.613	0.203	[0.156, 0.263]	1.132	0.544	0.078
BRACS	Aligned	Moderate	Diversity	1.280	0.608	0.193	[0.142, 0.256]	1.134	0.551	0.087
BRACS	Aligned	Moderate	Prevalence	1.036	0.510	0.053	[0.045, 0.084]	1.028	0.509	0.052
BRACS	Aligned	Severe	CE	1.903	0.658	0.252	[0.208, 0.310]	1.132	0.544	0.071
BRACS	Aligned	Severe	Nominal support	1.052	0.518	0.068	[0.053, 0.107]	1.048	0.520	0.073
BRACS	Aligned	Severe	Independent support	2.203	0.750	0.324	[0.251, 0.419]	1.203	0.573	0.104
BRACS	Aligned	Severe	Difficulty	2.395	0.766	0.336	[0.266, 0.424]	1.217	0.579	0.104
BRACS	Aligned	Severe	Diversity	2.045	0.748	0.320	[0.250, 0.410]	1.214	0.577	0.106
BRACS	Aligned	Severe	Prevalence	1.066	0.522	0.071	[0.055, 0.113]	1.050	0.519	0.066
BRACS	Reversed	Moderate	CE	1.843	0.652	0.231	[0.180, 0.297]	1.188	0.557	0.069
BRACS	Reversed	Moderate	Nominal support	1.062	0.518	0.060	[0.044, 0.104]	1.049	0.512	0.042
BRACS	Reversed	Moderate	Independent support	2.025	0.700	0.242	[0.193, 0.309]	1.264	0.596	0.075
BRACS	Reversed	Moderate	Difficulty	2.490	0.781	0.293	[0.234, 0.354]	1.338	0.636	0.079
BRACS	Reversed	Moderate	Diversity	1.496	0.668	0.189	[0.143, 0.247]	1.273	0.612	0.067
BRACS	Reversed	Moderate	Prevalence	1.055	0.513	0.065	[0.052, 0.101]	1.036	0.507	0.041
BRACS	Reversed	Severe	CE	2.642	0.754	0.317	[0.267, 0.380]	1.258	0.591	0.061
BRACS	Reversed	Severe	Nominal support	1.076	0.526	0.059	[0.046, 0.096]	1.078	0.525	0.059
BRACS	Reversed	Severe	Independent support	2.395	0.814	0.315	[0.259, 0.371]	1.387	0.667	0.083
BRACS	Reversed	Severe	Difficulty	3.695	0.927	0.415	[0.366, 0.471]	1.424	0.679	0.073
BRACS	Reversed	Severe	Diversity	3.860	0.927	0.414	[0.358, 0.480]	1.434	0.683	0.073
BRACS	Reversed	Severe	Prevalence	1.053	0.514	0.054	[0.044, 0.089]	1.052	0.513	0.055
BRACS	Native	Moderate	CE	1.311	0.570	0.159	[0.117, 0.214]	1.112	0.528	0.070
BRACS	Native	Moderate	Nominal support	1.034	0.504	0.061	[0.044, 0.099]	1.011	0.500	0.043
BRACS	Native	Moderate	Independent support	1.290	0.586	0.146	[0.109, 0.202]	1.120	0.544	0.068
BRACS	Native	Moderate	Difficulty	1.283	0.597	0.145	[0.102, 0.213]	1.156	0.562	0.067
BRACS	Native	Moderate	Diversity	1.147	0.555	0.095	[0.066, 0.150]	1.104	0.540	0.056
BRACS	Native	Moderate	Prevalence	1.041	0.507	0.062	[0.046, 0.100]	1.018	0.502	0.046
BRACS	Native	Severe	CE	1.865	0.655	0.240	[0.184, 0.308]	1.223	0.562	0.088
BRACS	Native	Severe	Nominal support	1.068	0.521	0.072	[0.053, 0.115]	1.047	0.514	0.054
BRACS	Native	Severe	Independent support	1.959	0.692	0.245	[0.190, 0.316]	1.260	0.597	0.075
BRACS	Native	Severe	Difficulty	2.155	0.728	0.272	[0.218, 0.341]	1.317	0.622	0.092
BRACS	Native	Severe	Diversity	1.942	0.704	0.247	[0.190, 0.319]	1.292	0.613	0.082
BRACS	Native	Severe	Prevalence	1.058	0.516	0.068	[0.050, 0.111]	1.043	0.511	0.056
BRACS	—	Balanced	CE	0.983	0.483	0.053	[0.041, 0.088]	0.982	0.482	0.053
BRACS	—	Balanced	Nominal support	1.048	0.513	0.056	[0.040, 0.103]	1.030	0.507	0.041
BRACS	—	Balanced	Independent support	1.047	0.513	0.058	[0.042, 0.104]	1.029	0.506	0.040
BRACS	—	Balanced	Difficulty	1.057	0.517	0.063	[0.043, 0.111]	1.036	0.508	0.040
BRACS	—	Balanced	Diversity	1.047	0.513	0.057	[0.042, 0.103]	1.029	0.506	0.040
BRACS	—	Balanced	Prevalence	1.048	0.513	0.056	[0.040, 0.103]	1.030	0.507	0.041
BRACS	—	Natural	CE	0.949	0.453	0.080	[0.051, 0.119]	0.908	0.441	0.041

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Dataset	Assignment	Condition	Method	NLL	Brier	ECE		Temp. NLL	Temp. Brier	Temp. ECE
CAMELYON16	Reversed	Moderate	CE	0.058	0.012	0.005	[0.003, 0.008]	0.034	0.012	0.004
CAMELYON16	Reversed	Moderate	Nominal support	0.054	0.022	0.007	[0.004, 0.011]	0.047	0.022	0.003
CAMELYON16	Reversed	Moderate	Independent support	0.101	0.035	0.010	[0.005, 0.016]	0.084	0.035	0.006
CAMELYON16	Reversed	Moderate	Difficulty	0.125	0.041	0.016	[0.010, 0.022]	0.087	0.039	0.005
CAMELYON16	Reversed	Moderate	Diversity	0.099	0.033	0.010	[0.006, 0.015]	0.085	0.033	0.007
CAMELYON16	Reversed	Moderate	Prevalence	0.079	0.026	0.008	[0.004, 0.013]	0.053	0.025	0.004
CAMELYON16	Reversed	Severe	CE	0.086	0.019	0.008	[0.005, 0.013]	0.038	0.018	0.002
CAMELYON16	Reversed	Severe	Nominal support	0.266	0.031	0.014	[0.011, 0.019]	0.055	0.027	0.003
CAMELYON16	Reversed	Severe	Independent support	0.172	0.057	0.021	[0.016, 0.027]	0.120	0.054	0.007
CAMELYON16	Reversed	Severe	Difficulty	0.320	0.087	0.030	[0.022, 0.039]	0.178	0.082	0.011
CAMELYON16	Reversed	Severe	Diversity	0.457	0.061	0.030	[0.021, 0.039]	0.139	0.058	0.017
CAMELYON16	Reversed	Severe	Prevalence	0.083	0.027	0.009	[0.005, 0.013]	0.062	0.027	0.004
CAMELYON16	Native	Moderate	CE	0.045	0.010	0.004	[0.002, 0.006]	0.024	0.011	0.004
CAMELYON16	Native	Moderate	Nominal support	0.037	0.015	0.004	[0.002, 0.006]	0.031	0.014	0.003
CAMELYON16	Native	Moderate	Independent support	0.042	0.012	0.004	[0.003, 0.006]	0.027	0.012	0.004
CAMELYON16	Native	Moderate	Difficulty	0.043	0.015	0.005	[0.003, 0.007]	0.032	0.015	0.002
CAMELYON16	Native	Moderate	Diversity	0.069	0.015	0.006	[0.003, 0.009]	0.037	0.015	0.004
CAMELYON16	Native	Moderate	Prevalence	0.107	0.022	0.008	[0.004, 0.013]	0.048	0.021	0.003
CAMELYON16	Native	Severe	CE	0.035	0.009	0.003	[0.002, 0.006]	0.019	0.009	0.002
CAMELYON16	Native	Severe	Nominal support	0.114	0.017	0.007	[0.004, 0.010]	0.042	0.016	0.004
CAMELYON16	Native	Severe	Independent support	0.033	0.009	0.004	[0.002, 0.007]	0.018	0.008	0.002
CAMELYON16	Native	Severe	Difficulty	0.035	0.009	0.003	[0.002, 0.006]	0.018	0.008	0.002
CAMELYON16	Native	Severe	Diversity	0.075	0.013	0.006	[0.004, 0.008]	0.030	0.013	0.007
CAMELYON16	Native	Severe	Prevalence	0.040	0.016	0.006	[0.003, 0.010]	0.038	0.016	0.004
CAMELYON16	—	Balanced	CE	0.018	0.007	0.002	[0.001, 0.004]	0.014	0.007	0.001
CAMELYON16	—	Balanced	Nominal support	0.040	0.016	0.004	[0.002, 0.006]	0.035	0.016	0.002
CAMELYON16	—	Balanced	Independent support	0.047	0.016	0.005	[0.003, 0.007]	0.035	0.016	0.004
CAMELYON16	—	Balanced	Difficulty	0.054	0.019	0.004	[0.003, 0.007]	0.051	0.019	0.005
CAMELYON16	—	Balanced	Diversity	0.049	0.017	0.005	[0.003, 0.009]	0.037	0.017	0.003
CAMELYON16	—	Balanced	Prevalence	0.041	0.016	0.004	[0.003, 0.006]	0.034	0.016	0.002
CAMELYON16	—	Natural	CE	0.024	0.010	0.004	[0.002, 0.009]	0.025	0.015	0.011
PANDA	Native	Moderate	CE	0.216	0.120	0.015	[0.014, 0.017]	0.209	0.120	0.014
PANDA	Native	Moderate	Nominal support	0.213	0.116	0.021	[0.018, 0.024]	0.200	0.115	0.010
PANDA	Native	Moderate	Independent support	0.214	0.116	0.022	[0.019, 0.024]	0.201	0.115	0.009
PANDA	Native	Moderate	Difficulty	0.215	0.116	0.023	[0.020, 0.026]	0.200	0.114	0.009
PANDA	Native	Moderate	Diversity	0.252	0.149	0.036	[0.030, 0.042]	0.243	0.145	0.011
PANDA	Native	Moderate	Prevalence	0.254	0.153	0.028	[0.026, 0.031]	0.254	0.151	0.024
PANDA	Native	Severe	CE	0.254	0.132	0.026	[0.023, 0.030]	0.229	0.132	0.020
PANDA	Native	Severe	Nominal support	0.355	0.162	0.063	[0.059, 0.067]	0.244	0.147	0.005
PANDA	Native	Severe	Independent support	0.397	0.172	0.072	[0.067, 0.076]	0.251	0.152	0.006
PANDA	Native	Severe	Difficulty	0.420	0.177	0.075	[0.071, 0.080]	0.254	0.154	0.008
PANDA	Native	Severe	Diversity	0.406	0.173	0.073	[0.069, 0.077]	0.252	0.152	0.006
PANDA	Native	Severe	Prevalence	0.254	0.152	0.026	[0.024, 0.029]	0.253	0.150	0.023

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Dataset	Assignment	Condition	Method	NLL	Brier	ECE		Temp. NLL	Temp. Brier	Temp. ECE
PANDA	—	Balanced	CE	0.185	0.105	0.007	[0.006, 0.009]	0.184	0.105	0.005
PANDA	—	Balanced	Nominal support	0.246	0.147	0.022	[0.020, 0.024]	0.246	0.147	0.021
PANDA	—	Balanced	Independent support	0.244	0.146	0.023	[0.021, 0.025]	0.244	0.145	0.021
PANDA	—	Balanced	Difficulty	0.242	0.144	0.022	[0.020, 0.023]	0.242	0.144	0.020
PANDA	—	Balanced	Diversity	0.246	0.147	0.022	[0.020, 0.024]	0.246	0.147	0.021
PANDA	—	Balanced	Prevalence	0.242	0.144	0.023	[0.021, 0.025]	0.242	0.144	0.021
PANDA	—	Natural	CE	0.194	0.106	0.011	[0.010, 0.012]	0.190	0.107	0.012
TCGA-UT	Aligned	Moderate	CE	0.797	0.329	0.049	[0.040, 0.059]	0.770	0.324	0.009
TCGA-UT	Aligned	Moderate	Nominal support	0.838	0.350	0.032	[0.024, 0.041]	0.829	0.348	0.010
TCGA-UT	Aligned	Moderate	Independent support	0.889	0.359	0.070	[0.060, 0.080]	0.840	0.350	0.013
TCGA-UT	Aligned	Moderate	Difficulty	0.861	0.354	0.054	[0.044, 0.065]	0.833	0.349	0.012
TCGA-UT	Aligned	Moderate	Diversity	0.871	0.356	0.060	[0.051, 0.070]	0.837	0.350	0.013
TCGA-UT	Aligned	Moderate	Prevalence	0.838	0.350	0.032	[0.023, 0.041]	0.829	0.349	0.010
TCGA-UT	Aligned	Severe	CE	0.960	0.361	0.095	[0.084, 0.107]	0.830	0.343	0.011
TCGA-UT	Aligned	Severe	Nominal support	0.850	0.355	0.033	[0.024, 0.042]	0.842	0.354	0.011
TCGA-UT	Aligned	Severe	Independent support	1.086	0.396	0.124	[0.114, 0.135]	0.890	0.366	0.018
TCGA-UT	Aligned	Severe	Difficulty	1.008	0.387	0.104	[0.091, 0.116]	0.887	0.367	0.017
TCGA-UT	Aligned	Severe	Diversity	1.050	0.389	0.113	[0.098, 0.127]	0.882	0.364	0.018
TCGA-UT	Aligned	Severe	Prevalence	0.849	0.355	0.030	[0.022, 0.039]	0.842	0.354	0.011
TCGA-UT	Reversed	Moderate	CE	0.839	0.337	0.073	[0.063, 0.083]	0.786	0.327	0.011
TCGA-UT	Reversed	Moderate	Nominal support	0.841	0.351	0.040	[0.032, 0.049]	0.828	0.348	0.011
TCGA-UT	Reversed	Moderate	Independent support	0.911	0.366	0.079	[0.068, 0.091]	0.852	0.355	0.013
TCGA-UT	Reversed	Moderate	Difficulty	0.960	0.375	0.095	[0.085, 0.106]	0.862	0.358	0.013
TCGA-UT	Reversed	Moderate	Diversity	0.922	0.368	0.083	[0.071, 0.096]	0.853	0.355	0.012
TCGA-UT	Reversed	Moderate	Prevalence	0.840	0.350	0.039	[0.031, 0.048]	0.827	0.347	0.011
TCGA-UT	Reversed	Severe	CE	1.214	0.388	0.144	[0.132, 0.156]	0.874	0.347	0.010
TCGA-UT	Reversed	Severe	Nominal support	0.846	0.354	0.035	[0.024, 0.047]	0.836	0.352	0.012
TCGA-UT	Reversed	Severe	Independent support	1.320	0.420	0.160	[0.147, 0.173]	0.913	0.371	0.013
TCGA-UT	Reversed	Severe	Difficulty	1.323	0.421	0.160	[0.148, 0.172]	0.919	0.374	0.012
TCGA-UT	Reversed	Severe	Diversity	1.256	0.415	0.152	[0.137, 0.168]	0.912	0.371	0.012
TCGA-UT	Reversed	Severe	Prevalence	0.840	0.352	0.034	[0.024, 0.046]	0.831	0.350	0.011
TCGA-UT	Native	Moderate	CE	0.893	0.346	0.087	[0.074, 0.100]	0.803	0.331	0.010
TCGA-UT	Native	Moderate	Nominal support	0.835	0.347	0.040	[0.030, 0.050]	0.821	0.344	0.009
TCGA-UT	Native	Moderate	Independent support	0.946	0.358	0.100	[0.090, 0.111]	0.823	0.338	0.011
TCGA-UT	Native	Moderate	Difficulty	0.939	0.358	0.097	[0.086, 0.109]	0.824	0.339	0.010
TCGA-UT	Native	Moderate	Diversity	0.936	0.356	0.097	[0.086, 0.108]	0.822	0.338	0.010
TCGA-UT	Native	Moderate	Prevalence	0.839	0.348	0.040	[0.030, 0.050]	0.825	0.346	0.009
TCGA-UT	Native	Severe	CE	1.244	0.395	0.142	[0.131, 0.153]	0.895	0.354	0.008
TCGA-UT	Native	Severe	Nominal support	0.821	0.342	0.031	[0.023, 0.041]	0.813	0.340	0.009
TCGA-UT	Native	Severe	Independent support	1.227	0.386	0.145	[0.132, 0.159]	0.852	0.343	0.012

Table 29: Raw and temperature-scaled expected calibration error, negative log likelihood, and Brier score per condition and method, with crossed patient-bootstrap intervals on the calibration error. Temperature is fitted on natural-prevalence validation data and never gates.

Temperature scaling removes a large share of the observed miscalibration on the cross-entropy reference wherever the calibration gate opened: on the BRACS difficulty-aligned moderate condition it takes expected calibration error from 0.169 to 0.048 and negative log likelihood from 1.397 to 1.066. It does not, however, substitute for mitigation, because a single global scalar cannot redistribute probability mass between classes and therefore leaves the tail-group deficit that the gate reads largely intact. This is the reason the protocol treats it as a diagnostic and never lets it gate.

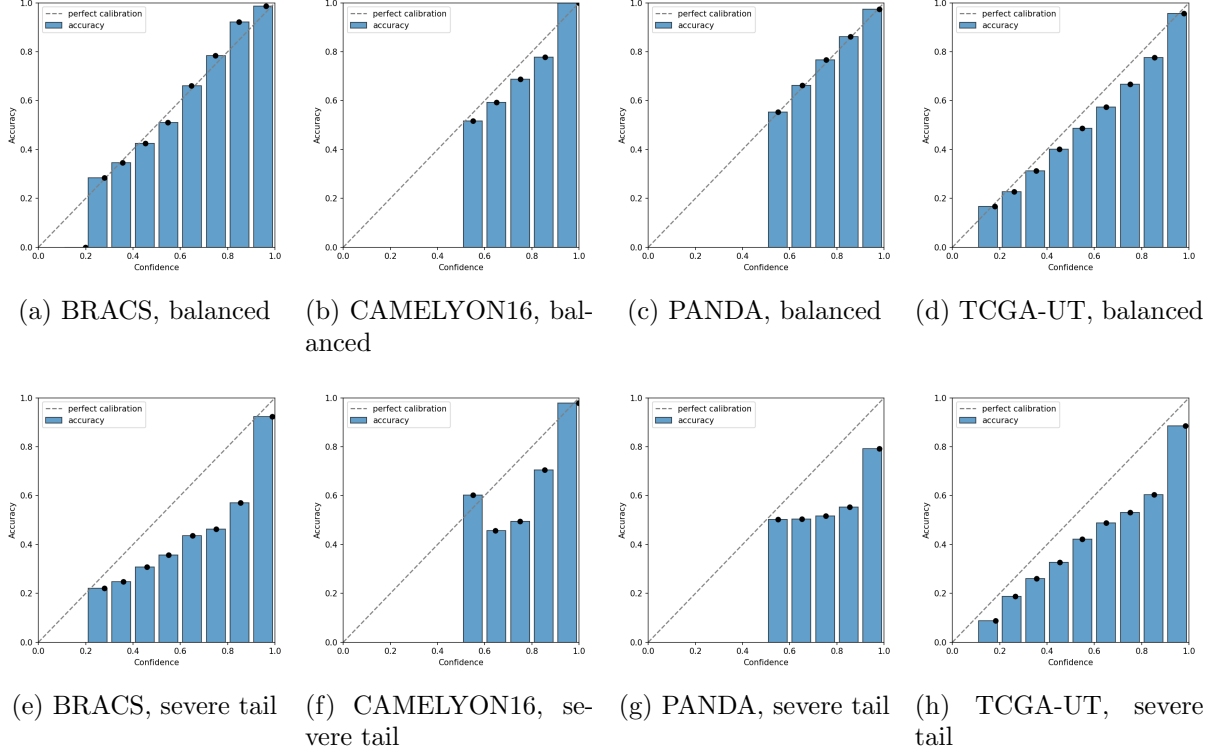


Figure 2: Reliability curves for the balanced reference and for the tail classes under the severe condition of the native assignment, with bin accuracy against mean predicted confidence over ten equal-width bins, on the first locked partition.

D Per-Partition Results

The main text averages the three partitions with equal weight, which is only defensible if no single partition drives a conclusion. This appendix reports the cross-entropy deficits separately for each partition, alongside the spread of achieved severity across them. A conclusion that holds in the equal-weight average but in only one partition individually is identified here and is qualified wherever it appears in the main text.

Dataset	Assignment	Severity	Gate	Split 0	Split 1	Split 2	Equal weight	ρ spread
BRACS	Native	Moderate	Discrimination	0.0367	0.0284	0.0323	0.0325	0.000
BRACS	Native	Moderate	Calibration	0.2806	0.1799	0.3822	0.2809	0.000
BRACS	Aligned	Moderate	Discrimination	0.0252	0.0400	0.0140	0.0264	0.000
BRACS	Aligned	Moderate	Calibration	1.3607	0.2323	0.5041	0.6990	0.000
BRACS	Reversed	Moderate	Discrimination	0.0917	0.0622	0.0889	0.0810	0.000
BRACS	Reversed	Moderate	Calibration	0.8221	0.0764	1.3195	0.7393	0.000
BRACS	Native	Severe	Discrimination	0.0639	0.0595	0.0579	0.0604	0.002
BRACS	Native	Severe	Calibration	0.6768	0.1712	1.4834	0.7771	0.002

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Dataset	Assignment	Severity	Gate	Split 0	Split 1	Split 2	Equal weight	ρ spread
BRACS	Aligned	Severe	Discrimination	0.0173	0.0725	0.0438	0.0445	0.002
BRACS	Aligned	Severe	Calibration	1.8838	0.6364	2.0714	1.5305	0.002
BRACS	Reversed	Severe	Discrimination	0.1359	0.1328	0.1323	0.1337	0.002
BRACS	Reversed	Severe	Calibration	1.2204	2.7078	2.2117	2.0466	0.002
CAMELYON16	Native	Moderate	Discrimination	-0.0006	0.0042	0.0030	0.0022	0.000
CAMELYON16	Native	Moderate	Calibration	-0.0373	0.0088	-0.0773	-0.0353	0.000
CAMELYON16	Reversed	Moderate	Discrimination	0.0024	0.0054	0.0020	0.0033	0.000
CAMELYON16	Reversed	Moderate	Calibration	0.0404	0.0638	0.0477	0.0506	0.000
CAMELYON16	Native	Severe	Discrimination	0.0006	0.0170	0.0334	0.0170	0.002
CAMELYON16	Native	Severe	Calibration	-0.0611	0.0571	0.4368	0.1443	0.002
CAMELYON16	Reversed	Severe	Discrimination	0.0222	0.0217	0.0106	0.0182	0.002
CAMELYON16	Reversed	Severe	Calibration	0.0249	0.2038	0.0336	0.0874	0.002
PANDA	Native	Moderate	Discrimination	0.0277	0.0262	0.0242	0.0260	0.000
PANDA	Native	Moderate	Calibration	-0.1363	-0.1468	-0.1957	-0.1596	0.000
PANDA	Native	Severe	Discrimination	0.1501	0.1410	0.1462	0.1458	0.000
PANDA	Native	Severe	Calibration	0.1803	0.0656	0.0164	0.0874	0.000
TCGA-UT	Native	Moderate	Discrimination	0.0068	0.0039	0.0022	0.0043	0.001
TCGA-UT	Native	Moderate	Calibration	-0.1785	-0.1895	-0.2336	-0.2005	0.001
TCGA-UT	Aligned	Moderate	Discrimination	0.0006	0.0018	-0.0001	0.0007	0.001
TCGA-UT	Aligned	Moderate	Calibration	-0.0892	-0.0705	-0.0262	-0.0620	0.001
TCGA-UT	Reversed	Moderate	Discrimination	0.0073	0.0071	0.0004	0.0049	0.001
TCGA-UT	Reversed	Moderate	Calibration	-0.0628	-0.0684	-0.0711	-0.0675	0.001
TCGA-UT	Native	Severe	Discrimination	0.0184	0.0166	0.0133	0.0161	0.009
TCGA-UT	Native	Severe	Calibration	-0.2002	-0.1486	-0.2066	-0.1851	0.009
TCGA-UT	Aligned	Severe	Discrimination	0.0095	0.0122	0.0112	0.0110	0.010
TCGA-UT	Aligned	Severe	Calibration	0.1613	0.0322	0.0701	0.0879	0.010
TCGA-UT	Reversed	Severe	Discrimination	0.0154	0.0175	0.0108	0.0146	0.009
TCGA-UT	Reversed	Severe	Calibration	0.0169	-0.0544	-0.0365	-0.0247	0.009

Table 31: Cross-entropy deficits by partition on both gate axes, with the equal-weight average and the relative achieved-severity spread across partitions, per dataset, condition, and tail assignment.

No gate decision rests on a single partition, but several rest on two of the three. On the discrimination axis every unit that opened the gate exceeds the 0.014,43 threshold in at least two partitions, and seven of the 12 exceed it in all three. The five that do not are the BRACS difficulty-aligned moderate condition, both CAMELYON16 severe conditions, and both TCGA-UT severe conditions; the widest departure is the CAMELYON16 native severe condition, whose smallest partition-level deficit is 0.000,6. On the calibration axis three of the six BRACS units exceed the 0.182,37-nat threshold in all three partitions and three exceed it in two, the smallest partition-level deficit being 0.076 nats under the difficulty-reversed moderate condition. Every conclusion drawn from those eight units is therefore a statement about the equal-weight average and is qualified accordingly wherever it appears in the main text. Achieved severity itself varies across partitions by at most 1.0% of the smallest realized value, well inside the 50% consistency guard.

E Computational Cost

The legacy table covers five methods only. The exposure-matched rerun reports the whole roster with exposure, time, memory, and model size; this table supports no roster-wide cost claim.

Cost bears on the design question because a method whose recovery requires substantially more computation than the reference is a different proposition from one that matches it. This appendix records processed examples, unique exposures, accelerator time, peak memory, and model size for each confirmatory-family method, expressed as a matched difference against cross-entropy in the same condition.

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Moderate	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Nominal support	Peak memory (bytes)	2.33e+09	[2.3e+09, 2.365e+09]
BRACS	Aligned	Moderate	Nominal support	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Nominal support	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Nominal support	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Independent support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Independent support	Peak memory (bytes)	2.33e+09	[2.3e+09, 2.365e+09]
BRACS	Aligned	Moderate	Independent support	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Independent support	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Independent support	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Difficulty	Peak memory (bytes)	2.33e+09	[2.3e+09, 2.365e+09]
BRACS	Aligned	Moderate	Difficulty	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Difficulty	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Difficulty	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Diversity	Accelerator hours	0.017	[0.017, 0.017]
BRACS	Aligned	Moderate	Diversity	Peak memory (bytes)	2.356e+09	[2.327e+09, 2.392e+09]
BRACS	Aligned	Moderate	Diversity	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Diversity	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Diversity	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Prevalence	Peak memory (bytes)	1.826e+09	[1.796e+09, 1.862e+09]
BRACS	Aligned	Moderate	Prevalence	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Prevalence	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Prevalence	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Nominal support	Peak memory (bytes)	2.246e+09	[2.195e+09, 2.306e+09]
BRACS	Aligned	Severe	Nominal support	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Nominal support	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Nominal support	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Independent support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Independent support	Peak memory (bytes)	2.246e+09	[2.195e+09, 2.306e+09]
BRACS	Aligned	Severe	Independent support	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Independent support	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Independent support	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Difficulty	Peak memory (bytes)	2.246e+09	[2.195e+09, 2.306e+09]
BRACS	Aligned	Severe	Difficulty	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Difficulty	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Difficulty	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Diversity	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Aligned	Severe	Diversity	Peak memory (bytes)	2.251e+09	[2.2e+09, 2.311e+09]
BRACS	Aligned	Severe	Diversity	Processed examples	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Severe	Diversity	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Diversity	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Prevalence	Peak memory (bytes)	1.743e+09	[1.692e+09, 1.803e+09]
BRACS	Aligned	Severe	Prevalence	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Prevalence	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Prevalence	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Nominal support	Peak memory (bytes)	2.307e+09	[2.3e+09, 2.316e+09]
BRACS	Reversed	Moderate	Nominal support	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Nominal support	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Nominal support	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Independent support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Independent support	Peak memory (bytes)	2.307e+09	[2.3e+09, 2.316e+09]
BRACS	Reversed	Moderate	Independent support	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Independent support	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Independent support	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Difficulty	Peak memory (bytes)	2.307e+09	[2.3e+09, 2.316e+09]
BRACS	Reversed	Moderate	Difficulty	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Difficulty	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Difficulty	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Diversity	Accelerator hours	0.024	[0.024, 0.024]
BRACS	Reversed	Moderate	Diversity	Peak memory (bytes)	2.343e+09	[2.336e+09, 2.352e+09]
BRACS	Reversed	Moderate	Diversity	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Diversity	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Diversity	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Prevalence	Peak memory (bytes)	1.804e+09	[1.796e+09, 1.812e+09]
BRACS	Reversed	Moderate	Prevalence	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Prevalence	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Prevalence	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Nominal support	Peak memory (bytes)	2.212e+09	[2.195e+09, 2.232e+09]
BRACS	Reversed	Severe	Nominal support	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Nominal support	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Nominal support	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Independent support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Independent support	Peak memory (bytes)	2.212e+09	[2.195e+09, 2.232e+09]
BRACS	Reversed	Severe	Independent support	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Independent support	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Independent support	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Difficulty	Accelerator hours	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Severe	Difficulty	Peak memory (bytes)	2.212e+09	[2.195e+09, 2.232e+09]
BRACS	Reversed	Severe	Difficulty	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Difficulty	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Difficulty	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Diversity	Accelerator hours	0.005	[0.005, 0.005]
BRACS	Reversed	Severe	Diversity	Peak memory (bytes)	2.219e+09	[2.203e+09, 2.239e+09]
BRACS	Reversed	Severe	Diversity	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Diversity	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Diversity	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Prevalence	Peak memory (bytes)	1.709e+09	[1.692e+09, 1.729e+09]
BRACS	Reversed	Severe	Prevalence	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Prevalence	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Prevalence	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Nominal support	Peak memory (bytes)	2.3e+09	[2.3e+09, 2.3e+09]
BRACS	Native	Moderate	Nominal support	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Nominal support	Parameters	0	[0, 0]
BRACS	Native	Moderate	Nominal support	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Independent support	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Independent support	Peak memory (bytes)	2.3e+09	[2.3e+09, 2.3e+09]
BRACS	Native	Moderate	Independent support	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Independent support	Parameters	0	[0, 0]
BRACS	Native	Moderate	Independent support	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Difficulty	Peak memory (bytes)	2.3e+09	[2.3e+09, 2.3e+09]
BRACS	Native	Moderate	Difficulty	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Difficulty	Parameters	0	[0, 0]
BRACS	Native	Moderate	Difficulty	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Diversity	Accelerator hours	0.028	[0.027, 0.028]
BRACS	Native	Moderate	Diversity	Peak memory (bytes)	2.343e+09	[2.343e+09, 2.343e+09]
BRACS	Native	Moderate	Diversity	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Diversity	Parameters	0	[0, 0]
BRACS	Native	Moderate	Diversity	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Prevalence	Peak memory (bytes)	1.796e+09	[1.796e+09, 1.796e+09]
BRACS	Native	Moderate	Prevalence	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Prevalence	Parameters	0	[0, 0]
BRACS	Native	Moderate	Prevalence	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Nominal support	Peak memory (bytes)	2.195e+09	[2.195e+09, 2.195e+09]
BRACS	Native	Severe	Nominal support	Processed examples	0	[0, 0]
BRACS	Native	Severe	Nominal support	Parameters	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Severe	Nominal support	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Independent support	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Independent support	Peak memory (bytes)	2.195e+09	[2.195e+09, 2.195e+09]
BRACS	Native	Severe	Independent support	Processed examples	0	[0, 0]
BRACS	Native	Severe	Independent support	Parameters	0	[0, 0]
BRACS	Native	Severe	Independent support	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Difficulty	Peak memory (bytes)	2.195e+09	[2.195e+09, 2.195e+09]
BRACS	Native	Severe	Difficulty	Processed examples	0	[0, 0]
BRACS	Native	Severe	Difficulty	Parameters	0	[0, 0]
BRACS	Native	Severe	Difficulty	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Diversity	Accelerator hours	0.005	[0.005, 0.005]
BRACS	Native	Severe	Diversity	Peak memory (bytes)	2.203e+09	[2.203e+09, 2.203e+09]
BRACS	Native	Severe	Diversity	Processed examples	0	[0, 0]
BRACS	Native	Severe	Diversity	Parameters	0	[0, 0]
BRACS	Native	Severe	Diversity	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Prevalence	Peak memory (bytes)	1.692e+09	[1.692e+09, 1.692e+09]
BRACS	Native	Severe	Prevalence	Processed examples	0	[0, 0]
BRACS	Native	Severe	Prevalence	Parameters	0	[0, 0]
BRACS	Native	Severe	Prevalence	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Nominal support	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Nominal support	Peak memory (bytes)	512	[512, 512]
BRACS	—	Balanced	Nominal support	Processed examples	0	[0, 0]
BRACS	—	Balanced	Nominal support	Parameters	0	[0, 0]
BRACS	—	Balanced	Nominal support	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Independent support	Accelerator hours	0	[0, 0.001]
BRACS	—	Balanced	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	—	Balanced	Independent support	Processed examples	0	[0, 0]
BRACS	—	Balanced	Independent support	Parameters	0	[0, 0]
BRACS	—	Balanced	Independent support	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Difficulty	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Difficulty	Peak memory (bytes)	512	[512, 512]
BRACS	—	Balanced	Difficulty	Processed examples	0	[0, 0]
BRACS	—	Balanced	Difficulty	Parameters	0	[0, 0]
BRACS	—	Balanced	Difficulty	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Diversity	Accelerator hours	0.048	[0.044, 0.052]
BRACS	—	Balanced	Diversity	Peak memory (bytes)	9.082e+07	[8.716e+07, 9.412e+07]
BRACS	—	Balanced	Diversity	Processed examples	0	[0, 0]
BRACS	—	Balanced	Diversity	Parameters	0	[0, 0]
BRACS	—	Balanced	Diversity	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Prevalence	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Prevalence	Peak memory (bytes)	512	[512, 512]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	—	Balanced	Prevalence	Processed examples	0	[0, 0]
BRACS	—	Balanced	Prevalence	Parameters	0	[0, 0]
BRACS	—	Balanced	Prevalence	Unique exposures	0	[0, 0]
CAMELYON16	Reversed	Moderate	Nominal support	Accelerator hours	0	[0, 0]
CAMELYON16	Reversed	Moderate	Nominal support	Peak memory (bytes)	512	[512, 512]
CAMELYON16	Reversed	Moderate	Nominal support	Processed examples	0	[0, 0]
CAMELYON16	Reversed	Moderate	Nominal support	Parameters	0	[0, 0]
CAMELYON16	Reversed	Moderate	Nominal support	Unique exposures	0	[0, 0]
CAMELYON16	Reversed	Moderate	Independent support	Accelerator hours	0	[0, 0]
CAMELYON16	Reversed	Moderate	Independent support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
CAMELYON16	Reversed	Moderate	Independent support	Processed examples	0	[0, 0]
CAMELYON16	Reversed	Moderate	Independent support	Parameters	0	[0, 0]
CAMELYON16	Reversed	Moderate	Independent support	Unique exposures	0	[0, 0]
CAMELYON16	Reversed	Moderate	Difficulty	Accelerator hours	0	[0, 0]
CAMELYON16	Reversed	Moderate	Difficulty	Peak memory (bytes)	512	[512, 512]
CAMELYON16	Reversed	Moderate	Difficulty	Processed examples	0	[0, 0]
CAMELYON16	Reversed	Moderate	Difficulty	Parameters	0	[0, 0]
CAMELYON16	Reversed	Moderate	Difficulty	Unique exposures	0	[0, 0]
CAMELYON16	Reversed	Moderate	Diversity	Accelerator hours	0.001	[0.001, 0.001]
CAMELYON16	Reversed	Moderate	Diversity	Peak memory (bytes)	1.197e+07	[1.197e+07, 1.197e+07]
CAMELYON16	Reversed	Moderate	Diversity	Processed examples	0	[0, 0]
CAMELYON16	Reversed	Moderate	Diversity	Parameters	0	[0, 0]
CAMELYON16	Reversed	Moderate	Diversity	Unique exposures	0	[0, 0]
CAMELYON16	Reversed	Moderate	Prevalence	Accelerator hours	0	[0, 0]
CAMELYON16	Reversed	Moderate	Prevalence	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
CAMELYON16	Reversed	Moderate	Prevalence	Processed examples	0	[0, 0]
CAMELYON16	Reversed	Moderate	Prevalence	Parameters	0	[0, 0]
CAMELYON16	Reversed	Moderate	Prevalence	Unique exposures	0	[0, 0]
CAMELYON16	Reversed	Severe	Nominal support	Accelerator hours	0.001	[0, 0.001]
CAMELYON16	Reversed	Severe	Nominal support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
CAMELYON16	Reversed	Severe	Nominal support	Processed examples	0	[0, 0]
CAMELYON16	Reversed	Severe	Nominal support	Parameters	0	[0, 0]
CAMELYON16	Reversed	Severe	Nominal support	Unique exposures	0	[0, 0]
CAMELYON16	Reversed	Severe	Independent support	Accelerator hours	0	[0, 0]
CAMELYON16	Reversed	Severe	Independent support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
CAMELYON16	Reversed	Severe	Independent support	Processed examples	0	[0, 0]
CAMELYON16	Reversed	Severe	Independent support	Parameters	0	[0, 0]
CAMELYON16	Reversed	Severe	Independent support	Unique exposures	0	[0, 0]
CAMELYON16	Reversed	Severe	Difficulty	Accelerator hours	0	[0, 0]
CAMELYON16	Reversed	Severe	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
CAMELYON16	Reversed	Severe	Difficulty	Processed examples	0	[0, 0]
CAMELYON16	Reversed	Severe	Difficulty	Parameters	0	[0, 0]
CAMELYON16	Reversed	Severe	Difficulty	Unique exposures	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
CAMELYON16	Reversed	Severe	Diversity	Accelerator hours	0.001	[0.001, 0.001]
CAMELYON16	Reversed	Severe	Diversity	Peak memory (bytes)	-1.562e+07	[-1.562e+07, -1.562e+07]
CAMELYON16	Reversed	Severe	Diversity	Processed examples	0	[0, 0]
CAMELYON16	Reversed	Severe	Diversity	Parameters	0	[0, 0]
CAMELYON16	Reversed	Severe	Diversity	Unique exposures	0	[0, 0]
CAMELYON16	Reversed	Severe	Prevalence	Accelerator hours	0	[0, 0]
CAMELYON16	Reversed	Severe	Prevalence	Peak memory (bytes)	512	[512, 512]
CAMELYON16	Reversed	Severe	Prevalence	Processed examples	0	[0, 0]
CAMELYON16	Reversed	Severe	Prevalence	Parameters	0	[0, 0]
CAMELYON16	Reversed	Severe	Prevalence	Unique exposures	0	[0, 0]
CAMELYON16	Native	Moderate	Nominal support	Accelerator hours	0	[0, 0]
CAMELYON16	Native	Moderate	Nominal support	Peak memory (bytes)	512	[512, 512]
CAMELYON16	Native	Moderate	Nominal support	Processed examples	0	[0, 0]
CAMELYON16	Native	Moderate	Nominal support	Parameters	0	[0, 0]
CAMELYON16	Native	Moderate	Nominal support	Unique exposures	0	[0, 0]
CAMELYON16	Native	Moderate	Independent support	Accelerator hours	0	[0, 0]
CAMELYON16	Native	Moderate	Independent support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
CAMELYON16	Native	Moderate	Independent support	Processed examples	0	[0, 0]
CAMELYON16	Native	Moderate	Independent support	Parameters	0	[0, 0]
CAMELYON16	Native	Moderate	Independent support	Unique exposures	0	[0, 0]
CAMELYON16	Native	Moderate	Difficulty	Accelerator hours	0	[0, 0]
CAMELYON16	Native	Moderate	Difficulty	Peak memory (bytes)	512	[512, 512]
CAMELYON16	Native	Moderate	Difficulty	Processed examples	0	[0, 0]
CAMELYON16	Native	Moderate	Difficulty	Parameters	0	[0, 0]
CAMELYON16	Native	Moderate	Difficulty	Unique exposures	0	[0, 0]
CAMELYON16	Native	Moderate	Diversity	Accelerator hours	0.001	[0.001, 0.001]
CAMELYON16	Native	Moderate	Diversity	Peak memory (bytes)	4.18e+06	[3.958e+06, 4.388e+06]
CAMELYON16	Native	Moderate	Diversity	Processed examples	0	[0, 0]
CAMELYON16	Native	Moderate	Diversity	Parameters	0	[0, 0]
CAMELYON16	Native	Moderate	Diversity	Unique exposures	0	[0, 0]
CAMELYON16	Native	Moderate	Prevalence	Accelerator hours	0	[0, 0]
CAMELYON16	Native	Moderate	Prevalence	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
CAMELYON16	Native	Moderate	Prevalence	Processed examples	0	[0, 0]
CAMELYON16	Native	Moderate	Prevalence	Parameters	0	[0, 0]
CAMELYON16	Native	Moderate	Prevalence	Unique exposures	0	[0, 0]
CAMELYON16	Native	Severe	Nominal support	Accelerator hours	0	[0, 0]
CAMELYON16	Native	Severe	Nominal support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
CAMELYON16	Native	Severe	Nominal support	Processed examples	0	[0, 0]
CAMELYON16	Native	Severe	Nominal support	Parameters	0	[0, 0]
CAMELYON16	Native	Severe	Nominal support	Unique exposures	0	[0, 0]
CAMELYON16	Native	Severe	Independent support	Accelerator hours	0	[0, 0]
CAMELYON16	Native	Severe	Independent support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
CAMELYON16	Native	Severe	Independent support	Processed examples	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
CAMELYON16	Native	Severe	Independent support	Parameters	0	[0, 0]
CAMELYON16	Native	Severe	Independent support	Unique exposures	0	[0, 0]
CAMELYON16	Native	Severe	Difficulty	Accelerator hours	0	[0, 0]
CAMELYON16	Native	Severe	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
CAMELYON16	Native	Severe	Difficulty	Processed examples	0	[0, 0]
CAMELYON16	Native	Severe	Difficulty	Parameters	0	[0, 0]
CAMELYON16	Native	Severe	Difficulty	Unique exposures	0	[0, 0]
CAMELYON16	Native	Severe	Diversity	Accelerator hours	0.001	[0.001, 0.001]
CAMELYON16	Native	Severe	Diversity	Peak memory (bytes)	-1.627e+07	[-1.628e+07, -1.627e+07]
CAMELYON16	Native	Severe	Diversity	Processed examples	0	[0, 0]
CAMELYON16	Native	Severe	Diversity	Parameters	0	[0, 0]
CAMELYON16	Native	Severe	Diversity	Unique exposures	0	[0, 0]
CAMELYON16	Native	Severe	Prevalence	Accelerator hours	0	[0, 0]
CAMELYON16	Native	Severe	Prevalence	Peak memory (bytes)	512	[512, 512]
CAMELYON16	Native	Severe	Prevalence	Processed examples	0	[0, 0]
CAMELYON16	Native	Severe	Prevalence	Parameters	0	[0, 0]
CAMELYON16	Native	Severe	Prevalence	Unique exposures	0	[0, 0]
CAMELYON16	—	Balanced	Nominal support	Accelerator hours	0	[0, 0]
CAMELYON16	—	Balanced	Nominal support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
CAMELYON16	—	Balanced	Nominal support	Processed examples	0	[0, 0]
CAMELYON16	—	Balanced	Nominal support	Parameters	0	[0, 0]
CAMELYON16	—	Balanced	Nominal support	Unique exposures	0	[0, 0]
CAMELYON16	—	Balanced	Independent support	Accelerator hours	0	[0, 0]
CAMELYON16	—	Balanced	Independent support	Peak memory (bytes)	512	[512, 512]
CAMELYON16	—	Balanced	Independent support	Processed examples	0	[0, 0]
CAMELYON16	—	Balanced	Independent support	Parameters	0	[0, 0]
CAMELYON16	—	Balanced	Independent support	Unique exposures	0	[0, 0]
CAMELYON16	—	Balanced	Difficulty	Accelerator hours	0	[0, 0]
CAMELYON16	—	Balanced	Difficulty	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
CAMELYON16	—	Balanced	Difficulty	Processed examples	0	[0, 0]
CAMELYON16	—	Balanced	Difficulty	Parameters	0	[0, 0]
CAMELYON16	—	Balanced	Difficulty	Unique exposures	0	[0, 0]
CAMELYON16	—	Balanced	Diversity	Accelerator hours	0.001	[0.001, 0.001]
CAMELYON16	—	Balanced	Diversity	Peak memory (bytes)	2.181e+07	[2.023e+07, 2.337e+07]
CAMELYON16	—	Balanced	Diversity	Processed examples	0	[0, 0]
CAMELYON16	—	Balanced	Diversity	Parameters	0	[0, 0]
CAMELYON16	—	Balanced	Diversity	Unique exposures	0	[0, 0]
CAMELYON16	—	Balanced	Prevalence	Accelerator hours	0	[0, 0]
CAMELYON16	—	Balanced	Prevalence	Peak memory (bytes)	512	[512, 512]
CAMELYON16	—	Balanced	Prevalence	Processed examples	0	[0, 0]
CAMELYON16	—	Balanced	Prevalence	Parameters	0	[0, 0]
CAMELYON16	—	Balanced	Prevalence	Unique exposures	0	[0, 0]
PANDA	Native	Moderate	Nominal support	Accelerator hours	-0.014	[-0.044, 0.017]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
PANDA	Native	Moderate	Nominal support	Peak memory (bytes)	1.252e+10	[-3.132e+09, 2.816e+10]
PANDA	Native	Moderate	Nominal support	Processed examples	0	[0, 0]
PANDA	Native	Moderate	Nominal support	Parameters	0	[0, 0]
PANDA	Native	Moderate	Nominal support	Unique exposures	0	[0, 0]
PANDA	Native	Moderate	Independent support	Accelerator hours	-0.007	[-0.031, 0.017]
PANDA	Native	Moderate	Independent support	Peak memory (bytes)	6.264e+09	[-9.386e+09, 2.19e+10]
PANDA	Native	Moderate	Independent support	Processed examples	0	[0, 0]
PANDA	Native	Moderate	Independent support	Parameters	0	[0, 0]
PANDA	Native	Moderate	Independent support	Unique exposures	0	[0, 0]
PANDA	Native	Moderate	Difficulty	Accelerator hours	-0.022	[-0.046, 0.003]
PANDA	Native	Moderate	Difficulty	Peak memory (bytes)	1.878e+10	[9.383e+09, 2.817e+10]
PANDA	Native	Moderate	Difficulty	Processed examples	0	[0, 0]
PANDA	Native	Moderate	Difficulty	Parameters	0	[0, 0]
PANDA	Native	Moderate	Difficulty	Unique exposures	0	[0, 0]
PANDA	Native	Moderate	Diversity	Accelerator hours	0.04	[0.017, 0.062]
PANDA	Native	Moderate	Diversity	Peak memory (bytes)	5.093e+07	[-1.562e+10, 1.573e+10]
PANDA	Native	Moderate	Diversity	Processed examples	0	[0, 0]
PANDA	Native	Moderate	Diversity	Parameters	0	[0, 0]
PANDA	Native	Moderate	Diversity	Unique exposures	0	[0, 0]
PANDA	Native	Moderate	Prevalence	Accelerator hours	0.015	[0, 0.03]
PANDA	Native	Moderate	Prevalence	Peak memory (bytes)	-9.389e+09	[-1.878e+10, -3.129e+09]
PANDA	Native	Moderate	Prevalence	Processed examples	0	[0, 0]
PANDA	Native	Moderate	Prevalence	Parameters	0	[0, 0]
PANDA	Native	Moderate	Prevalence	Unique exposures	0	[0, 0]
PANDA	Native	Severe	Nominal support	Accelerator hours	-0.021	[-0.044, 0.005]
PANDA	Native	Severe	Nominal support	Peak memory (bytes)	2.191e+10	[3.139e+09, 3.756e+10]
PANDA	Native	Severe	Nominal support	Processed examples	0	[0, 0]
PANDA	Native	Severe	Nominal support	Parameters	0	[0, 0]
PANDA	Native	Severe	Nominal support	Unique exposures	0	[0, 0]
PANDA	Native	Severe	Independent support	Accelerator hours	-0.013	[-0.043, 0.018]
PANDA	Native	Severe	Independent support	Peak memory (bytes)	1.879e+10	[1.326e+07, 3.445e+10]
PANDA	Native	Severe	Independent support	Processed examples	0	[0, 0]
PANDA	Native	Severe	Independent support	Parameters	0	[0, 0]
PANDA	Native	Severe	Independent support	Unique exposures	0	[0, 0]
PANDA	Native	Severe	Difficulty	Accelerator hours	-0.002	[-0.025, 0.022]
PANDA	Native	Severe	Difficulty	Peak memory (bytes)	1.565e+10	[-3.129e+09, 3.442e+10]
PANDA	Native	Severe	Difficulty	Processed examples	0	[0, 0]
PANDA	Native	Severe	Difficulty	Parameters	0	[0, 0]
PANDA	Native	Severe	Difficulty	Unique exposures	0	[0, 0]
PANDA	Native	Severe	Diversity	Accelerator hours	0.011	[-0.015, 0.036]
PANDA	Native	Severe	Diversity	Peak memory (bytes)	2.506e+10	[9.414e+09, 3.758e+10]
PANDA	Native	Severe	Diversity	Processed examples	0	[0, 0]
PANDA	Native	Severe	Diversity	Parameters	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
PANDA	Native	Severe	Diversity	Unique exposures	0	[0, 0]
PANDA	Native	Severe	Prevalence	Accelerator hours	0.003	[-0.011, 0.018]
PANDA	Native	Severe	Prevalence	Peak memory (bytes)	9.199e+04	[-9.386e+09, 9.386e+09]
PANDA	Native	Severe	Prevalence	Processed examples	0	[0, 0]
PANDA	Native	Severe	Prevalence	Parameters	0	[0, 0]
PANDA	Native	Severe	Prevalence	Unique exposures	0	[0, 0]
PANDA	—	Balanced	Nominal support	Accelerator hours	0.025	[0.008, 0.045]
PANDA	—	Balanced	Nominal support	Peak memory (bytes)	3.424e+06	[-1.251e+10, 1.251e+10]
PANDA	—	Balanced	Nominal support	Processed examples	0	[0, 0]
PANDA	—	Balanced	Nominal support	Parameters	0	[0, 0]
PANDA	—	Balanced	Nominal support	Unique exposures	0	[0, 0]
PANDA	—	Balanced	Independent support	Accelerator hours	-0.019	[-0.042, 0.005]
PANDA	—	Balanced	Independent support	Peak memory (bytes)	1.253e+10	[6.808e+06, 2.505e+10]
PANDA	—	Balanced	Independent support	Processed examples	0	[0, 0]
PANDA	—	Balanced	Independent support	Parameters	0	[0, 0]
PANDA	—	Balanced	Independent support	Unique exposures	0	[0, 0]
PANDA	—	Balanced	Difficulty	Accelerator hours	0.021	[0.001, 0.042]
PANDA	—	Balanced	Difficulty	Peak memory (bytes)	-3.122e+09	[-1.564e+10, 9.396e+09]
PANDA	—	Balanced	Difficulty	Processed examples	0	[0, 0]
PANDA	—	Balanced	Difficulty	Parameters	0	[0, 0]
PANDA	—	Balanced	Difficulty	Unique exposures	0	[0, 0]
PANDA	—	Balanced	Diversity	Accelerator hours	0.052	[0.022, 0.081]
PANDA	—	Balanced	Diversity	Peak memory (bytes)	1.769e+10	[4.585e+09, 3.078e+10]
PANDA	—	Balanced	Diversity	Processed examples	0	[0, 0]
PANDA	—	Balanced	Diversity	Parameters	0	[0, 0]
PANDA	—	Balanced	Diversity	Unique exposures	0	[0, 0]
PANDA	—	Balanced	Prevalence	Accelerator hours	0.015	[-0.007, 0.038]
PANDA	—	Balanced	Prevalence	Peak memory (bytes)	-1.251e+10	[-2.503e+10, 512]
PANDA	—	Balanced	Prevalence	Processed examples	0	[0, 0]
PANDA	—	Balanced	Prevalence	Parameters	0	[0, 0]
PANDA	—	Balanced	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Moderate	Nominal support	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Aligned	Moderate	Nominal support	Peak memory (bytes)	-1.647e+07	[-1.647e+07, -1.647e+07]
TCGA-UT	Aligned	Moderate	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Moderate	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Moderate	Independent support	Accelerator hours	0	[0, 0]
TCGA-UT	Aligned	Moderate	Independent support	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Aligned	Moderate	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Moderate	Independent support	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Moderate	Difficulty	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Aligned	Moderate	Difficulty	Peak memory (bytes)	-1.647e+07	[-1.647e+07, -1.647e+07]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Aligned	Moderate	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Moderate	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Moderate	Diversity	Accelerator hours	0	[0, 0]
TCGA-UT	Aligned	Moderate	Diversity	Peak memory (bytes)	1.421e+07	[1.414e+07, 1.433e+07]
TCGA-UT	Aligned	Moderate	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Moderate	Diversity	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Moderate	Prevalence	Accelerator hours	0	[0, 0]
TCGA-UT	Aligned	Moderate	Prevalence	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Aligned	Moderate	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Moderate	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe	Nominal support	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Aligned	Severe	Nominal support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Aligned	Severe	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe	Independent support	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Aligned	Severe	Independent support	Peak memory (bytes)	-1.647e+07	[-1.647e+07, -1.647e+07]
TCGA-UT	Aligned	Severe	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	Independent support	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe	Difficulty	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Aligned	Severe	Difficulty	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Aligned	Severe	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe	Diversity	Accelerator hours	0.002	[0.001, 0.002]
TCGA-UT	Aligned	Severe	Diversity	Peak memory (bytes)	-1.398e+07	[-1.4e+07, -1.395e+07]
TCGA-UT	Aligned	Severe	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	Diversity	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe	Prevalence	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Aligned	Severe	Prevalence	Peak memory (bytes)	1.691e+07	[1.691e+07, 1.691e+07]
TCGA-UT	Aligned	Severe	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Moderate	Nominal support	Accelerator hours	0.001	[0, 0.001]
TCGA-UT	Reversed	Moderate	Nominal support	Peak memory (bytes)	-1.647e+07	[-1.647e+07, -1.647e+07]
TCGA-UT	Reversed	Moderate	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	Nominal support	Unique exposures	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Reversed	Moderate	Independent support	Accelerator hours	0	[0, 0]
TCGA-UT	Reversed	Moderate	Independent support	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Reversed	Moderate	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	Independent support	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Moderate	Difficulty	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Reversed	Moderate	Difficulty	Peak memory (bytes)	-1.647e+07	[-1.647e+07, -1.647e+07]
TCGA-UT	Reversed	Moderate	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Moderate	Diversity	Accelerator hours	0	[0, 0]
TCGA-UT	Reversed	Moderate	Diversity	Peak memory (bytes)	1.516e+07	[1.513e+07, 1.518e+07]
TCGA-UT	Reversed	Moderate	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	Diversity	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Moderate	Prevalence	Accelerator hours	0	[0, 0]
TCGA-UT	Reversed	Moderate	Prevalence	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Reversed	Moderate	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe	Nominal support	Accelerator hours	0.001	[0, 0.001]
TCGA-UT	Reversed	Severe	Nominal support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Reversed	Severe	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe	Independent support	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Reversed	Severe	Independent support	Peak memory (bytes)	-1.647e+07	[-1.647e+07, -1.647e+07]
TCGA-UT	Reversed	Severe	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe	Independent support	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe	Difficulty	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Reversed	Severe	Difficulty	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Reversed	Severe	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe	Diversity	Accelerator hours	0.001	[0.001, 0.002]
TCGA-UT	Reversed	Severe	Diversity	Peak memory (bytes)	-1.404e+07	[-1.408e+07, -1.401e+07]
TCGA-UT	Reversed	Severe	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe	Diversity	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe	Prevalence	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Reversed	Severe	Prevalence	Peak memory (bytes)	1.691e+07	[1.691e+07, 1.691e+07]
TCGA-UT	Reversed	Severe	Prevalence	Processed examples	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Reversed	Severe	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Native	Moderate	Nominal support	Accelerator hours	0.001	[0, 0.001]
TCGA-UT	Native	Moderate	Nominal support	Peak memory (bytes)	-1.647e+07	[-1.647e+07, -1.647e+07]
TCGA-UT	Native	Moderate	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	Native	Moderate	Independent support	Accelerator hours	0	[0, 0]
TCGA-UT	Native	Moderate	Independent support	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Native	Moderate	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	Independent support	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Native	Moderate	Difficulty	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Native	Moderate	Difficulty	Peak memory (bytes)	-1.647e+07	[-1.647e+07, -1.647e+07]
TCGA-UT	Native	Moderate	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Native	Moderate	Diversity	Accelerator hours	0	[0, 0]
TCGA-UT	Native	Moderate	Diversity	Peak memory (bytes)	1.737e+07	[1.734e+07, 1.74e+07]
TCGA-UT	Native	Moderate	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	Diversity	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Native	Moderate	Prevalence	Accelerator hours	0	[0, 0]
TCGA-UT	Native	Moderate	Prevalence	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Native	Moderate	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe	Nominal support	Accelerator hours	0	[0, 0.001]
TCGA-UT	Native	Severe	Nominal support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Native	Severe	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe	Independent support	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Native	Severe	Independent support	Peak memory (bytes)	-1.647e+07	[-1.647e+07, -1.647e+07]
TCGA-UT	Native	Severe	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe	Independent support	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe	Difficulty	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Native	Severe	Difficulty	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Native	Severe	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe	Diversity	Accelerator hours	0.001	[0.001, 0.001]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Native	Severe	Diversity	Peak memory (bytes)	-1.356e+07	[-1.359e+07, -1.353e+07]
TCGA-UT	Native	Severe	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe	Diversity	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe	Prevalence	Accelerator hours	0.001	[0, 0.001]
TCGA-UT	Native	Severe	Prevalence	Peak memory (bytes)	1.691e+07	[1.691e+07, 1.691e+07]
TCGA-UT	Native	Severe	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	Nominal support	Accelerator hours	0.002	[0.001, 0.002]
TCGA-UT	—	Balanced	Nominal support	Peak memory (bytes)	-1.657e+07	[-1.657e+07, -1.657e+07]
TCGA-UT	—	Balanced	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	Nominal support	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	Independent support	Accelerator hours	0	[0, 0]
TCGA-UT	—	Balanced	Independent support	Peak memory (bytes)	1.659e+07	[1.659e+07, 1.659e+07]
TCGA-UT	—	Balanced	Independent support	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	Independent support	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	Difficulty	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	—	Balanced	Difficulty	Peak memory (bytes)	1.179e+05	[1.179e+05, 1.179e+05]
TCGA-UT	—	Balanced	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	Difficulty	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	Diversity	Accelerator hours	0	[0, 0]
TCGA-UT	—	Balanced	Diversity	Peak memory (bytes)	8.946e+07	[8.943e+07, 8.95e+07]
TCGA-UT	—	Balanced	Diversity	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	Diversity	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	Prevalence	Accelerator hours	0.002	[0.002, 0.002]
TCGA-UT	—	Balanced	Prevalence	Peak memory (bytes)	-1.657e+07	[-1.657e+07, -1.657e+07]
TCGA-UT	—	Balanced	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	Prevalence	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	Prevalence	Unique exposures	0	[0, 0]

Table 33: Legacy matched computational-cost comparisons for five methods. The exposure-matched rerun covers the whole roster.

This legacy five-method subset cannot support a roster-wide cost conclusion. The exposure-matched rerun supplies that comparison.

References

- Yannik Queisler. Diagnosing mitigation need: Controlled shortages and signal-matched interventions in histopathology patch classification. Technical report, TU Berlin, 2026a.
- Yannik Queisler. Computational pathology and benchmark datasets: An introduction to the domain and data. Technical report, TU Berlin, 2026b.
- Yannik Queisler. Beyond prevalence: Mitigation signals and interventions for histopathology patch classification and WSI-level MIL. Technical report, TU Berlin, 2026c.